# Are short-term rentals a short-term scapegoat? Mixed evidence on the effect of vacation rentals on housing in Hawaii *

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**Abstract**

A growing literature examines the effect of transient vacation rentals on housing prices, motivated by policy debates in high-cost housing markets. Hawaii provides a particularly salient setting for this question, as worsening housing affordability has fueled intense debate over the regulation of vacation rentals. Using zoning policy changes in Hawaii county and Honolulu in 2019 as an instrument for vacation rental activity in an instrumented difference in differences design, we find several context-relevant results. Our first stage analyses reveal the policies had significant, negative effects on vacation rental activity in newly prohibited zones: in Honolulu, transient vacation rentals (as a share of housing stock) declined by 20-22%. Critically, our Wald difference in difference estimates amount to limited and inconclusive evidence vacation rentals increase housing prices. We attribute these findings to poor enforcement of the zoning restrictions and find limited evidence of displacement of rentals in disallowed areas to always-allowed areas.

***Keywords:*** *Short-term rentals; housing prices; tourism.*

## 1 Introduction

The rapid expansion of peer-to-peer short-term rental platforms (STRs) over the last decade has opened a question in urban and housing economics: how does the reallocation of residential housing stock to transient use affect the cost and composition of local housing markets? The scale of the phenomenon, both in reach and in intensity, is striking. Airbnb alone has listed more than 200 million distinct units worldwide since its founding in 2008, and policymakers–with little evidence to ground their regulatory framework–have responded with the kitchen sink: licensing caps, hosting restrictions, outright prohibitions on unhosted short-term rentals, and so on. Whether those restrictions deliver the relief proponents advertise begs an answer to an empirical question—the equilibrium response of local prices and rents to a policy-induced contraction in STR supply—on which the quasi-experimental literature has reached mixed conclusions. Estimates of the price and rent response span roughly $\pm 5\%$ across different settings and methods, and nearly every one of those estimates is recovered from a large, industrially diversified metropolitan economy. This paper brings new evidence to the debate from a setting in which its stakes are considerably higher: Hawaii.

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Hawaii has long been among the most expensive places to live in the United States. Its island geography, climate, and amenities—the same features that make Hawaii a dream destination for millions of visitors each year—impose a hard constraint on developable land. The combination of limited supply, restrictive zoning, strong amenity demand, and economic dependence on a narrow set of industries has produced persistently high housing costs.[1] The same amenity profile that generates these costs also generates considerable demand from non-resident buyers and short-term visitors (Glaeser et al., 2001), and the interaction between that demand and the state's binding supply constraint makes housing market research in Hawaii policy-relevant and economically interesting.

In this environment, transient vacation rentals have proliferated at a pace and scale that is extraordinary by any national or international benchmark. In cities that have attracted the most regulatory attention for STR proliferation (e.g., San Francisco, New York, Los Angeles, Las Vegas, Chicago, etc.), Airbnb listings account for between 0.6-1.4% of the housing stock. The 2023 statewide share in Hawaii was near 4.7%, more than three times that of San Francisco and more than seven times that of New York City. The figures are far more extreme on the neighbor islands, with Maui and Kauai having nearly 12% and 14% of their residential housing stock used as Airbnbs, respectively (Inafuku and Tyndall, 2023). In some areas, shares are as high as almost 90% (Napili-Honokawai, Maui), and recent estimates from Bonham et al. (2025) suggest 1 in 5 Maui homes are TVRs. This dynamic has led to considerable frustration among Hawaii residents, particularly among Native Hawaiians and long-time local families who view the continued and accelerating expansion of TVRs as yet another instance of the State prioritizing the interests of the visitor industry at the expense of its people–political pressure to which the 2019 zoning ordinances we exploit for identification are a direct response.

The gap between Hawaii and the settings that dominate the STR literature is a concern about external validity. Tourism-dependent economies differ from diversified metropolitan settings in ways that directly affect how an STR restriction transmits to prices. Most of the existing quasi-experimental STR papers study Los Angeles (Koster et al., 2021), Boston (Horn and Merante, 2017), Barcelona (Àngel Garcia-López et al., 2020), London (Shabrina et al., 2022a,b; Todd et al., 2022), New York (Calder-Wang, 2021; Sheppard and Udell, 2016; Almagro and Dominguez-Iino, 2024), Berlin (Duso et al., 2020), Portugal (Franco and Santos, 2021), Sydney (Thackway et al., 2022), and a broad cross-section of U.S. cities (Barron et al., 2021). In each of these settings, STRs absorb a small share of the housing stock, visitors substitute imperfectly between STRs and hotels (Zervas et al., 2017, 2021), and a meaningful share of local income is independent of the visitor economy, so an STR restriction interacts in a less unpredictable way with the city's general equilibrium. Housing and rental

[1] A driving hypothesis for high housing costs in Hawaii is the impact regulation and zoning have on housing supply, and therefore on the cost of housing. Glaeser and Gyourko (2003), Hsieh and Moretti (2017), and others have outlined the effect of zoning and land use regulation on housing prices, finding cities with the most restrictive zoning, such as San Francisco and New York, to have some of the highest housing prices in the country. Indeed, Tyndall and Kim (2024) estimate nearly 60% of the high price of condominiums in Hawaii to be attributable to the "regulatory tax" identified by Glaeser and Gyourko. Further discussion and exploration of these channels are beyond the scope of this work.

markets in Hawaii, and many other locations with tourism-dependent economies, differ substantially from those in large cities. Visitor industry spending accounts for roughly a quarter of the state's private sector GDP, tourism employment is concentrated in a handful of subsectors whose demand is in some way tied to visitor arrivals, and TVRs constitute a substantial share of visitor accommodations outside the State's resort zones.[2] A restriction on vacation rentals in Hawaii is thus a non-trivial shock to the island economy as a whole, with consequences which propagate through labor, tax, and housing markets simultaneously. To our knowledge, no study has attempted original causal identification of the effect of TVRs on housing costs in a setting where tourism and tourism-adjacent industries dominate economic activity and where TVRs account for a large share of the housing stock.[3] A causal estimate of the price response in Hawaii therefore generalizes naturally to the class of tourism-dependent locales where similar policy debates are active but comparable quasi-experimental evidence is absent.

Using the staggered adoption of Hawaii County's Bill 108 (April 2019) and Honolulu's Bill 89 (August 2019) as instruments for TVR supply in an instrumented difference in differences design in the spirit of de Chaisemartin and D'Haultfœuille (2018) and Miyaji (2024), we recover three headline facts. First, the ordinances bite on TVR supply in Honolulu but not in Hawaii county: the Bill 89 first stage reduces non-resort TVR density by approximately 22-24%, while the Bill 108 first stage is weak and statistically indistinguishable from zero at quarterly frequency. Second, the reduced form response of log transaction prices in treated non-resort zones is flat in both cohorts, and the resulting Wald-DID point estimates–though signed in the policymaker-anticipated direction in Honolulu–do not allow us to reject the null under weak instrument robust inference. We read the ensemble as a qualified null: the data are consistent with a moderate negative price response of economically meaningful magnitude, and equally consistent with no effect. Third, we estimate a property-type and spatial heterogeneity analysis of the pooled Wald-DID estimates. By property type, the Honolulu's effects separate into a negative-signed condominium price effect and a positive-signed single family residence price effect whose magnitudes net to roughly the pooled coefficient, consistent with condominium units bearing the externality channel most heavily and single family homes capitalizing the investor-demand and supply-reallocation channels. By distance to the nearest resort boundary, the first stage crunch concentrates within one kilometer of the resort line and decays with distance, so the instrument's identifying power lives almost entirely in resort-adjacent cells. Neither cut delivers a precise rejection of the policymaker's null, however, and the narrowness of

[2]Anecdotally, we conjecture a majority of Hawaii's employment is some two degrees of separation from the visitor industry or less. Including the defense industry (e.g., the Department of Defense/US military), which is also export-intensive, makes us more confident in our estimate.

[3]Similar contexts arise in tourism-forward locations such as Indonesia, the Algarve, coastal Mexico, and the Caribbean, where TVRs account for comparable proportions of the housing stock in some areas. TVR bans or restrictions in these settings would plausibly generate substantial general equilibrium effects (e.g., lost jobs, reduced visitor spending, wage and rent ripple effects) by virtue of the large share of the local economy catering to tourism. An example of these general equilibrium effects in a Hawaii sub-market is captured by Bonham et al. (2025), though external validity of such estimates are limited given the UHERO model extrapolates effects using an existing GE model rather than developing or adapting one specifically accounting for TVR market dynamics as does Calder-Wang (2021).

the realized bite traces to four specific institutional enforcement shortfalls we document in detail: broad grandfather provisions, under-resourced and complaint-driven enforcement, a hosted/unhosted reclassification margin that both ordinances left open, and the absence of platform liability teeth within our estimation window.

Our findings contributes to three active literatures. First, it adds a causally identified price estimate to the applied housing economics literature on short-term rentals, with the distinctive feature that our setting sits in the tails of both the STR share distribution and the tourism-dependence distribution. Where existing estimates cluster around small positive effects on rents and mixed effects on prices, our estimates are consistent with a modest negative price effect the policy instrument is statistically underpowered to detect, a pattern whose institutional interpretation we develop at length in section 6.3. Second, we engage the externality and welfare literature on peer-to-peer short-term rental platforms, which has emphasized the observed net price effect is a reduced form summary of three offsetting channels: investor demand, owner-occupier disamenity (i.e., externalities), and effective supply reallocation. Farronato and Fradkin (2022) show, in a structural model of competition between peer hosts and hotels, that the welfare gains from STR entry are concentrated in locations and periods in which hotel capacity is constrained–a margin that is particularly salient in Hawaii, where visitor arrivals exhibit pronounced seasonality and where resort zone hotel capacity is itself supply-constrained by the same land use regime that governs TVRs. We embed this decomposition in a toy partial equilibrium model adapted from Calder-Wang (2021) and use it to discipline a property type decomposition of our reduced form estimates, which reveals sign heterogeneity–condos up, single family residences flat-to-down–consistent with condominium units bearing the Farronato-Fradkin externality channel most heavily and single family homes capitalizing the investor demand and supply reallocation channels. Third, the paper contributes briefly to the methodological literature on DID-IVs with heterogeneous treatment effects under staggered adoption (de Chaisemartin and D'Haultfœuille, 2018; Miyaji, 2024), by implementing and comparing the stacked two-stage least squares (STS) estimator against cohort-specific Wald-DIDs and a contaminated two-way fixed effects IV benchmark. The pattern in our data–a strong first stage in TWFE-IV that inverts the sign of the reduced form because it double-counts already treated comparisons–is an illustration of the Goodman-Bacon (2021) bad comparison problem in an instrumental variables setting discussed by Miyaji (2024).

Finally, and most importantly, the paper speaks directly to the applied Hawaii-specific literature. Inafuku and Tyndall (2023) and Smith et al. (2018) document the scale and spatial pattern of STR activity in Hawaii, Bonham et al. (2025) estimate the general equilibrium effects of a hypothesized Maui apartment zone phase-out with an emphasis on visitor spending leakage, and Tyndall and Kim (2024) attribute the majority of Hawaii's condominium price premium to regulatory constraints on supply. These studies have, whether out of necessity or convenience, imported TVR price estimates from mainland US or European settings and applied them to Hawaii's policy debate. We provide causally identified estimates of the TVR price effect from Hawaii itself, and, in doing so, both question and qualify the extrapolations prior work has relied on.

The remainder of the paper is organized as follows. Section 2 introduces the institutional background of transient vacation rentals in Hawaii, the centralized two-tier land use regime under which county zoning is nested, and the legislative history of Bill 108 and Bill 89 to qualify our methodology. Section 3 describes the transaction, TVR listing, zoning, and parcel-level data. Section 4 develops the conceptual framework and the instrumented DID estimation strategy, including the stacked two-stage aggregator for staggered adoption. Section 5 presents the headline results and the ensemble reading of the four main specifications. Section 6 stress-tests the results against additional selection on observables, spatial displacement and other SUTVA violations, and the institutional enforcement gap we argue is the binding constraint on the first stage signal. Section 7 probes heterogeneity of our estimates across property types and space to understand the driving mechanisms of price change we pose in section 4.1. Finally, section 8 concludes with a discussion of implications for policy, for the broader tourism-dependent locales to which our setting generalizes, and for future research this design could support.

# 2 Background

## 2.1 Transient vacation rentals in Hawaii

Transient vacation rentals (TVR), commonly referred to as short-term rentals or short-term vacation rentals, are a type of short-term rental (STR), a broad term which encompasses both transient vacation rentals, bed and breakfasts (B&Bs), motels, and hostels. TVRs are often defined as units continuously rented to individuals not related by blood to the property owner for less than a minimum continuous occupancy period. In public discourse, the term transient vacation rental (and also short-term rental) refer almost exclusively to units listed on peer-to-peer sharing platforms like Airbnb or VRBO. For policy relevance, we focus our analysis on peer-to-peer TVRs, and specifically those listed on Airbnb. Table 1 provides a concise comparison of definitional differences across jurisdictions in the State of Hawaii; Appendix A.1 covers formal definitions in greater detail.

The official launch of Airbnb in 2008 made TVR operation an accessible form of alternative (and later, primary) income, and the entry of competitors like VRBO, Flipkey, and Homeaway further lowered the barriers to short-term rental operation (Inafuku and Tyndall, 2023). In Hawaii, where tourism and tourism-adjacent industries account for the single largest share of the state's economy, incentives to convert residential housing stock to vacation rentals are particularly appealing. Indeed, a 2014 study for the Honolulu Office of Community Services found "at 80 percent occupancy, the average Airbnb unit brings in about 3.5 times more revenue than a long-term rental."[4]. The number of vacation rental

[4]Both SMS Research & Marketing Services, Inc. (2016) and Smith et al. (2018) attribute this conclusion to Cassiday (2014), though the finding is not explicitly highlighted in Cassiday's 2014 report. We suspect SMS Research analyzed the data provided by Cassiday, and that Smith and colleagues cited SMS Research's findings.

| Jurisdiction | Statutory Definition | Minimum Occupancy | Owner-Presence Rule |
|---|---|---|---|
| **State** | "Furnishing of a room, apartment, suite, single family dwelling... to a transient" | $<$ 180 days | Unspecified |
| **Honolulu** | "A dwelling or lodging unit... for compensation to transient occupants" | $<$ 30 days | Owner/operator need not be present |
| **Hawaii County** | "A dwelling unit of which the owner or operator does not reside on the lot site... no more than five bedrooms" | $\leq$ 30 days | Owner/operator must not reside on-site |
| **Maui** | "Occupancy of a dwelling or lodging unit by transients" | $<$ 180 days | Owner/operator proximity requirements ($<$ 1 hr/30 mi) for STRHs |
| **Kauai** | "A dwelling unit... provided to transient occupants for compensation or fees" | $\leq$ 180 days | Not part of relevant TVR definition |

Table 1: Comparison of TVR definitions by counties in Hawaii

units across the state grew by 35 percent between 2015 and 2017 alone, and by 2023 approximately 30,000 of the state's 565,000 housing units–roughly some 5%–were listed as rentals on platforms like Airbnb.

The vacation rental industry in Hawaii is not primarily a matter of local homeowners renting out spare rooms, as Airbnb suggests is common for operators on their platform. Commercial operators dominate the market: as of 2018, 73.5% of Hawaii hosts operated multiple listings, and 84.8% of listings were entire homes or apartments rather than shared spaces. At least 52% of vacation rentals in the state are owned by non-residents, and among the apartment-zoned TVRs in Maui County that would be affected by recent legislative proposals, 85% have out-of-state mailing addresses Smith et al. (2018); Bonham et al. (2025). Additionally, Hawaii's low property tax rates and the shifting profile of the tourism industry make vacation rentals particularly attractive to outside speculators, who can capture substantial returns while bearing little risk. This ownership pattern means income generated by TVRs disproportionaltely flows to out of the state, while the costs, often in the form of reduced housing supply, higher rents, and neighborhood disruption, are borne by locals.

Despite the intensity of public debate and the scale of regulatory action, the relationship between TVRs and housing market outcomes in Hawaii remains poorly studied and understood. Policymakers and advocacy organizations have relied primarily on extrapolations from studies conducted on the US mainland or European cities by replicating methods and borrowing estimates from other settings without leveraging Hawaii's unique economic context (e.g., Inafuku and Tyndall 2023; Bonham et al. 2025).

### 2.2 Land use regulation in Hawaii

Hawaii operates under a centralized, two-tier land use system that is uncommon in other states. Most states delegate all zoning authority to local governments; conversely, the State of Hawaii retains considerable control over statewide land classification, which it layers on top of county zoning designations. The State broadly groups the 4.1 million acres of land in the islands into four categories: urban, rural, agricultural, and conservation districts.[5] Agricultural and conservation districts are under the primary jurisdiction of state agencies, while rural districts are under shared state-county jurisdiction. Within state-designated urban districts, each of the four counties–Honolulu, Hawaii, Maui, and Kauai–controls more granular zoning through their own land use or comprehensive zoning ordinances.[6] These county codes define the specific uses permitted on a parcel (e.g., residential, apartment, commercial, resort, industrial, etc.) and are the primary policy instruments regulating STRs. This dual structure means property owners seeking to operate an STR must comply with both the state-level district classification and county-level zoning designation. In practice, however, county-level zone designations are the primary relevant regulation property owners adhere to.

Critically for our identification strategy, the foundations for zoning and land use regulation today were established decades before the rise of TVRs or peer-to-peer vacation rental platforms. The Hawaii State Legislature passed Act 187 in 1961, establishing one of the first statewide zoning systems in the country, to address resident concerns uncontrolled development would convert prime agricultural land into scattered residential subdivisions. Indeed, the Harland Bartholomew & Associates (1961) implementation report established initial district boundaries based on geographic conditions, exisiting land use, terrain, population, and climate which still dictate land use and zoning decisions today.[7] The 1968 constitutional convention subsequently granted counties the power of self-governance and the authority to regulate land use within state designated districts. Despite amendments to state and county zoning codes over the last half-century, the foundation for those district boundaries, and their geographic and agricultural motivations, have remained largely unchanged.

[5]For more on state-level land use districts, see Appendix A.2.

[6]There are technically five counties in the State of Hawaii, the fifth being Kalawao county (consisting of a portion of the island of Molokai corresponding to the former leper colony). Politically, it is a special judicial district of Maui county. We exclude it from our discussion due to its small size and the absence of variation in variables of interest over time. For more on county land use regulation, refer to A.3.

[7]The impact of the "outdated" 1961 report on zoning today is emphasized by the discussion of a 2024 report commissioned by the State Legislature in a *2025 Hawaii Business Magazine article*, which notes: "Most remarkably, soil data collected in the 1930s, released in the 1950s, and incorporated into Land Study Bureau ratings in the 1960s continues to govern critical land use decisions–such as where solar energy projects will be established on agricultural lands in the 2030s."

### 2.3 2019 zoning ordinances

Driven by longstanding–and accelerating, with the rise of peer-to-peer platforms enabling greater access to TVRs–concerns over the unsupervised and relatively unregulated proliferation of unhosted vacation rentals in residential and rural neighborhoods, the Honolulu and Hawaii city and county councils enacted TVR-specific ordinances in 2019. [8] Hawaii county passed Bill 108 in November 2018, which went into effect as Ordinance 18-114 on April 1, 2019. The ordinance restricted new unhosted STRs, defined as whole-unit rentals in which the host is not present, to areas zoned for resort use, explicitly prophibited new TVRs in single family residential and agricultural zones, and required existing rental owners to acquire an annually renewed non-conforming use certificate (NUC). In an effort to compromise with both community members and rental owners and operators, the ordinance extended TVR elligibility to certain residential mixed-use zones within resort designated areas while explicating prohibition everywhere else. Owners of rentals operating in prohibited zones prior to April 2019 had 6 months from the ordinance enactment date to apply for and acquire an NUC. If acquired, owners were required to adhere to a set of operating rules outlined by the county's planning department. TVR-permitted and unpermitted zones are shown in the right panel of Figure 1.

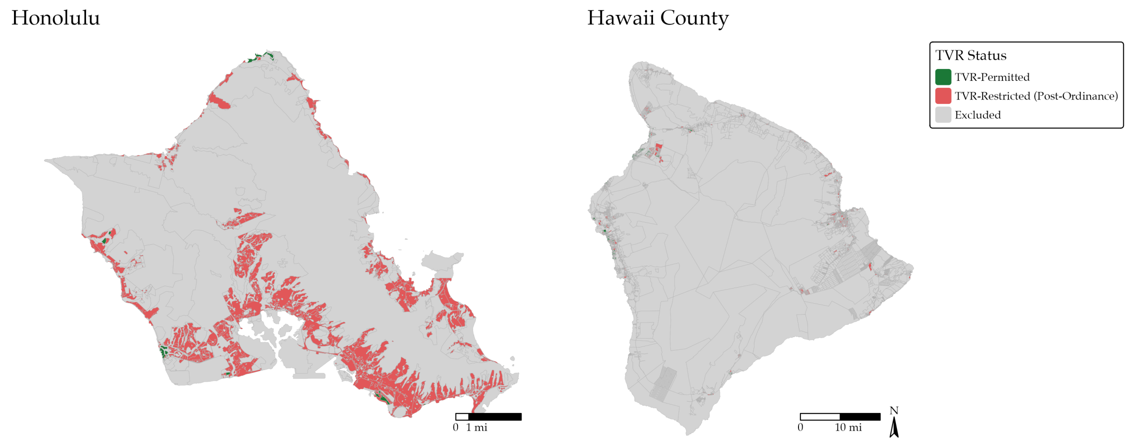



Figure 1: TVR-permitted and unpermitted zones after 2019 ordinances[9]

Honolulu's ordinance had a longer and more contentious legislative history. Despite the city's pause on NUC issuance in September 1990, TVRs continued to proliferate throughout Honolulu, and at accelerating rates due to Airbnb and VRBO. By 2019, Oahu had nearly 10,000 TVRs: nearly a third of all the state's STR stock and an approximated 3% of the city's total housing stock (Smith et al., 2018; Inafuku and Tyndall, 2023). Legislative

[8]Both counties' ordinances built on a longer history of TVR-related regulation dating to the late 1980s. Honolulu first restricted TVRs outside resort zones in 1989 (Ordinance 89-154) and stopped issuing non-conforming use certificates in September 1990. Maui similarly restricted TVRs from apartment districts in 1989 and Kauai enacted comparable restrictions in 2008. We refer the reader to A.3 for detailed county-by-county regulatory histories.

[9]Excluded zones are those primarily zoned or designated for agriculture or conservation. Before the ordinances, they already had zero or low TVR density, and thus the ordinances did not have much, if any, effect on their TVR density. As noted in A.2, most of the state's land falls into these two categories ($\sim$ 95%).

attempts in both the State Legislature and City Council to address the rapid growth of TVRs accelerated began in 2018, with several being vetoed by Rick Blangiardi, the Mayor of Honolulu, and David Ige, the then Governor of Hawaii.

On June 25, 2019, Honolulu passed Bill 89, which similarly prohibited STRs in residentially zoned areas and prohibited the advertisement of unpermitted rentals. TVRs were permitted to continue operating in resort districts, apartment zones within a 3,500 foot buffer zone of the Ko Olina resort district, and a special precinct of Waikiki. A small number of B&Bs were allowed to continue operating through a new registration system, as were grandfathered TVRs holding NUCs. It, along with specific guidelines outlined by the city's Department of Planning and Permitting, also introduced a multi-pronged enforcement approach targeting both owners/operators and peer-to-peer listing platforms to address weaknesses of prior regulatory efforts. Bill 89 went into effect as Ordinance 19-18 on August 1, 2019. Enforcement of the ordinance began almost immediately, with the city mailing some 5,000 letters to suspected violators with fines of $1,000-$10,000 per day. TVR-permitted and unpermitted zones in Honolulu are shown in the left panel of Figure 1.

# 3 Data

## 3.1 Housing market outcomes data

We leverage microdata on housing market transactions from the ATTOM recorder dataset compiled from records filed with the State of Hawaii Bureau of Conveyences. It contains administrative, structural, and land information on all real estate market transactions from Janaury 2013 through December 2025.[10] Supplementary land covariate data are retrieved from the Hawaii Statewide GIS Program and the American Community Survey, and addresses are geocoded using the US Census Geocoder with a Google Maps API fallback.[11]

Summary statistics for housing market transactions are presented in Table 2, partitioned by whether the transaction took place inside or outside of a TVR-permitted zone based on county-specific guidelines covered in sections 2.2 and 2.3. Properties in restricted zones differ most notably from properties in permitted zones in that they are slightly more expensive with a mean transaction price of $832,197 relative to $753,387 and are substantially larger by around 400 square feet. The larger proportion of condominium sales (97% relative to 49%) explains this difference. This pattern shifts over time across Hawaii county and Honolulu, and we comment on this pattern in more detail in section 7.

[10] We exclude non-market instruments (e.g., quitclaim deeds, court orders, tax sale certificates) to ensure our data do not reflect arm's-length transactions. Such transactions are identifiable both by the recorded document type and the transaction price. We defer further discussion of transaction data processing to Appendix A.5.1.

[11] The ATTOM dataset provides coordinates for 8.3% of properties, so some properties may have two separate geocoding variants. The deduplication step discussed in Appendix A.5.2 covers the selection process for consistently selecting the appropriate geocoding for each property.

Table 2: Descriptive statistics of housing transactions by TVR permissibility status

| *Panel A: TVR-Permitted Zones* ($n = 6,486$) | Mean | SD | Min | Max |
|---|---|---|---|---|
| House price (*in \$*) | 753,387.76 | 668,115.86 | 50,000 | 5,000,000 |
| House price per sqft (*in \$*) | 738.04 | 553.80 | 20.48 | 6,973.91 |
| Post-restriction (Bill 89) | 0.18 | 0.38 | 0 | 1 |
| STR listings rate | 0.13 | 0.48 | 0 | 14.33 |
| Property size (*in sqft*) | 1,054.02 | 541.13 | 500 | 6,075 |
| Parcel size (*in sqft, excl. condos*) | 200,664.68 | 410,139.07 | 1,949 | 1,103,495.06 |
| Condo | 0.97 | 0.16 | 0 | 1 |
| Number of bedrooms | 1.70 | 0.80 | 0 | 7 |
| Number of bathrooms | 1.84 | 0.98 | 0 | 20 |
| *Panel B: TVR-Restricted Zones* ($n = 27,917$) | Mean | SD | Min | Max |
| House price (*in \$*) | 832,197.12 | 657,703.06 | 50,000 | 5,000,000 |
| House price per sqft (*in \$*) | 633.24 | 423.72 | 10.70 | 8,514.29 |
| Post-restriction (Bill 89) | 0.32 | 0.47 | 0 | 1 |
| STR listings rate | 0.05 | 0.23 | 0 | 6.07 |
| Property size (*in sqft*) | 1,410.95 | 876.88 | 500 | 6,075 |
| Parcel size (*in sqft, excl. condos*) | 10,652.72 | 20,212.42 | 680 | 1,103,495.06 |
| Condo | 0.49 | 0.50 | 0 | 1 |
| Number of bedrooms | 2.74 | 1.77 | 0 | 76 |
| Number of bathrooms | 2.15 | 1.33 | 0 | 37 |

*Notes:* Sample restricted to residential properties (SFR, Condo, Multi-Family) with winsorization in line with Koster et al. (2021). The number of transactions for TVR-permitted zones is 6,486. TVR-restricted zones it is 27,917.

### 3.2 TVR data

We use data scraped roughly quarterly from Airbnb's website by Inside Airbnb as a proxy for data on TVRs from 2013 to 2025, which is summarized in Table 3. Despite some concerns about data quality (e.g., Alsherief 2020), it is common to employ open data sources like Inside Airbnb or AirDNA, in addition to self-designed web scrapers, in the housing economics and urban planning literature (e.g., Jiao and Bai 2020; Almagro and Dominguez-Iino 2024; Gurran and Phibbs 2017). Data of review timestamps, also from Inside Airbnb, provide a lower bound on the listing activity of a TVR unit. For consistency, we use an imputed span review measure (i.e., a TVR is assumed to be active in all quarters between its first review and last review) to impute the activity of TVR listings in our main specification. We discuss our selection choice with respect to different activity measures in Appendix A.4, where we find different measures of density to be comparable in their summarization of TVR activity. The quantity of interest is TVR density per zone (a similar, though different, measure to the listings rate, the probability a property is an active listing, which is also common in the literature), which is the number of TVR units in the zone computed via spatial join as a share of the zone's imputed total housing stock.

Because we use a doubly approximated measure of TVRs in Hawaii, we may expect our data to consistently underestimate the total number of TVRs, ultimately resulting in biased estimates of our causal effects. While warranted, we do not expect the downward bias to be particularly large, if at all present, for the following reasons. First, most properties listed on Airbnb are also cross-listed on other peer-to-peer sharing platforms like

VRBO or TripAdvisor (Zervas et al., 2021); second, we expect the proportions of properties exclusively listed on alternative peer-to-peer sharing platforms to account for an insignificant proportion of overall TVRs in Hawaii. Figure 3 from Inafuku and Tyndall (2023) appears to support this hypothesis. Use of Airbnb listing data as a proxy for TVRs is also a common measure of TVR count and density in similar studies to our own (e.g., Koster et al. 2021; Shabrina et al. 2022a,b; Horn and Merante 2017)

Table 3: Descriptive Statistics: Airbnb Listings by TVR Permissibility Status

| *Panel A: TVR-Permitted Zones* ($n = 9,419$) | mean | sd | min | max |
|---|---|---|---|---|
| Price per night (*in \$*) | 317.65 | 208.71 | 26.00 | 1,000.00 |
| Entire home/apt | 0.93 | 0.25 | 0.00 | 1.00 |
| Home sharing | 0.06 | 0.24 | 0.00 | 1.00 |
| Accommodation size (*in persons*) | 4.58 | 1.87 | 1.00 | 16.00 |
| Number of bedrooms | 1.40 | 1.13 | 0.00 | 48.00 |
| Minimum nights | 4.06 | 15.36 | 1.00 | 365.00 |
| Number of reviews | 35.19 | 64.41 | 0.00 | 1,012.00 |
| Superhost | 0.59 | 0.49 | 0.00 | 1.00 |
| *Panel B: TVR-Restricted Zones* ($n = 11,842$) | mean | sd | min | max |
| Price per night (*in \$*) | 253.70 | 162.78 | 25.00 | 1,000.00 |
| Entire home/apt | 0.93 | 0.25 | 0.00 | 1.00 |
| Home sharing | 0.06 | 0.24 | 0.00 | 1.00 |
| Accommodation size (*in persons*) | 4.44 | 1.99 | 1.00 | 16.00 |
| Number of bedrooms | 1.42 | 1.02 | 0.00 | 24.00 |
| Minimum nights | 7.61 | 21.32 | 1.00 | 365.00 |
| Number of reviews | 49.41 | 81.83 | 0.00 | 1,506.00 |
| Superhost | 0.55 | 0.50 | 0.00 | 1.00 |

*Notes:* Sample restricted to listings in zones included in the main specifications (zones with non-zero STR density in at least one pre-treatment quarter and non-missing treatment indicators) with winsorization in line with Koster et al. (2021).

Table 3 presents descriptive statistics for Airbnbs in Hawaii. Listings in resort zones and non-resort zones are similar in many dimensions, with comparable proportions of listings being hosted by superhosts, number of bedrooms, accommodation size, and proportion of whole-property rental units. The minimum night requirement is nearly twice as long in TVR-restricted zones but nowhere near the zoning restriction imposed minimum of 30 days in Honolulu or Hawaii county. TVRs in permitted zones also charge more per night on average, which we hypothesize to be the owner or operator pricing in tourist-specific amenities like proximity to ABC stores (a chain of convenience stores catering exclusively to tourists), transportation, and travel activities.

# 4 Empirical strategy

## 4.1 Conceptual framework

The sign of the causal effect of TVRs on housing prices is ambiguous ex-ante. For intuition on mechanisms of effect, we adapt a toy version of the rental market model proposed by Calder-Wang (2021) to a transactory housing market.[12] Let the transaction price in neighborhood $n$ at time $t$ be $P_{Tnt}$, and let local TVR activity be captured by some measure $A_{nt}$. Define $\boldsymbol{d}_n \equiv (d_n^R, d_n^B, d_n^C)' \in \mathbb{R}_+^3$ to the location vector characterizing the neighborhood $n$'s distance to the nearest resort zone centroid $d_n^R$, beach access point $d_n^B$, and the county's nearest urban core $d_n^C$.[13]

Total demand for properties of type $T$ in neighborhood $n$ at time $t$ are defined to be:

$$D_{Tnt}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) \equiv D_{Tnt}^{\text{own}}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) + D_{Tnt}^{\text{inv}}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n),$$

and assume $\frac{\partial D^{\text{own}}}{\partial P_{Tnt}}, \frac{\partial D^{\text{inv}}}{\partial P_{Tnt}} < 0$. Note differentiation of demand by property type is important as we reasonably expect the strength of substitution effects to differ across property types. We decompose owner demand into two parts:

$$D_{Tnt}^{\text{own}}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) = \bar{D}_{Tnt}^{\text{own}}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) - E_T(A_{nt}; \boldsymbol{d}_n).$$

$\bar{D}_{Tnt}^{\text{inv}}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n)$ captures baseline residential demand for type $T$ properties in neighborhood $n$, perhaps as a function of structural or land characteristics, while $E(A_{nt}; \boldsymbol{d}_n)$ captures the disamenity of greater TVR activity to local buyers attributable to congestion, noise, or loss of the neighborhood's character. Assume $E_T(A_{nt}; \boldsymbol{d}_n) \geq 0$ and $\frac{\partial E_T}{\partial A_{nt}} > 0$. Suppose also that $E_T(A_{nt}; \boldsymbol{d}_n)$ is increasing in $\boldsymbol{d}_n$ under the assumption the disamenity of the marginal TVR in residential areas far from resort or beachfront areas where transience may be expected by local residents. Similarly, we decompose investor demand into two parts:

$$D_{Tnt}^{\text{inv}}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) = \bar{D}_{Tnt}^{\text{inv}}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) + I_T(A_{nt}; \boldsymbol{d}_n).$$

$I_T(A_{nt}; \boldsymbol{d}_n)$ corresponds to the additional investor interest in rentals of property type $T$ in the neighborhood, which we assume to be informed by current TVR activity and proximity of visitor amenities. As in the Calder-Wang model, assume neighborhood housing stock for type $T$ properties to be fixed at $S_{Tn}^F$ with some being reallocated to use for rentals:

$$S_{Tnt}(A_{nt}) = S_{Tn}^F - R_T(A_{nt}; \boldsymbol{d}_n),$$

where we assume $R_T(A_{nt}; \boldsymbol{d}_n) \geq 0$.[14] Equilibrium transaction prices clear the market:

[12] I develop an alternative auction-style model of the transaction market–which is more directly reminiscent of the Calder-Wang model–with identical predictions to the toy model presented here in Appendix A.6. It conceptually identifies a general equilibrium spillover which is unobserved in our toy model and is a promising area for future research.

[13] We specify "nearest" urban core for cases like the City and County of Honolulu, which have two possible urban "cores" (Kapolei on the West side; Honolulu on the East side).

[14] Trivially, we can choose $R_T(A_{nt}; \boldsymbol{d}_n) = A_{nt}$ if there is no homesharing in the TVR industry in neighborhood $n$ and TVR operation decisions are independent of location. However, if homesharing is common (e.g., if most of the TVRs are B&Bs) then $R_T(A_{nt}) < A_{nt}$. Rigorously, we might consider a functional form for reallocation $R_T(A_{nt}; \boldsymbol{d}_n) = g(\boldsymbol{d}_n) \cdot A_{nt}$, where $g(\cdot)$ is an increasing real-valued function akin to a "propensity to homeshare" as a function of amenity proximity (resort distance) and beach access.

$$D^{\text{own}}_{Tnt}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) + D^{\text{inv}}_{Tnt}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) = S^F_{Tn} - R_T(A_{nt}; \boldsymbol{d}_n)$$
$$\bar{D}^{\text{own}}_{Tnt}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) + \bar{D}^{\text{inv}}_{Tnt}(P_{Tnt}, A_{nt}; \boldsymbol{d}_n) = S^F_{Tn} - R_T(A_{nt}; \boldsymbol{d}_n) + E_T(A_{nt}; \boldsymbol{d}_n) - I_T(A_{nt}; \boldsymbol{d}_n).$$

The above expression identifies three mechanisms of effect. First, $-R_T(A)$ reveals reallocation lowers supply of properties, raising prices; second, $+I_T(A; \boldsymbol{d}_n)$ suggests investor interest raises demand, raising prices; third, $-E_T(A; \boldsymbol{d}_n)$ shows disamenity of TVRs depresses demand among local buyers, lowering prices. The ambiguity in sign is explicated in the comparative statics, where differentiating with respect to $A_{nt}$ and isolating the marginal change in transaction prices in response to a marginal change in TVR activity yields:

$$\frac{\partial P_{Tnt}}{\partial A_{nt}} = \frac{R'_T(A_{nt}; \boldsymbol{d}_n) + I'_T(A_{nt}; \boldsymbol{d}_n) - E'_T(A_{nt}; \boldsymbol{d}_n)}{-\left(\frac{\partial \bar{D}^{\text{own}}_{Tnt}}{\partial P_{Tnt}} + \frac{\partial \bar{D}^{\text{own}}_{Tnt}}{\partial P_{Tnt}}\right)\Big|_{\boldsymbol{d}_n}}, \tag{1}$$

The net effect of the TVRs on transacted prices is positive if the following condition holds:

$$\frac{\partial P_{Tnt}}{\partial A_{nt}} > 0 \iff R'_T(A_{nt}) + I'_T(A_{nt}) > E'_T(A_{nt}).$$

Thus the reduced form coefficient on TVR density in our econometric framework captures the net partial equilibrium effect of three mechanisms: an outward shift in investor demand, an inward shift in effective housing supply, and an inward shift in owner-occupier demand due to TVR-driven externalities. Importantly, observe our simply model for properties of type $T$ predicts two neighborhoods with the same $A_{nt}$ but different $\boldsymbol{d}_n$ will have different price responses to the same TVR shock.

### 4.2 Econometric framework

Our primary interest is in the effect of transient vacation rentals on transacted home prices, though we may also be interested in the effect of the zoning ordinances on the density and spatial distribution of TVRs. The ideal experiment to answer the causal question of interest is the following. Randomly assign TVR conversion status, whether or not a home in a neighborhood is converted into a vacation rental, to a small set of residential properties in two non-contiguous neighborhoods with a comparable housing stock decomposition and similar demographic and locational characteristics.[15] In the treated neighborhood, the randomly selected set of properties are converted to full-time transient rental use. In the control neighborhood, the selected set of homes are left unconverted. Property values in both neighborhoods are observed at regular intervals before and after the forced conversion to materialize all three effect channels as the housing market for each neighborhood adjusts to the new supply of TVRs.

[15]The non-contiguity of the two neighborhoods minimizes spatial spillover effects: both direct externalities from TVR activity and displacement of investor demand from the control neighborhood–where the set of randomly selected prospective TVR properties remain owner-occupied–to the treated neighborhood–where the set of randomly selected prospective TVR properties convert to vacation rentals.

Even in this idealized setting, the experiment faces noncompliance as some assigned homeowners may decline to operate their home as a TVR, and some control group homeowners may begin renting their property to transients on their own initiative. The cleanest estimand would then be the local average treatment effect among compliers, or those whose TVR status is actually changed by the experimental intervention.

This experiment is trivially infeasible. We cannot legally mandate the conversion of private residences, and treatment, even at moderate scale, would generate general equilibrium effects contaminating the treatment neighborhood through housing market dynamics. A natural and parsimonious quasi-experimental alternative to estimating both causal effects is a linear instrumental variables TSLS approach, in line with the methodology of Koster et al. (2021). Yet, such an approach is problematic in this policy setting for several reasons. First, the staggered roll out of the ordinances create many potential comparisons both across and within counties, so the relevant source of identifying variation is not immediately clear. Absent a careful definition of the causal estimand of interest, different comparisons may also recover different objects or fail to identify a well-defined causal effect. Second, some candidate comparisons are likely contaminated by interference. For instance, comparing resort (control) and non-resort (treated) zones in Honolulu may produce biased estimates if zoning restrictions displace TVR activity from newly disallowed non-resort zones to always-allowed resort zones, violating SUTVA. Third, selection bias in policy rollout may violate conditional exogeneity, and we may find non-zero covariance between the policy and TVR density despite there being a non-zero causal relationship between the two. We address these threats to identification in greater detail in section 6.

We employ the Wald-DID estimator first popularly used by Duflo (2001) and formally described by de Chaisemartin and D'Haultfœuille (2018), which scales the outcome difference in difference contrast by the difference in difference contrast of the endogenous regressor. Under common trends of the regressor and outcome variables, conditional exclusion, and monotonicity, the Wald-DID identifies the local average treatment effect on the treated, or LATET (Hudson et al., 2017). In our setting, the LATET estimand corresponds to the average causal effect of TVR density on log transaction price among complier properties whose TVR exposure changed due to the zoning ordinances. Complier properties are those that would have been TVRs absent the ordinance (e.g., purchased by an investor and converted into a rental) but were not TVRs because of it.

Aggregating LATET estimates may improve precision or interpretability. However, a naive instrumented two-way fixed effects (TWFE) specification contaminates cohort LATETs with the bad comparisons identified by Goodman-Bacon (2021) because Hawaii and Honolulu enact their ordinances at different times. We therefore first separately estimate cohort-specific Wald-DIDs–cohort 1 with Hawaii County (with treatment corresponding to the fourth quarter of 2018) against a Honolulu, Maui, and Kauai pooled control group and Cohort 2 with Honolulu (with treatment in the third quarter of 2019) against a Maui and Kauai pooled control group–then aggregate them using the stacked TSLS estimator proposed by Miyaji (2024) to construct a weighted average of cohort LATETs with guaranteed non-negative weights.

Formally, let $P_{it}$ be the transacted price of property $i$, which is of type $T(i)$, at time $t$ with land and structural characteristics $\boldsymbol{X}_{j(i)t}$ and $\boldsymbol{W}_{it}$, respectively.[16] Define $D_{j(i)t}$ to be the density of TVRs in zone $j$, in which property $i$ is located, and $Z_{j(i)t}$ be an indicator for whether zone $j$ is disallowed from hosting TVRs after the zoning restriction is enacted at time $t^\star$. To ensure a standard interpretation of our TSLS estimates, we residualize the log transacted price with respect to structural covariates and property-type fixed effects in line with hedonic methodology before running the outcome regressions specified below.[17] Let $\widehat{\log \ell_{it}}$ be the estimated log land value for property $i$. Each property is then left with a vector of excluded instruments containing only hedonic land controls $X_{j(i)t}$, which are the same for all properties in zone $j$. For each cohort, we estimate the three regressions:

$$\log P_{it} = \rho_{T(i)} + \boldsymbol{X}'_{it}\boldsymbol{\zeta} + \log \ell_{it} \tag{2}$$

$$D_{j(i)t} = \theta_{j(i)} + \eta_t + \boldsymbol{X}_{j(i)t}\boldsymbol{\xi} + \pi Z_{j(i)t} + u_{it} \tag{3}$$

$$\widehat{\log \ell_{it}} = \alpha_{j(i)} + \delta_t + \beta \hat{D}_{j(i)t} + \boldsymbol{X}'_{j(i)t}\boldsymbol{\gamma} + \varepsilon_{it}, \tag{4}$$

where we explicate the first stage and reduced form regressions for clarity. As is convention, let $\hat{D}_{j(i)t}$ be the residualized TVR density from the first stage, and define $\{\theta_j, \alpha_j\}$, $\{\eta_t, \delta_t\}$, and $\{\rho_T\}$ to be the sets of zone, time, and property type fixed effects, respectively. Then the parameter estimate corresponding to the Wald-DID estimator is $\hat{\beta}$, one for each cohort, which we denote $\hat{\beta}^c$ with $c \in \{1, 2\}$.

In the staggered time setting, denote $t^\star_c$ to be cohort $c$'s treatment date and $\mathcal{S}_c$ to be the subset of the sample including transactions in the treated county of cohort $c$ and in all counties not yet treated by time $t^\star_c$, restricted to non-resort zones.[18] We stack these subsets, defining the cohort instrument $Z^c_{j(i)t} \equiv Z_{j(i)t}\mathbf{1}\{c(i,t) = c, j \text{ in cohort } c\text{'s treated county}\}$. The stacked two-stage difference in difference regressions estimated are:

$$D_{j(i)t} = \theta_{c,j(i)} + \eta_{c,t} + \boldsymbol{X}'_{j(i)t}\xi + \pi Z^c_{j(i)t} + v_{it} \tag{5}$$

$$\widehat{\log \ell_{it}} = \alpha_{c,j(i)} + \delta_{c,t} + \beta^{STS}\hat{D}_{j(i)t} + \boldsymbol{X}'_{j(i)t}\gamma + \epsilon_{it} \tag{6}$$

[16] We crucially do not include a "distance to CBD" measure in the vector of structural covariates due to the across-county geography of the Hawaii islands (e.g., the only "true" CBD is Honolulu, which is on Oahu) and the within-county problem of there being more than one potential urban center. (e.g., Oahu has two possible urban cores: Honolulu CBD and Kapolei).

[17] This procedure is not required to obtain the headline estimates we present in tables 4 and 5. Indeed, residualizing log transacted price with respect to structural controls $\boldsymbol{W}_{it}$ and running the outcome regression with structural controls are algebraically equivalent under the Frisch-Waugh. However, this procedure is necessary to produce to the ancillary results we present in appendix A.9 as the two operations are not equivalent under mean panel aggregation. Additionally, property-type fixed effects are removed from residualized log land value to enable our heterogeneity analysis in section 7.

[18] The window for $\mathcal{S}_1$ is truncated at $t^\star_2$, the enactment date of Honolulu's zoning restriction, to keep Honolulu in the control group without contaminating it with its own later treatment; similarly, Hawaii county is excluded from $\mathcal{S}_2$ because all zones have already been treated by $t > t^\star_1$ and therefore cannot serve as a clean control for Honolulu. This holds only for the Miyaji design; the Wald-DID specifications only use Maui and Kauai as controls. We construct the stacked dataset $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2$, and let $c(i,t)$ denote the cohort index of observation $(i,t)$. An observation can appear in both sub-datasets: a transaction on Maui from 2017 appears in both $\mathcal{S}_1$ and $\mathcal{S}_2$ but with different cohort labels $c$, so it enters the stacked data twice.

Note now fixed effects interact with cohort indicators $c$ to ensure each cohort's comparison is self-contained (e.g., a Maui zone appearing in both cohorts get separate zone fixed effects in each estimation so the cohort 1 comparison and cohort 2 comparison do not contaminate each other).Standard errors are clustered at the zone level $j(i)$. Under the identifying assumptions, $\beta^{STS}$ is a weighted average of the cohort-specific LATETs: $\beta^{STS} = w_1\beta^1 + w_2\beta^2$ with $w_1, w_2 \geq 0$ and $w_1 + w_2 = 1$, proportional to the share of compliers in each cohort.

# 5 Results

This section presents our headline results, leading with Table 4, which contains baseline estimates for the regressions in equations 4 and 6 using transaction data at the zone-quarter level, and Table 5, which contains estimates of the same regressions using transaction data at the zone-year level. We present three key findings. First, the first stage is negative and statistically significant in Honolulu across all specifications and frequencies, but weak and noisy in Hawaii county. Second, the Wald-DID estimate on residual log land value is generally inconclusive, with point estimates varying in sign across cohorts, and Anderson-Rubin confidence intervals including zero in every case. Third, because Honolulu is a considerably larger sample (i.e., $N = 12,049$ transactions against $N = 3,047$ in Hawaii county) and because its first stage is credibly strong, we treat Honolulu's results as the primary case study and Hawaii county's as being mostly uninformative due to insufficient power. Taken together, these results provide limited evidence vacation rentals affect transacted housing prices in the direction policymakers and residents hypothesize.

One specification not shown in Tables 4 and 5, the Honolulu zone-year panel reported in Appendix A.9, returns a positively signed estimate whose Anderson-Rubin interval excludes zero. We defer its interpretation and further discussion to Appendix A.9, but flag it here to avoid a misrepresentation of our findings.

## 5.1 First stage: zoning restrictions on TVR density

Honolulu's first stage estimates are statistically significant at the 0.01 level and hover between $-0.2$ and $-0.22$ across frequencies and specification structures with a first stage, cluster-robust $F$ statistic between 7 and 8. Figure 2 plots the first stage dynamic path of TVR density in non-resort zones of Honolulu relative to the control counties' non-resort zones, normalized to zero at the quarter immediately preceding treatment. TVR density in non-resort zones in Honolulu began to decline in the quarter of enactment and settles approximately 20-22% below its pre-treatment trend within a year and a half. Note the pooled first stage coefficient of $-0.2042$ is statistically significant at the 0.01 level. This response is consistent with the city's initial aggressive enforcement strategy of notifying some 5,000 suspected violators with daily fines within days of the ordinance going into effect. However, we caveat the sharp decline in the first stage yearly event study plot by nothing the aggressive increase in TVR activity on Maui in the years immeidately following the pandemic and preceding the 2023 Lahaina wildfire. The pooled first stage coefficient in the yearly specification of $-0.22$ (significant at the 0.01 level) then reflects both the enforcement of the zoning ordinance and the rising TVR density in Maui. Our pre-trends discussion is not affected by this, but the interpretation of the post-period estimates is.

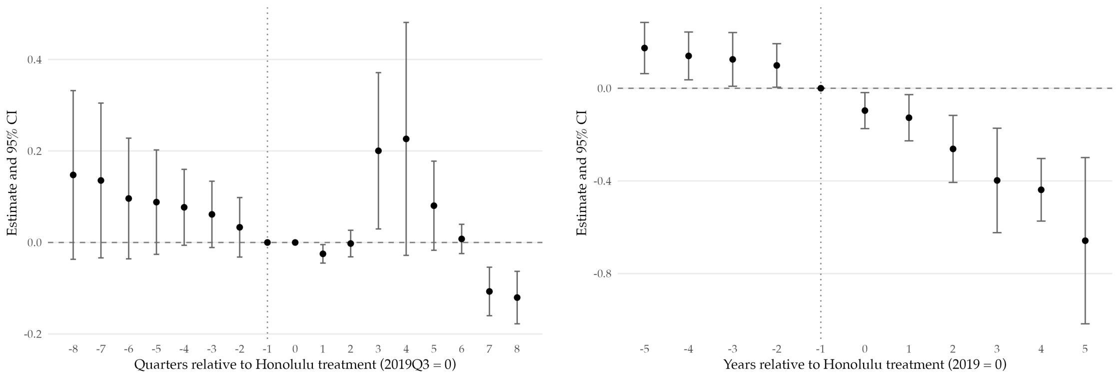



Figure 2: Event study plot of first stage coefficients for Honolulu on TVR density (Cohort 2)

The positive yet declining pre-trends from 2014 through 2018 for cohort 2's yearly event study plot are admittedly anomalous and challenge our double difference identification strategy under common trends. Indeed, non-resort zones on Kauai and Maui were on different trajectories than non-resort zones in Honolulu, with TVR activity booming on the neighbor islands while remaining relatively stagnant on Oahu. One possible explanation for this follows the intuition we pose in the one-shot auction model discussed in appendix A.6, where beach proximity drives expected vacation rental profits. Maui and Kauai non-resort housing stock is far more beach-adjacent than Oahu's, which is dominated by suburban single family communities which do not appeal heavily to TVR investors. Thus, while significant, we caution against overinterpreting our property-type pooled Honolulu results as being strictly causal due to possible violations of pre-trends testing in the cohort 2 sample and advise, in line with our qualified null narrative, interpreting our findings as being suggestively causal. We attempt more credible causal estimation in section 7, where we disentangle property-type heterogeneity.

Hawaii county's first stage estimates are, in contrast, weak and in the quarterly specification statistically indistinguishable from zero ($-0.078$ with cluster-robust standard error 0.09 and first stage $F$-statistic $F = 0.8$). The yearly specification recovers a first stage coefficient that is marginally significant at the 0.1 level ($-0.192$ with cluster-robust standard error 0.115 and first stage $F$-statistic $F = 2.8$). Figure 3 plots the corresponding first stage event study. At first inspection the figure appears to contradict the pooled estimate reported in Table 5 that individual post-treatment coefficients are small and statistically indistinguishable from zero, while the pooled yearly estimate is marginally significant. This tension resolves once results with quarterly granularity are taken into account. Indeed, the data at yearly frequency, which carries approximately five years of post-treatment data, recovers the negative and (marginally) significant coefficient estimate consistent with eventual first stage TVR responses to the zoning restriction. We therefore contend it is the appropriate point of reference for interpreting cohort 1's first stage sign.[19]

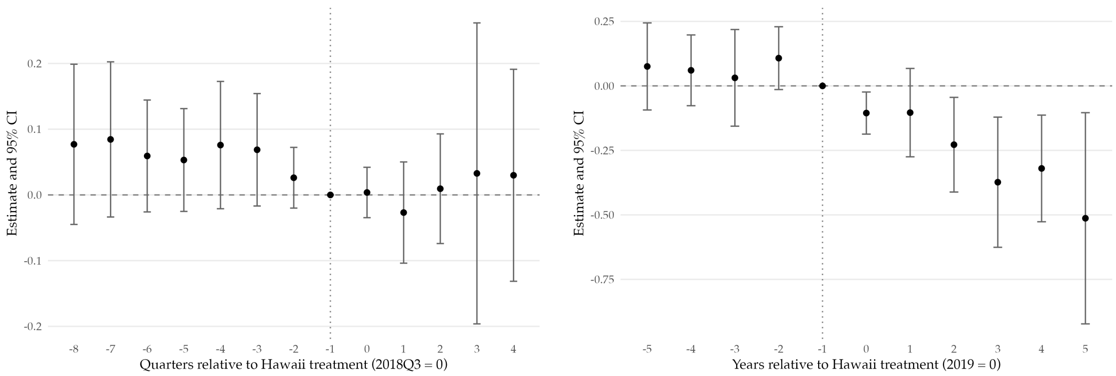



Figure 3: Event study plot of first stage coefficients for Hawaii on TVR density (Cohort 1)

The substantive question raised by the first stage results is why cohort 1's first stage remains weak even under aggregation, in marked contrast to the cohort 2 bite documented above. We hypothesize these responses to the institutional design of Hawaii county's Bill 108 relative to Honolulu's Bill 89, which we take up in detail in section 6.3.

The naive TWFE-IV (column 4 in Tables 4 and 5) yields first stage coefficients nearly twice the magnitude of the stacked estimate, with a correspondingly higher first stage $F$-statistic. Despite this apparently strong first stage, the second stage estimate flips sign relative to the stacked TSLS, from $-1.1442$ to 0.1852 and from $-0.5754$ to 0.3238 for the quarterly and yearly estimates, respectively. This pattern is consistent with Miyaji's 2024 critique that the TWFE-IV's strong first stage may be an artifact of pooling variation from the contaminated comparison (e.g., already-treated Hawaii county as control for Honolulu and thus "double counting") and that reduced form attenuation from this same comparison reverses the sign of the Wald ratio. A large $F$-statistic in this contaminated specification points away from the causal interpretation of the first stage we are interested in and toward the more conventional relevance condition of the first stage in linear instrumental variables models.

## 5.2 Reduced form: zoning effects on transacted prices

The reduced form regression asks whether the zoning instrument variation in TVR legality across zones and cohorts moved log transacted prices directly, independent of whether it moved TVR density. Figure 4 plots the reduced form dynamic path of log transacted price for both cohorts and the stacked sample. In neither cohort does the post-enactment series of event study coefficients exhibit a discrete level break at the enactment quarter, and in neither cohort does the series drift cumulatively in either direction away from zero.

The Honolulu reduced form path hugs zero throughout, and the Hawaii county form

[19]For further commentary on the deviation between the first stage event study plot coefficients and the pooled Wald-DID estimates, refer to Appendix A.7.

is noisy and statistically indistinguishable from its pre-treatment trend. We contend the flat reduced form to be the proximate reason the Wald-DID point estimates reported are inconclusive. When the numerator is statistically indistinguishable from zero, the ratio either takes a small magnitude (if the denominator is sufficiently strong, as in Honolulu's case) or explodes with unbounded confidence bands (if the denominator is near zero, as in Hawaii county's case). The two patterns visible across columns 1 and 2 of Table 4–a moderate magnitude estimate with a wide but finite AR interval for Honolulu, and a large magnitude estimate but unbounded AR interval for Hawaii county–are both attributable to the reduced form coefficients in Figure 4 being pooled together.

Critically, the flat reduced form is material on its own, independent of what it implies statistically. Under the exclusion restriction maintained throughout our study, that the zoning ordinances affected transaction prices only through their effect on TVR density, the reduced form captures the total effect on prices of being assigned to a non-resort zone in a treated county after the polices are enacted. We propose three possible interpretations for the total effect being statistically indistinguishable from zero in both cohorts. First, the ordinances may have moved TVR density by far less than policymakers intended. We discuss this further in section 6.3. Second, the ordinances may have moved TVR density as intended but with an elasticity smaller than policymakers assumed. Lastly, the observed first stage may have redirected operators out of disallowed zones and into always-allowed zones rather than forcing them to exit the TVR market. We test the plausibility of this SUTVA violation in the dispalcement test implemented in section 6.2.[20]

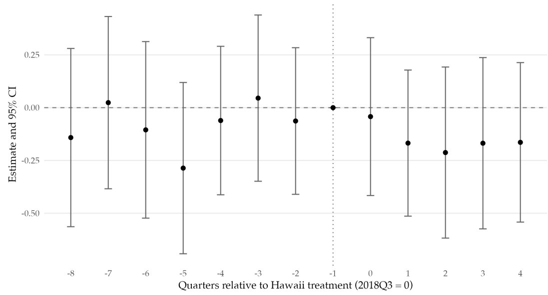
In the context of our empirical strategy, the pre-treatment coefficients in cohort 1 are jointly indistinguishable from zero at the 5% level in the first stage and reduced form event studies at both quarterly and annual frequency. For cohort 2, the joint test rejects the null of zero pre-trends at conventional levels in one or more of the corresponding specifications. We treat the cohort 2 rejection as a substantive identification concern, but we take it less as a wholesale failure of common trends more than it is evidence of small, precisely estimated pre-trends the enlarged cohort 2 sample has the power to detect.[21]

### 5.3 Two stages together: linking TVRs to housing market pressures

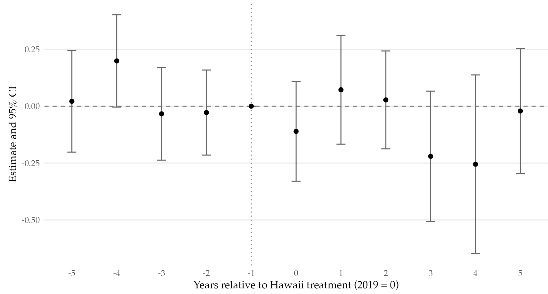
Table 4 reports the four headline specifications: the Wald-DIDs for Hawaii county and Honolulu, the Miyaji (2024) stacked TSLS aggregation, and a naive TWFE-IV for diagnostic purposes only. Table 5 reproduces the same specifications at annual frequency. The four columns in each table should be read as a single test of the policymaker's null rather than

[20]Note the flat reduced form trend is consistent with any combination of the three, not just one. An attempt to separate and interpret them is the task of the sections to follow.

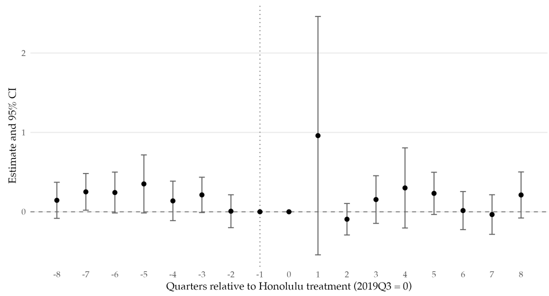
[21]Appendix A.8 report the joint Wald $F$-statistics and $p$-values for all size cohort-specific event students and the stacked TSLS aggregator. Critically, it develops and advances the identification robustness argument under which the qualified null interpretation of our cohort 2 headline Wald-DID survives rejection. That is: (1) the estimated pre-period coefficients are small in economic magnitude relative to the post-enactment response, (2) the sign of the implied bias runs against rather than toward the policymaker's hypothesized effect, and (3) the identifying variation of the Wald-DID localizes to the near-treatment region of the event window where common trends remains plausibly credible.



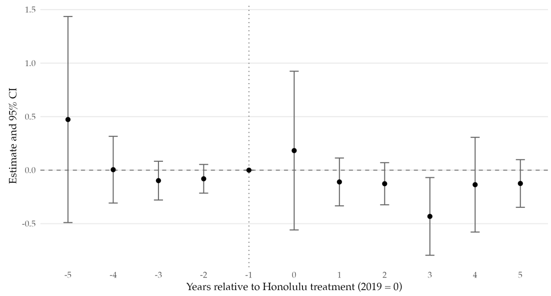

(a) Cohort 1 (Hawaii) — quarterly



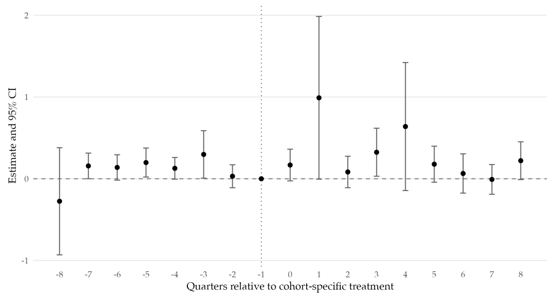

(b) Cohort 1 (Hawaii) — yearly



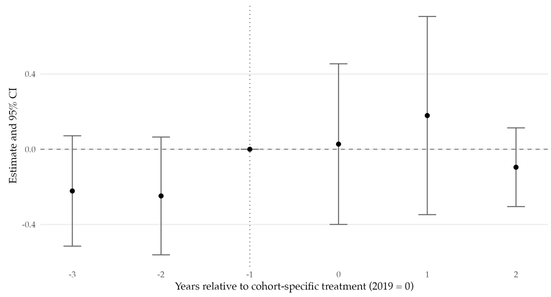

(c) Cohort 2 (Honolulu) — quarterly




(d) Cohort 2 (Honolulu) — yearly




(e) Stacked (Hawaii + Honolulu) — quarterly




(f) Stacked (Hawaii + Honolulu) — yearly

Figure 4: Event study plot of reduced form coefficients on residualized log land value

four independent tests. Each isolates a different margin of identifying variation, and the pattern across them is more informative than any single point estimate.

The Wald-DID coefficient on TVR density for Honolulu is $-0.404$ with Anderson-Rubin (AR) 95% confidence interval $[-2.583, 1.01]$. The sign is directionally consistent with the policymaker's hypothesis that Bill 89 should have reduced non-resort TVR supply and thereby softened transacted prices in treated zones, and the negative and significant results from the first stage, despite the $F$-statistic being suggestive of instrument weakness, reflects the genuine bite of Bill 89 documented in the event study evidence from section 5.1. The AR interval, however, comfortably contains zero. We read this as limited evidence for the policymaker's hypothesis rather than as confirmation of it. The annual analogue attenuates further toward zero ($-0.053$ with AR CI $[-2.246, 1.544]$), a pattern consistent with quarterly frequency preserving the sharp enforcement spikes in Honolulu the annual aggregation averages out.

Table 4: Effect of TVR Density on Housing Prices (Transaction-Level) — Quarterly, Residualized Outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | | *Panel A: Two-Stage Least Squares* | | |
| TVR density | -0.4044 | 1.2908 | -1.1442 | 0.1852 |
| | (0.7607) | (1.6661) | (0.7004) | (0.2741) |
| AR 95% CI | [-2.5833, 1.0100] | $[-\infty, +\infty]$ | [-3.8544, 0.2286] | [-0.3520, 1.0255] |
| | | *Panel B: First Stage for TVR Policy* | | |
| Zoning policy | -0.2042*** | -0.0790 | -0.2381*** | -0.4631*** |
| | (0.0722) | (0.0900) | (0.0848) | (0.1234) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 8.0 | 0.8 | 7.9 | 14.1 |
| $R^2$ | 0.237 | 0.465 | 0.333 | 0.282 |
| N observations | 12,049 | 3,047 | 27,626 | 44,318 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. The 5 structural variables (building area, age, pool, baths, bedrooms) are absorbed by the residualization step that constructs $\widetilde{\ell}$ and so are omitted from the second-stage RHS by Frisch–Waugh–Lovell. Stacked 2SLS omits all hedonic regressors; cohort $\times$ property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + year-quarter. Property-type fixed effects are absorbed by the residualization step that constructs $\widetilde{\ell}$. Fixed effects for (3): cohort $\times$ tract + cohort $\times$ time + cohort $\times$ property type. Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

Conversely, the coefficient for Hawaii is 1.291 with an empirically unbounded AR confidence interval. We attribute this to the weak and underpowered first stage with $F$-statistic $F = 0.8$, which is informative as it serves as the anchor for the compliance narrative we develop in section 6.3 by tracing the weakness of the first stage in cohort 1 to specific institutional failures of Bill 108 distinguishing it from Honolulu's Bill 89. While the annual specification yields a first stage coefficient that is marginally significant ($-0.192$, $p < 0.1$) with $F = 2.8$, the Wald-DID AR interval remains unbounded. We thus take column 2 as uninformative about the structural effect of TVR density on prices, and treat the first stage weakness itself as a substantive finding.

While the Miyaji stacked estimator returns a coefficient of $-1.14$ on TVR density (95% AR CI $[-3.854, 0.229]$) with $F = 7.9$, two features of this estimate warrant interpretative caution. Because the stacked design re-weights observations to enforce clean treated-control comparisons with cohort-specific event time windows, it is not a simple average of columns 1 and 2 but instead a weighted average of the cohort LATET estimates which, in this sample, leans on the Honolulu-driven identifying variation due the Hawaii county sample contributing no first stage signal. Additionally, both the quarter and annual stacked aggregation in tables 4 and 5 contain zero in bounded AR intervals. We view these estimates, then, as contributing to our headline result–negative, economically large, directionally consistent with the Honolulu channel–and do not take them to be statistically

Table 5: Effect of TVR Density on Housing Prices (Transaction-Level) — Yearly, Residualized Outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | -0.0533 | 0.0173 | -0.5754 | 0.3238 |
| | (0.7394) | (0.3021) | (0.5598) | (0.2626) |
| AR 95% CI | [-2.2456, 1.5444] | $[-\infty, +\infty]$ | [-3.1916, 1.7595] | [-0.1645, 1.1024] |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.2178*** | -0.1919* | -0.2947** | -0.6681*** |
| | (0.0816) | (0.1153) | (0.1337) | (0.1769) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 7.1 | 2.8 | 4.9 | 14.3 |
| $R^2$ | 0.217 | 0.380 | 0.318 | 0.254 |
| N observations | 11,548 | 4,757 | 29,025 | 43,025 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. The 5 structural variables (building area, age, pool, baths, bedrooms) are absorbed by the residualization step that constructs $\widetilde{\ell}$ and so are omitted from the second-stage RHS by Frisch–Waugh–Lovell. Stacked 2SLS omits all hedonic regressors; cohort $\times$ property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + year. Property-type fixed effects are absorbed by the residualization step that constructs $\widetilde{\ell}$. Fixed effects for (3): cohort $\times$ tract + cohort $\times$ time + cohort $\times$ property type. Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

decisive.

Indeed, the TWFE-IV coefficient of 0.1852 in Table 4 (with AR CI $[-0.352, 1.026]$) and 0.324 (with AR CI $[-0.1645, 1.1024]$) in table 5 are the counterparts to the contaminated first stage we diagnose in section 5.1. An uninformed reader encountering only the results in columns 4 may wrongly conclude TVR density raises prices due to these estimates appearing most precise, which is exactly the opposite of what the more carefully identified specifications suggest. That this misinterpretation is produced twice by exactly the bad comparison contamination the staggered adoption literature attempts to resolve is, in our view, the cleanest possible justification for the design choices we lay out in section 4.2.

### 5.4 All together: reading the ensemble

Where our zoning policy instrument has identifying power (e.g., Honolulu Wald-DID and the Miyaji STS estimates in Table 4), the price response to TVR density is signed in the direction opposite policymakers anticipated and is large enough to matter economically, but it is not statistically indistinguishable from zero under inference robust to weak instruments. Where the first stage collapses (e.g., Hawaii Wald-DID in 4 and the Miyaji STS estimates in Table 5), the IV estimates are uninformative in either direction.

The headline claim we advance is therefore a qualified null: the data are consistent with a moderate negative effect of TVR density on transacted prices at an economically consequential magnitude, but they are also consistent with no effect, and a confident rejection of the null would require a stronger first stage, which we argue in section 6.3 a tighter, realized enforcement regime may have been able to produce, or tighter identifying variation than the short policy window, made additionally noisy by COVID-19, provides. This framing motivates three investigations that follow. Within section 6, we trace first stage heterogeneity to specific institutional features of the two ordinances in 6.3 and test whether a SUTVA violation is attenuating the headline results toward zero in 6.2. Finally, we conduct a property-type and spatial heterogeneity analysis in section 7 to better understand the mechanisms we pose in section 4.1.

# 6 Robustness checks

## 6.1 Selection bias in policy rollout

Identification of the cohort-specific Wald-DIDs and of the Miyaji STS aggregator rests on the exogeneity of the zoning instrument to potential outcomes conditional on the fixed effects: the treated county post indicator must be as good as randomly assigned after netting out zone, time, and property type means. The institutional record gives us pause on this assumption. Bill 89 was enacted following a well-documented expansion of Oahu STR activity through 2016-2019 and sustained political mobilization by housing and neighborhood groups (Honolulu Star-Advertiser, 2019; Imanaka Asato LLLC, 2019). Bill 108 followed a similar rhythm in Hawaii county, compounded by the *Rosehill* agricultural zone litigation (West Hawaii Today, 2024). The two treated counties' TVR stocks entering the treatment window were already substantially larger than Maui's or Kauai's, as Figure 6 makes visually apparent, and the policy decision itself is plausibly a response to that observable level. If pre-existing TVR density both predicts treatment assignment and carries a non-causal loading onto log transacted price (e.g., through an amenity sorting channel), then our exogeneity assumption collapses even under the maintained exclusion restriction, and the Wald ratio is again a contaminated LATET.

We address this threat directly by augmenting the first stage and reduced form with the zone's own TVR density measured $L$ periods earlier, $D_{j(i),t-L}$, entered as an exogenous control in both equations:

$$D_{j(i)t} = \theta_{j(i)} + \eta_t + \lambda_1 D_{j(i),t-L} + \boldsymbol{X}'_{j(i)t}\boldsymbol{\xi} + \pi Z_{j(i)t} + u_{it} \tag{7}$$

$$\log P_{it} = \alpha_{j(i)} + \delta_t + \lambda_2 D_{j(i),t-L} + \beta \hat{D}_{j(i)t} + \boldsymbol{X}'_{j(i)t}\gamma + \varepsilon_{it}, \tag{8}$$

with $L = 4$ quarters in the quarterly panel and $L = 1$ year in the yearly panel, both computed from the balanced zone-time panel and chosen to predate the treatment window so their inclusion does not block a post-treatment channel through which the ordinances operate. Economically, the lag absorbs the cross-sectional variation in pre-existing TVR stock that counties are most plausibly reacting to when they legislate. The identifying assumption tightens: now, treatment is conditionally exogenous given the zone-county-cohort pre-treatment TVR exposure, not merely given its time-invariant zone characteristics. Note the lag is a proxy for observable selection on levels and does not address unobservable selection on trend, which we attempt to address separately in the pretrends tests of Appendix A.8.

Table 6: Effect of TVR Density on Housing Prices (Transaction-Level, Lagged TVR Density Control) — Quarterly, Residualized Outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | -5.6362 | 1.3252 | -30.2820 | 0.6886 |
| | (7.8600) | (1.2357) | (62.3042) | (3.1312) |
| AR 95% CI | [-46.3188, 8.1880] | [-0.8482, 13.8032] | $[-\infty, +\infty]$ | $[-\infty, +\infty]$ |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.0208** | -0.0770** | -0.0072 | -0.0372* |
| | (0.0085) | (0.0372) | (0.0139) | (0.0197) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 6.0 | 4.3 | 0.3 | 3.5 |
| $R^2$ | 0.238 | 0.465 | 0.334 | 0.244 |
| N observations | 12,049 | 3,047 | 27,626 | 38,950 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. The 5 structural variables (building area, age, pool, baths, bedrooms) are absorbed by the residualization step that constructs $\widetilde{\ell}$ and so are omitted from the second-stage RHS by Frisch–Waugh–Lovell. Stacked 2SLS omits all hedonic regressors; cohort × property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + year-quarter. Property-type fixed effects are absorbed by the residualization step that constructs $\widetilde{\ell}$. Fixed effects for (3): cohort × tract + cohort × time + cohort × property type. Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

Tables 6 and 7 report the transaction-level estimates at quarterly and yearly frequency. Tables A6 and A7, reported in Appendix A.9, give the corresponding zone-panel results. In this selection on observables specification, the first stage coefficient attenuates. Honolulu's quarterly headline $-0.204$ compresses to $-0.021$ once the lag is included, and the cluster-robust $F$-statistic falls from 8.0 to 6.0 in Honolulu and from 7.9 to 0.3 in the stacked column. This attenuation is expected as post-treatment density is correlated with its own lag through within-zone persistence, and it is precisely the variation we want to strip out when concerned about possible selection on levels. The reduced form also does not recover an effect not captured before. Every 95% AR interval for the transaction-level Wald-DIDs contains zero, the stacked and TWFE-IV intervals collapse to unbounded at both frequencies, and the point estimates for the Wald-DID coefficient remain inconclusive. The single qualitative anomaly (the Honolulu zone-year Wald-DID whose AR interval excludes zero in column 1 of Table A5) persists under the lagged specification and does not flip sign after conditioning on pre-existing density, a point we develop further in Appendix A.9.

Together, selection on observable confounding is consistent with our headline results

Table 7: Effect of TVR Density on Housing Prices (Transaction-Level, Lagged TVR Density Control) — Yearly, Residualized Outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| *Panel A: Two-Stage Least Squares* | | | | |
| TVR density | -2.3331 | -0.1932 | 2.9521 | 5.8071 |
| | (8.8802) | (2.0351) | (5.2158) | (10.2343) |
| AR 95% CI | [-46.5108, 26.6725] | $[-\infty, +\infty]$ | $[-\infty, +\infty]$ | $[-\infty, +\infty]$ |
| *Panel B: First Stage for TVR Policy* | | | | |
| Zoning policy | -0.0190** | -0.0290* | 0.0368 | -0.0225 |
| | (0.0082) | (0.0151) | (0.0489) | (0.0342) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 5.4 | 3.7 | 0.6 | 0.4 |
| $R^2$ | 0.218 | 0.380 | 0.321 | 0.215 |
| N observations | 11,548 | 4,757 | 29,025 | 37,781 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. The 5 structural variables (building area, age, pool, baths, bedrooms) are absorbed by the residualization step that constructs $\widetilde{\ell}$ and so are omitted from the second-stage RHS by Frisch–Waugh–Lovell. Stacked 2SLS omits all hedonic regressors; cohort $\times$ property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + year. Property-type fixed effects are absorbed by the residualization step that constructs $\widetilde{\ell}$. Fixed effects for (3): cohort $\times$ tract + cohort $\times$ time + cohort $\times$ property type. Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

and interpretation. Once the cross-sectional margin of observable selection is removed, the remaining policy-induced variation is both smaller in magnitude and less informative about the price response, and the zero null on our estimates remains unrejected. The exercise bounds rather than eliminates selection. It handles the component of selection captured by the level of pre-existing TVR density but leaves untouched selection on unobservable trend, and the reduction in first stage power is the cost our design pays for the tighter identifying assumption. Accordingly, we believe the results presented in Tables 6 and 7 reinforce the interpretation of the Wald-DID estimates we advance in Section 5.

### 6.2 Displacement tests

Our Wald-DID design assumes no violation of the Stable Unit Treatment Value Assumption (SUTVA), as is the case with most popular quasi-experimental methods. In our policy setting, SUTVA requires a property's potential outcomes depend only its own treatment status and not on the treatment assignment of any other unit. However, if Bill 89 and Bill 108 pushed investors out of newly disallowed non-resort zones and into always-allowed resort zones, both arms of the Wald ratio are contaminated by interference. The first stage denominator overstates the true supply contraction because resort zones (control units) are gaining the TVRs non-resort zones (treated units) are losing; the reduced form numerator is pushed by a price response in the zones receiving the displaced stock. The sign of the net bias on the Wald estimand is generally unclear, though the denominator channel attenuates the magnitude toward zero (de Chaisemartin and D'Haultfœuille, 2018). A similar concern applies across counties, where operators priced out of Honolulu or Hawaii county may relocate to Maui or Kauai, though this admittedly depends on the substitutability of TVRs across and within counties.

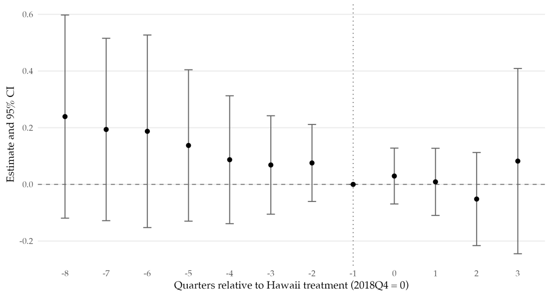
We attempt to test the plausibility of this spatial displacement hypothesis. The specification mirrors the main cross-county Wald-DID but reverses the role of resort and non-resort zones. Restricting the sample to resort-zoned properties in all four counties, we regress the zone-quarter log transaction price on TVR density instrumented by a treated county, post-treatment indicator. Under the displacement hypothesis, the first stage on resort-zone TVR density should be positive, in contrast to the negative first stage our main specification identifies on non-resort zone density. Under the null of no displacement (i.e., SUTVA is not violated in the way we hypothesize), the coefficient is zero. A positive pooled first stage indicates displaced operators in fact flock to resort zones to maintain their footing in the TVR industry. A positive post-treatment trajectory in the event study would time-locate the reallocation to the ordinance. We implement the test as a cohort-specific Wald-DID with the same Maui-Kauai resort control group used for the main specification and use event study estimates of the first stage and reduced form as diagnostics.[22]

Figure 5 plots four event study series and table 8 reports the pooled Wald-DID estimates. Note the first stage event study plots in the top row of the figure do not exhibit the post-treatment step up in resort zone TVR density the displacement hypothesis predicts. In cohort 2, resort zone density in Honolulu actually falls relative to the Maui-Kauai resort pooled control group after treatment, falling by roughly 0.2 units by event time $E_t = 6$. In cohort 1, the point estimates are negative but small and the confidence bands straddle zero for every post-treatment quarter within the truncated five-quarter window. The pooled resort zone first stage coefficients in table 8 are all negative, and none is directionally consistent with displacement absorbing regulated non-resort stock.

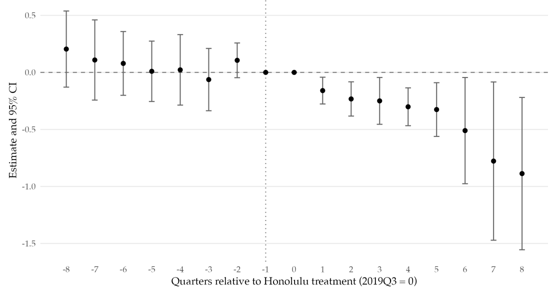
Additionally, the reduced form event study plots in the bottom row of the figure show no clear level break or sustained drift at either ordinance's enactment date. The plot for cohort 2 drifts upward by event time $E_t = 3$ to $E_t = 4$, but the movement is within the pre-treatment dispersion and by event time $E_t = 6$ the coefficient series reverts to zero. As has been the case, the series for cohort 1 is flat outside of a single period in the fourth quarter in 2018, whose wide confidence interval reflects a thinly populated event time bin. The pooled resort zone Wald-DID point estimates in table 8 are directionally positive across all specifications, but all 95% AR confidence intervals contain zero and most are unbounded, which is attributable to the underpowered and weak first stage.

[22] We formalize the spatial-displacement logic and bias direction in the Wald ratio denominator in Appendix A.12.

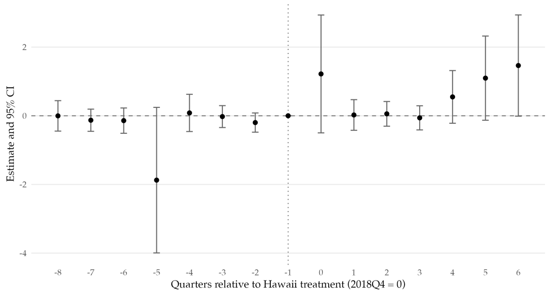
[23] Displacement test event studies estimated on the resort zone sample, by cohort. The dashed vertical line marks the cohort-specific treatment date (2018Q4 for Hawaii county, 2019Q3 for Honolulu). Under the cross-county displacement hypothesis, the post-treatment first stage path in the top row should be positive; observed point estimates are negative or near zero in both cohorts. Reduced form coefficients in the bottom row do not exhibit a level break at enactment in either cohort.



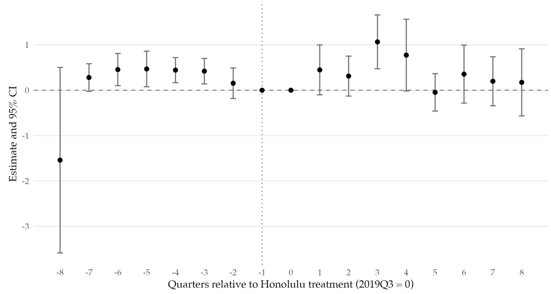

(a) Cohort 1 first stage event study on TVR density




(b) Cohort 2 first stage event study on TVR density




(c) Cohort 1 reduced form event study on residual log land value




(d) Cohort 2 reduced form event study on residual log land value

Figure 5: Displacement test event studies, resort-zone sample[23]

We present this evidence as consistent with the null of no detectable spatial displacement conditional on two material caveats we emphasize heavily.[24] Firstly, the cohort 2 post-treatment window coincides almost perfectly with the COVID-19 tourism collapse in the first quarter of 2020 through the second quarter of 2021 and the asymmetric recovery that followed. While our ensemble of fixed effects absorbs some pandemic-related shocks, the sharp negative resort zone first stage in panel (b) of figure 5 is, in our reading, a residual pandemic-induced tourism demand shock that fell on all resort zones across the state (all counties) rather than a reverse displacement effect, and it thus pushes the cohort 2 resort zone first stage toward zero, independent of whether any displacement actually occurred. Additionally, the post-pandemic surge in remote work and arrival of "digital nomads" from the mainland produced significant disruption of real estate markets for single family homes on the neighbor islands and moderate disruption on Oahu (Honolulu). We defer the discussion of compositional differences to section 7, but note here the asymmetric pressure across property types and counties from housing market dynamics (e.g., upward pressures on homes on neighbor islands relative to Oahu) may further attenuate our findings beyond the tourism-demand shock induced by the pandemic.

Secondly, the cohort 1 post-treatment window is shorter than the five quarter window

[24]We place heavy emphasis on *detectable*. The disproportionate ownership of vacation rentals by out-of-state investors would imply a greater margin of displacement than is actually observed, so we are careful to present our findings as evidence of no *detectable* displacement at the zone-level while acknowledging there is likely *undetectable* (in our specification) displacement at the individual operator/unit level.

Table 8: Effect of TVR Density on Housing Prices: Resort Zones, Residualized Outcome (Displacement Analysis)

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | -0.3785 | -2.5501 | -1.0911 | 0.0867 |
| | (0.3359) | (1.5673) | (0.7199) | (0.6263) |
| AR 95% CI | $[-1.4757, 2.2040]$ | $[-\infty, +\infty]$ | $[-13.7958, 1.3328]$ | $[-\infty, +\infty]$ |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.5915*** | -0.3323 | -0.4987** | -0.4943 |
| | (0.1699) | (0.2127) | (0.2328) | (0.3614) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 12.1 | 2.4 | 4.6 | 1.9 |
| $R^2$ | 0.574 | 0.616 | 0.560 | 0.417 |
| N observations | 2,154 | 2,929 | 5,083 | 8,930 |

*Notes:* Fixed effects for (1), (2), (4): zone + year-quarter. Property-type fixed effects are absorbed by the residualization step that constructs $\widetilde{\ell}$. Fixed effects for (3): cohort × zone + cohort × time + cohort × property-type (cohort × property-type retained as a stricter FE not redundant with the residualization). Columns (1), (2), (4) include 6 locational/demographic hedonic controls (`lot_area`, `tsu_tez`, `tsu_xtez`, `elevation_ft`, `bach_pct`, `mean_hh_income`). The 5 structural variables `bldg_area`, `age`, `pool_ind`, `baths`, `bedrooms` are absorbed by the residualization step that constructs $\widetilde{\ell}$ and so are omitted from the second-stage RHS by Frisch–Waugh–Lovell. Column (3) Stacked 2SLS omits all hedonics to preserve Miyaji (2024) identification under staggered treatment, matching the convention in the main headline table. Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument (`fixest::fitstat(m, "ivwald")`). Brackets report 95% Anderson–Rubin confidence intervals via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization.

in the main specification and covers only the pre-pandemic interval, but the event study series is imprecise for the same sparsely populated cell reasons documented in A.7 and cannot discriminate small displacement effects from zero. The appropriate interpretation is then that under the realized enforcement regime we critique in section 6.3, the test has almost no power to detect a small displacement margin and not that a zero finding supports its absence. In sum, the evidence is consistent with the null of no detectable cross-county displacement, and the headline qualified-null interpretation in section 5 is robust to spatial displacement of the magnitude the test is powered to detect.

A perhaps puzzling result is the negative and statistically significant (at the 0.01 level) first stage coefficient for Honolulu and the stacked specifications. Under enforcement of Honolulu's zoning restriction, one might reasonably expect an opposite signed effect of the policy on TVR density in resort zones. We attribute the negative coefficients to the contemporaneous crackdown on illegal (unpermitted) vacation rentals in treated (non-resort) zones in addition to control (resort) zones. Around the same time Honolulu n began enforcing Bill 89 on newly unpermitted vacation rentals, it simultaneously issued citations to already illegal rentals (e.g., units without transient accommodations licenses).Yerton (2019) The sign and statistical significance of this effect suggests enforcement of the zoning restriction in non-resort zones also spurred additional effective enforcement of past policies to eliminate already illegal units in permitted areas.

### 6.3 Compliance diagnostics

The headline results rest on reduced forms being statistically indistinguishable from zero and first stages that, outside of Honolulu at quarterly frequency, are weak. Before interpreting the null as structural, we must first rule out the alternative the null is institutional: that the ordinances did not reduce TVR density in non-resort zones because the realized enforcement regime was insufficient to make them bind. We argue the second reading is more plausible. Using the administrative permit record, the entire home share of listings by cohort and zone, and the spatial distribution of Airbnb listings before and after the two ordinances, we trace the weak cohort 1 first stage and the only partially-binding cohort 2 first stage to four institutional features distinguishing Bill 108 and Bill 89 from the clean supply restriction instrument our Wald-DID design needs for identification.

The first is a grandfather provision under Bill 108 large enough the ordinance had little scope to bind at any observable margin. Hawaii county's NUC program issued approximately 988 permits to TVR operators seeking grandfathered legal status in the six-month application window following Bill 108's enactment (Hawai'i County Planning Department, 2024).[25] In the same year, the Inside Airbnb span panel observes only 381 active listings in the county's non-resort zones, an Airbnb-to-NUC ratio of approximately 0.39. The ex ante legal pool alone already exceeds the set of listings our first stage could plausibly move, so even an enforcement regime removing every observable Airbnb unit would have left roughly 600 legally operating TVRs in non-resort zones not listed or observed on in our best-case measure of Airbnb activity. Moreover, because Airbnb undercounts total TVR activity by a non-insignificant margin, our measured first stage should be an upper bound on the true post-ordinance decline in non-resort TVR density, biased toward zero by the platform coverage gap. Both channels suggest the non-resort stock was substantially insulated from the ordinance before the first inspector was ever dispatched. Honolulu's Bill 89 grandfathered a much smaller pool, which is part of why its first stage bites at all.[26]

There is also considerable and persistent enforcement asymmetry between the two counties. Honolulu's Department of Planning and Permitting mailed some 5,000 pre-enforcement warning letters within days of Bill 89's August 2019 effective date, with a fine schedule of $1,000-$10,000 per day, and had issued 1,957 Notices of Violation and 880 Notices of Order by March 2024 (Honolulu Department of Planning and Permitting, 2024; Imanaka Asato LLLC, 2019). Hawaii county's statutory fine ceiling reaches $10,000 per day under Hawaii County Code Chapter 25 (an updated limit after more recent legislative changes), but the aggravated tier is rarely reached. Hawaii county's Planning Department has no dedicated enforcement funding stream and thus relies on complaint-driven investigation requiring resident reporting. It is also fiscally under-resourced relative to its Oahu counterpart (Hawai'i County Planning Department, 2024). A long running agricultural zone challenge by the *Rosehill* petitioners added further interpretive friction in the interim.

[25] NUC counts are sourced from the Hawaii County Planning Department's public data release and corroborated against the department's annual reports; see Appendix A.11 for the exact matching procedure.

[26] Honolulu stopped issuing new NUCs in September 1990, and by 2019 the surviving pre-1989 grandfathered feature count had fallen below one thousand for the entire island. See Appendix A.11 for the full roster.

The Hawaii Supreme Court did not uphold Bill 108's agricultural zone prohibition until a unanimous September 2024 ruling (West Hawaii Today, 2024), and administrative action on agricultural zone STVRs was correspondingly cautious in the intervening years. Neither ordinance imposed a reliable working platform liability regime within our estimation window. Bill 89 contained nominal platform liability and monthly reporting provisions, but the reporting requirement was suspended as a condition of the *Kokua Coalition v. DPP* stipulation (October 2019). Additionally, the city's November 2020 voluntary agreement with Airbnb and Expedia produced, per DPP's own admission, zero monthly reports through 2025 (Honolulu Civil Beat, 2025; Honolulu Star-Advertiser, 2019; Imanaka Asato LLLC, 2019). Bill 108 contained no platform liability provision at all. Honolulu's Ordinance 25-50 (June 2025) is the first enforceable platform liability regime in the state, and it lies outside our estimation window.[27] In short, Bill 89 waas a fat dog with dull and missing teeth; Bill 108 was a quadruple amputee mutt with nothing but gums.

The third feature is a scope gap common to both ordinances. Both restrict unhosted TVRs but leave hosted B&B home-style rentals in non-resort zones largely untouched, creating a low-cost exit option that allows operators to nominally comply by reclassification rather than actual withdrawal. We observe this reclassification response directly in the Airbnb panel. Post-2019Q3 Honolulu non-resort listings show a sharp drop in the entire-home share, consistent with operators shifting to hosted or private-room listings to remain compliant on paper. The Hawaii county non-resort series shows no such shift, consistent with the lighter enforcement pressure that made reclassification unnecessary.[28] A genuine supply restriction would show up as a fall in the total listings count; a scope-driven reclassification shows up as a fall in one listing type and a rise in another. Our first stage is only able to identify the former and not the latter.

Figure 6 documents the resulting time path. In both counties, total TVR density in non-resort zones continues to rise after ordinance enactment in 2019, and in Honolulu the post-2019 non-resort expansion is roughly four-fold through 2025. The restriction that was intended to halt the non-resort trajectory appears, on aggregate, to have at most reshaped it. Resort zone density rises in parallel, a pattern that is equally consistent with the general secular recovery in Hawaii's tourism post-2021 perhaps with a modest displacement margin. The displacement test in section 6.2 is designed to discriminate between them, but as we have just shown, the test is uninformative under the realized enforcement regime. A fourth feature, sticky operator behavior reinforced by the COVID-era tourism rebound in 2021-2022, is the most parsimonious interpretation of the post-pandemic non-resort rise: lifted TVR profitability rewarded operators who did not exit and new entrants who entered under nominal non-compliance, while the counties' enforcement capacity (or cynically, interest) remained roughly fixed.

[27] We enumerate seven Bill 108 enforcement-gap factors and a parallel set for Bill 89–including the Kokua Stipulation, under-utilization of the Bed and Breakfast Home permit cap, and evidence of majority non-compliance on Oahu as of 2023–in A.11.

[28] A full figure of the entire-home share trajectory is included in Appendix A.11. The directional result is also apparent in the ZIP-level panels of Figure 7, in which the surviving non-resort listings in 2025 concentrate in configurations compatible with hosted operation.

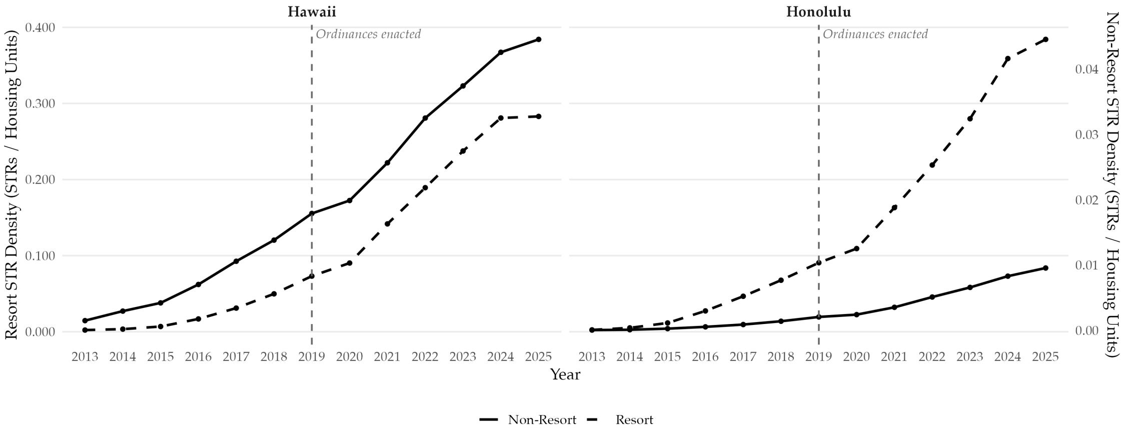



Figure 6: Within-county TVR density over time, resort versus non-resort, by county[29]

The spatial distribution of the non-compliance is equally suggestive. Figure 7 contrasts Airbnb listing locations in 2018 and 2025 across three adjacent Honolulu ZIP codes–96815 (Waikiki, resort), 96814 (Ala Moana/Kaka'ako, non-resort), and 96826 (McCully/Moiliili, non-resort). Under a strictly enforced Bill 89, we would expect the 2025 panel to show listings concentrated within 96815, with 96814 and 96826 largely emptied out. Instead, the non-resort ZIPs exhibit a dense post-ordinance expansion, with 96814 in particular densifying substantially beyond its 2018 footprint. This pattern is precisely what the aggregate first stage cannot cleanly separate. The regulated stock persists in the regulated zones, and the margin on which an ordinance was supposed to operate is narrower (and in some places, absent) in practice than it is on paper.

Taken together, these four shortcomings exhibit action-last political showmanship consistent with the weak cohort 1 first stage and the partial cohort 2 first stage that is difficult to reconcile with a purely statistical null. They also sharpen the interpretation of the Wald-DID itself. The LATET we identify is defined over the complier properties whose TVR exposure actually changed because of the ordinance, and the four features above make that subsample incredibly narrow.

# 7 Heterogeneity analysis

The pooled Wald-DID results presented in section 5 average over a transaction panel what the toy model in section 4.1 suggests ought to be heterogeneous across two dimensions: property type $T$ and distance $\boldsymbol{d}$. In pooling properties of different types, namely condominiums and single family homes, we may entangle properties whose substitutionary

[29]Plot uses span activity measure, housing-stock weighted mean. The dashed vertical line marks the 2019 ordinances. Note the dual vertical axes: in both panels, resort-zone density is read from the left axis and non-resort zone density from the right. The post-ordinance trajectory in non-resort zones is flat to rising in both counties.

[30]Airbnb listing locations in Honolulu ZIP codes 96814, 96815, and 96826 in 2018 and 2025. ZIP 96815 (Waikiki) is Bill 89-permitted; 96814 (Ala Moana/Kaka'ako) and 96826 (McCully/Moilili) are non-resort. Listings are plotted at their geocoded coordinates for any listing with at least one review in the indicated year.

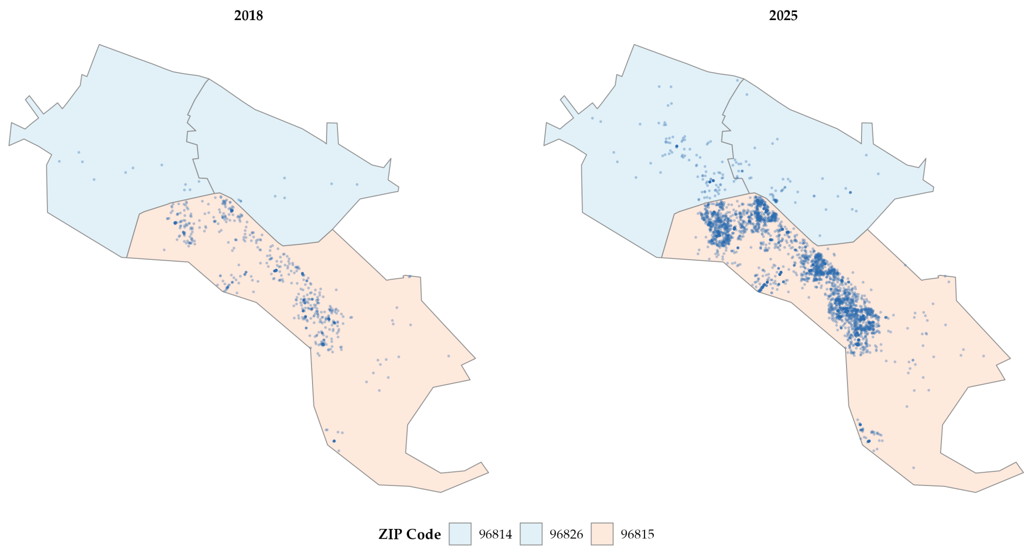



Figure 7: Airbnb listing locations, Honolulu ZIPs 96814/96815/96826, 2018 vs. 2025[30]

relationships with TVR units are, according to the priors we have discussed, very different from each other. Moreover, in pooling properties distributed non-uniformly across space (with different distance vectors $\boldsymbol{d}_n$), we muddle the spatial effect the model attempts to separate for neighborhoods with the same TVR density $A_n$ but different amenity access $\boldsymbol{d}_n$.

Here we attempt to test heterogeneity along these dimensions. Methodologically, we use the same four column estimator set up of section 5 and re-estimate the Wald-DID estimand within finer cells, partitioning by property type and, separately, within distance bands to the nearest resort zone boundary.[31] The Wald-DID estimate is then interpretable as a cell-specific LATET: the average causal effect of TVR density on residualized log land value among the complier subset whose TVR exposure changed because of the ordinance, restricted to that property type or distance band. We preserve the four column set up to maintain comparability with the headline tables, and our heterogeneity analyses can be boiled down to changes in interpretation of the underlying estimand and to changes in identifying power across cells.

### 7.1 Group heterogeneity: property-type decomposition

The sign of the price response to a change in TVR density in equation 1 is dependent on the property type $T$. One may reasonably hypothesize condominium units to be close substitutes for the type of units listed on peer-to-peer homesharing platforms in Hawaii (whole-home apartment rentals) and, in a typical Honolulu high-rise, a TVR unit and an owner-

[31]One reviewer of an earlier draft kindly advised an additional partition allowing for spatial dependence on the property type decomposition under the hypothesis resort zones contain more "hotel room"-like units (e.g., condominiums), which might result in more drastic price responses to TVR density changes. In running these estimates, however, we encountered a bin sparsity problem where there was insufficient signal for the first stage regressions in many of the partitions or regressor interactions, producing uninformative results. We exclude this additional investigation for this reason.

occupied unit share walls, elevators, pools, amenities, and much more. The disamenity channel $E_T$ is concentrated along this margin, where noise, transient corridor and automobile traffic, building wear, etc. manifest. Single family residences (SFR), by contrast, are imperfect TVR substitutes both physically and contractually as TVR quality short-term rental demand for whole homes is generally thinner far from resort zones and mortgage, insurance, or other pseudo-legal stipulations may bind harder for SFR investors.[32] What investor interest exists for SFR conversion to TVR sits more squarely in the $I_T$ channel, speculative purchase by out-of-state buyers anticipating amenity-driven returns, and its withdrawal should propagate through $R_T$ (released supply) and $I_T$ (compressed investor demand).

Recall the pooled coefficient in tables 4 and 5 are, by the Wald-DID construction at the neighborhood level, a complier-share weighted average of these two type-specific causal objects. If the two arms have opposite signs, the observed neighborhood-level estimates are uninformative not because of failed identification but because the average of two channels lacks structural interpretation. Re-estimation after partitioning by property type enables a more granular interpretation of our findings.

We re-estimate equation 4 and 6 after partitioning our sample by property type $T \in \{\text{SFR}, \text{Condo}\}$, the results of which are presented in tables 9 and 10.[33] To avoid collinearity with property type fixed effects, we drop $\{\rho_{T(i)}\}$ in our estimation of the residualized log land value. The additional estimation row labeled TVR density (+ dist control) presents the Wald-DID estimate when distance to the nearest resort zone boundary is added to the vector of covariates.

The Honolulu Wald-DID estimate for condominiums is $-0.467$ (AR interval $[-5.36, 1.95]$) with cluster-robust first stage $F$-statistic $F = 5.8$, while the estimate for SFRs is 0.437 (AR interval $[-1.258, 2.155]$) with first stage $F$-statistic $F = 21.5$. While the signs invert across types, the estimates are comparable in magnitude, and the SFR cell is the only cohort 2 cell across our heterogeneity analysis whose first stage liberally clears most bias-adjusted weak instrument thresholds. As has been the theme throughout our paper, neither estimate delivers a clean rejection of the policymaker's null on its own. However, the signs are, individually, what the toy model intuition from earlier predicts: lower TVR density produces appreciatory effects on condo units due to decreased externalities and depreciatory effects on single family residences from less supply reallocation and investor interest. Together, the pooled $-0.40$ headline estimate is an aggregation of two opposing channels which average to mostly noise.

Per our hypothesis, we also observe Bill 89 constricting condo TVR density more aggressively than single family residence TVR density. In percentage terms, the condo first stage point estimate is more than twice the SFR estimate ($-0.087$ for SFRs against $-0.223$ for condos), but the noise is roughly five times larger, and the resulting Wald-ratio noise on

[32]In cities like Chicago, for example, there are mortgage limitations for apartment units anticipated for use in the short-term rental market. Additionally, home owner association rules may prohibit use of SFRs as vacation rentals under the threat of financial penalty.

[33]Yearly counterparts to tables 9 and 10 can be found in appendix A.13 with supplementary discussion.

Table 9: Effect of TVR density on residualized log land value (transaction-level) — quarterly — Condo, residualized outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | -0.4673 | -0.2025 | -0.2545 | 0.7537 |
| | (1.1733) | (0.4609) | (0.9165) | (0.5372) |
| AR 95% CI | [-5.3608, 1.9500] | $[-\infty, +\infty]$ | [-5.3665, 2.0021] | [-0.1911, 2.5622] |
| TVR density (+ dist control) | -0.0879 | -16.3343 | -0.6013 | 0.2617 |
| | (0.5072) | (122.6083) | (0.7718) | (0.3066) |
| AR 95% CI | [-1.3878, 0.9062] | $[-\infty, +\infty]$ | [-5.4493, 0.7561] | [-0.4624, 1.3246] |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.2227** | 0.2527 | -0.2491** | -0.3684*** |
| | (0.0921) | (0.2103) | (0.1223) | (0.1291) |
| Fixed effects | Y | Y | Y (no coh×type) | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 5.8 | 1.4 | 4.1 | 8.1 |
| $R^2$ | 0.282 | 0.387 | 0.263 | 0.354 |
| N observations | 7,415 | 1,772 | 13,707 | 19,470 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: building area, age, pool, baths, bedrooms, lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. Stacked 2SLS omits hedonic regressors; cohort × property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + time. Fixed effects for (3): cohort × tract + cohort × time (cohort × property-type FE omitted because sample is a single property type). Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

the condo arm is what ultimately kills its precision. However, we present this as evidence the ordinance has a sharper compliance effect on the SFR margin than the condo margin where, anecdotally, Bill 89's official and complaint-driven enforcement mechanisms enabled easier targeting of single-property operators rather than building-level condo stacks (where ownership is opaque and grandfathering rumored to be widespread). The condo first stage measures the policy's intent more cleanly than its compliance effect, which is what ultimately identifies the second stage.

Our qualitative reading remains intact after controlling for resort zone proximity. Condo point estimates attenuate toward zero (e.g., cohort 2 moves from $-0.467$ to $-0.088$) but Anderson-Rubin intervals tighten (e.g., intervals tighten from $[-5.36, 1.95]$ to $[-1.39, 0.91]$), which we interpret as the ordinance variation in condo prices being partially absorbed by zone-level resort proximity (a feature of the urban geography of Honolulu, where condo stock concentrates in resort-adjacent areas). The fact this pattern does not carry over for SFRs and that sign separation across property types persists suggests proximity to resort amenities is not the observable confounder responsible for property type-level heterogeneity.

Table 10: Effect of TVR density on residualized log land value (transaction-level) — quarterly — SFR, residualized outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | 0.4370 | -6.3442 | 0.0054 | 0.6793* |
| | (0.7598) | (13.1380) | (0.9649) | (0.4010) |
| AR 95% CI | [-1.3576, 2.1552] | $[-\infty, +\infty]$ | [-2.8552, 1.9933] | [-0.0664, 1.7474] |
| TVR density (+ dist control) | 0.4467 | -9.4400 | -0.0614 | 0.7009* |
| | (0.7464) | (15.2147) | (0.9500) | (0.3926) |
| AR 95% CI | [-1.1661, 2.0595] | $[-\infty, +\infty]$ | [-2.4007, 1.8959] | [-0.0686, 1.6676] |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.0873*** | -0.0096 | -0.0529*** | -0.1408*** |
| | (0.0188) | (0.0134) | (0.0148) | (0.0350) |
| Fixed effects | Y | Y | Y (no coh×type) | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 21.5 | 0.5 | 12.8 | 16.2 |
| $R^2$ | 0.158 | 0.613 | 0.235 | 0.214 |
| N observations | 4,446 | 1,120 | 8,176 | 15,408 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: building area, age, pool, baths, bedrooms, lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. Stacked 2SLS omits hedonic regressors; cohort $\times$ property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + time. Fixed effects for (3): cohort $\times$ tract + cohort $\times$ time (cohort $\times$ property-type FE omitted because sample is a single property type). Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

## 7.2 Spatial heterogeneity: distance band decomposition

The same toy model predicts two neighborhoods with identical TVR density $A_{nt}$ but different $\boldsymbol{d}_n$ will respond differently to a TVR shock. For simplicity, we focus on one component of $\boldsymbol{d}_n$, the distance to the nearest resort zone boundary $d^R$, for an institutional and economic reason. Institutionally, Bill 89's text grants several explicit carve-outs around resort zones perimeters, most notably the 3,500 foot apartment zone buffer around Ko Olia and the precinct-level exception within Waikiki. The properties closest to the resort boundary are then those for which the legal status of TVR operation changes least sharply at enactment (i.e., they are more similar to the control zones in TVR permissibility). Economically, $E'_T$ is plausibly increasing in resort proximity since beachfront-adjacent residential properties are exposed to the highest transient turnover and bear the largest disamenity per TVR unit. The supply reallocation channel $R'_T$, on the other hand, should be stronger in the same cells since TVR profitability (via the marginal value of regulatory exit, which we present in the auction model in appendix A.6) is highest where TVR demand concentrates.

We restrict our investigation here exclusively to the cohort 2 sample and partition transactions into three distance bands using the property-to-nearest-resort-boundary distance defined:

$$b(i) = \begin{cases} \leq 1 \text{ km} & : d_i^R \leq 1 \\ 1-5 \text{ km} & : 1 < d_i^R \leq 5 \\ > 5 \text{ km} & : d_i^R > 5 \end{cases}$$

where $d_i^R \equiv \min_{r \in \mathcal{R}} \|x_i - r\|$ is the kilometer distance from the location of property $i$, $x_i$, to the nearest resort zone boundary, $r^\star \in \mathcal{R}$. Within each pair $(T, b) \in \{\text{Condo, SFR}\} \times \{\leq 1 \text{ km}, 1-5 \text{ km}, > 5 \text{ km}\}$, we re-estimate the cohort 2 Wald-DID on the subsample (defined using the cohort notation from section 4.2) $\mathcal{S}_2 \cap \{i : T(i) = T, b(i) = b\}$ while excluding the most distant group. The specifications are identical to the original Wald-DID specifications with the exception of a new vector of structural covariates $\boldsymbol{X}_{j(i)t}^{-TEZ}$, which excludes tsunami and extreme tsunami warning zones (recall we use these markers to proxy beach access $d^B$) as they are linearly dependent on the band restriction. We do not report cohort 1 or stacked variants of the band cut as the cohort 1 first stage is too weak to support disaggregation, and the stacked aggregator is not designed to be informative about within-cohort spatial heterogeneity. Table 11 reports specific cohort 2 Wald-DID estimates at quarterly frequency (we defer the yearly counterpart to appendix A.13).

Table 11: Cohort 2 Wald-DID on TVR density (residualized log land value), by property type and distance band — quarterly

| | Condo | | SFR | |
|---|---|---|---|---|
| | ≤1 km (1) | 1–5 km (2) | ≤1 km (3) | 1–5 km (4) |
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | -1.7694 | +0.1787 | +3.1628 | -0.2304 |
| | (3.0954) | (0.1928) | (3.2557) | (1.3501) |
| AR 95% CI | [-11.8798, 3.6747] | [-0.6448, 1.0021] | [-5.5081, 12.8154] | [-6.2684, 3.6365] |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.1152*** | -0.8038** | -0.0553*** | -0.0741*** |
| | (0.0323) | (0.3261) | (0.0174) | (0.0282) |
| First-stage F | 12.8 | 6.1 | 10.1 | 6.9 |
| $R^2$ | 0.310 | 0.177 | 0.249 | 0.141 |
| N observations | 5,813 | 1,194 | 732 | 1,571 |
| Clusters (zones) | 76 | 29 | 96 | 71 |

*Notes:* Fixed effects: zone + year-quarter. Resort-zone observations are excluded from all cells. Sample restricted to the listed distance band; > 5 km observations are intentionally excluded (chronically weak first stage and dominated by interior non-resort variation). Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument (`fixest::fitstat(m, "ivwald")`). Cells with first-stage F < 16.38 (Stock-Yogo 2005, 5% IV-bias threshold) should be interpreted under weak-instrument inference; bracketed rows report 95% Anderson–Rubin CIs constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization (grid size 200, seed 42). TVR density measure: `str_density_span` (quarterly Inside Airbnb listings per zone, span-weighted).

A pattern from table 11 we highlight is that the cluster-robust first stage $F$-statistic is higher at the boundary than in the intermediate band for both property types. That is, condos go from $F = 12.8$ at the nearest band (including $N = 5.813$ properties across 76 zones) to $F = 6.1$ at the intermediate band (including $N = 1,194$ properties across 29 zones) and

SFRs go from $F = 10.1$ at the nearest band (including $N = 732$ properties across 96 zones) to $F = 6.9$ at the intermediate band (including $N = 1,571$ properties across 71 zones). These findings suggest the instrument has the most identifying power where the legal status discontinuity in Bill 89 text is most ambiguous. The geographic pattern of compliance also dovetails with the narrative Honolulu's DPP's complaint-driven enforcement concentrated on the most visible non-compliant operators who are disproportionately located in resort-adjacent areas.

The first stage coefficients in panel B of table 11 add nuance beyond the initial reading presented above. For condos, the per-unit decline in TVR density is much larger in the intermediate band than at the boundary, despite the lower cluster-robust $F$-statistic in the intermediate band. We attribute this to the small sample noise in the intermediate band asserting itself. That is, the policy moves a high-baseline density per zone in the boundary band by a small percentage and a low baseline density per zone in the intermediate band by a larger percentage, but with much fewer effective clusters (29 zones against 76 zones), and the resulting cluster-robust Wald statistic on the excluded instrument is correspondingly weaker. For the SFRs, first stage coefficients are similar in magnitude across bands (e.g., $-0.055$ versus $-0.074$), both significant in their own right, and driven by the dispersion of cluster sizes rather than a genuine difference in policy effectiveness.

As has been the case for most of our second stage estimates, they are noisier than the first stage and are likewise cautious about pushing our results as causal evidence in the heterogeneous case. Despite the width of our AR intervals and thef act they all straddle zero, the interpretation of our second stage point estimates is not entirely intuitive at first read. The negative effect on resort-adjacent condominium prices is perhaps the only expected result as it reaffirms our prior that condominium or resort-adjacent units (much less both) are most strongly affected by the transience-based externalities which accompany the visitor and vacation rental industry. But the other three estimates (positive effects on intermediate distanced condos and resort-adjacent SFRs and a negative effect on intermediate distanced SFRs) do not immeidately align with this logic.

One possible explanation for the inflationary effect of TVR density on SFR relative to condo prices is that condo units, as a whole, tend to be located nearer to resort zones than are SFRs. Condos, then, bear externalities at point-blank range. Per the logic we advance in appendix A.6, some of these condo units may be located in the same building as or share their elevators, parking spaces, or HOA fees with owners of vacation rentals in resort-adjacent areas (many of which may be illegal under Bill 108 or 89). Conversely, single-family homes are more likely to be detached from neighborhing units and, as a result, they may not experiene the immediate neighbor nuisance externalities which penalize condo units in zones with low compliance. Still, they sit in a tourism-amenity ecosystem (e.g., restaurants, services, walkability, beaches) that are highly correlated with TVR activity. Under that mappping, resort-adjacent condo units bear an externality concentrated locally on adjacent residents (manifesting as a negative price effect) while resort-adjacent SFRs benefit from tourism agglomeration amenities (manifesting as a positive price effect).

Both of these affects attentuate with distance, though this interpretation may benefit from additional unit and distance granularity.

An alternative explanation for the more sizable resort-adjacent SFR estimate is the selection in who the Wald-DID estimate applies to (recall complier zones differ both by property type and distance in this specification). We only observe 96 resort-adjacent zones with 732 transacted single-family units, most of which are located near boundaries of the Waikiki resort zones. The "near-Waikiki SFR whose zone TVR density was changed by Bill 89's restriction" is plausibly a small outlier group including mostly luxury properties in wealthy neighborhoods like Diamond Head and Kahala. The esimate of 3.1628 (though not statistically significant under weak-instrument-robust inference) is the LATET for the compliers in that set of properties, not for SFRs across Honolulu's non-resort zones generally. The fact luxury SFR prices in wealthy neighborhoods may track tourism demand for reasons unrelated to the narrative constructed in section 4.1 warrants additional caution in reading our estimates in this section as being purely causal.

We do not push the spatial decomposition estimates beyond the question they are positioned to answer. The band cut is only more informative about where the policy was effective in reducing TVR density and less about how much it moved prices anywhere. It does, however, reveal that boundary properties are those which likely drive the identification for cohort 2, and that the type-level sign separation we document in section 7.1 is preserved spatially across both property types.

### 7.3 Concluding remarks

The two decompositions point to different predictions of the structural model and should be interpreted as one might interpret two comparative statics analyses. The property-type decomposition reveals the pooled Wald-DID averages across two opposite signed structural objects: the condo group is consistent with a regime in which the externality channel dominates and the SFR arm with a regime in which the supply-reallocated and investor-demand channels dominate. Each individual arm fails to reject the null of zero effect under weak-instrument robust inference, and we are deliberate about not reading the sign separation as confirmation of the policymaker's null. What it does allow us to do, however, is reach the interesting economic conclusion that a uniform effect interpretation of the pooled coefficient is not consistent with the within-Hawaii data (this is precisely the prediction of the model in section 4.1). A counterfactual analysis ignoring the type partition and assuming a single Wald-DID estimate applying to every property in the treated post-period would, at best, average two non-trivial channels of opposite sign and muddy an informative investigation into mechanisms of effect.

The spatial decomposition also sharpens the institutional interpretation we advance in section 4.1. The policy bite is concentrated within roughly one kilometer of the resort boundary, which is consistent with the geography of Bill 89's effective reach (e.g., resort-adjacent buffer properties are where the legal status of TVR operation changes most ambiguously at enactment) and with the geography of Honolulu's DPP's complaint-driven enforcement (i.e., resort-adjacent listings are the most visible to neighbors). A version of the policy with enforcement extending into the residential interior of Honolulu, consistent with the legislative aims of Bill 89, would, on this evidence, deliver tighter first stage results in the 1-5 kilometer bands and a real prospect of moving the Wald-DID intervals away from zero.

Under the realized regime, the effectiveness of the policy stops where the boundary-adjacent cells stop, and the headline qualified null is, in a meaningful sense, a statement about the inability of the policy instrument to bite down on TVR density in residential areas. In particular, the economic content of the qualified null shifts from "the policy did nothing" to "the policy moved opposing channels by amounts the realized enforcement regime is not powerful enough to identify separately."

We close with the methodological observation that the Wald-DID with one binary instrument and one continuous endogenous regression cannot, within a single product cell, separately identify the three structural channels outlined in the supply and demand model outlined in section 4.1. The cleanest way to recover those objects in our setting would be a fully general equilibrium model akin to the model and estimation exercise in Calder-Wang (2021). The most informative contribution of our partial equilibrium approach is our parititoning of the sample into cells where one channel is a priori dominant and confirm the observed signs are consistent with that prior. The heterogeneity analysis presented in this section attempts this exercise, and, in doing so, provides a structural narrative more aligned to the toy model than the pooled headline is capable of advancing.

# 8 Conclusion

In this paper, we investigate whether Hawaii's 2019 zoning restrictions on transient vacation rentals reduced housing prices in newly prohibited non-resort zones, and through what mechanisms. Using Hawaii county's Bill 108 and Honolulu's Bill 89 as staggered instruments for TVR supply in an instrumented difference in differences design, our answer is a carefully qualified null. Honolulu's Bill 89 moved non-resort TVR density by $20-22\%$, but the reduced form response of log transaction prices is statistically flat and the corresponding Wald-DID point estimates, while signed in the policymaker-anticipated direction, is not distinguishable from zero under weak instrument robust inference. Hawaii county's Bill 108 did not appreciably bite on TVR density at all, and its Wald-DID is correspondingly uninformative. The Miyaji (2024) stacked aggregator, our preferred resolution to the Goodman-Bacon (2021) bad comparison problem in a staggered setting, inherits the Honolulu identifying signal and returns a marginally significant negative coefficient at quarterly frequency that does not survive temporal aggregation. The disaggregated LATET estimates from our heterogeneity anlaysis tell a richer story than the pooled coefficient. Condominium and single family home prices move in opposite directions under TVR restriction (i.e., condominiums up as the external-disamenity channel is released, single family homes flat to slightly down as the investor-demand and supply-reallocation channels recede), and the first stage effect identifying either channel concentrates within one kilometer of the resort boundary. Neither partitioning rejects the policymaker's null with the precision political rhetoric demands, however, and we trace the weak reduced forms to four specific institutional features of the two ordinances we outlined in section 6.3. We argue ex post the null is at least as consistent with underenforcement as it is with a small structural price elasticity.

Three lines of extension follow naturally from the limitations of the present design. The first is a full general equilibrium model in the style of Calder-Wang (2021) or Almagro and Dominguez-Iino (2024), extended to a tourism-dependent setting in which visitor welfare, labor-market effects, and tax incidence are priced explicitly. Our partial equilibrium imputation holds the rental market, the long-term labor market, and amenity distributions fixed at realized values. However, a tourism-dependent economy's response to a large non-resort TVR removal would work through each of these channels simultaneously, and the sign and magnitude of the resulting welfare change are ambiguous. Bonham et al. (2025) take a first pass at this exercise for a Maui apartment zone phase-out using a calibrated computable general equilibrium framework. A natural next step is to couple a similar framework to micro-founded estimates of the price channel of the type we advocate. Distributionally, the fact that a substantial share of Hawaii's workforce and landowners come from different populations with non-trivially overlapping interests deserves treatment more careful than the existing literature on the mainland provides.

A second area is a more thorough treatment of SUTVA violations than our design can afford. Our cross-zone displacement test is underpowered against small, realistic displacement margins, and our cross-county test is further compromised by the COVID-19 tourism shock that arrives immediately inside the cohort 2 post-treatment window. A spatially explicit model that takes the substitutability of TVRs across resort and non-resort zones seriously, potentially in combination with the boundary-discontinuity design along county and zoning borders employed by Koster et al. (2021), would discipline the SUTVA assumption more tightly. A complementary exercise would estimate the operator exit margin, using the Airbnb data's host-level information to decompose the first stage contraction into exit, reclassification, and geographic displacement.

A third and largely untouched channel concerns the effect of TVR activity on assessed property values and, through them, on the property tax bills of Hawaii residents. If TVR-driven capitalization flows disproportionately into resort zoned or TVR-adjacent non-resort cells, and if county assessors apply near-uniform valuation procedures across zones and property types, then long-time residents in TVR-adjacent parcels could bear a regressive tax-side spillover on top of any direct housing-price effect—a second-order consequence of vacation rentals the TVR debate has almost entirely ignored. Our assessor panel, developed in a companion analysis, is tailored to exactly this question: an annual property-year DID-IV framework nearly identical to the one developed here, applied to log assessed total value, log market value, and log tax bill, with a regressivity supplement in the spirit of Berry (2021). The policy stakes of this extension are, in our view, considerably underappreciated in the current Hawaii discourse.

For Hawaii and its policymakers, the implications are blunt. The applied Hawaii literature has been forced to import estimates from mainland and European settings that differ from Hawaii in precisely the dimensions that matter most for our context, and that importation has quietly disciplined a policy discourse whose stakes are considerably higher than the evidence base has warranted. Credible causal estimates recovered from the Hawaii setting itself are essential to any data-driven regulatory design, and building that evidence base requires sustained investment in administrative data access and cross-agency cooperation. Second, and more uncomfortably, the evidence we do produce is useless if the policies we evaluate go unenforced. Bill 108 and Bill 89 were passed with fanfare and justified with exactly the type of housing-affordability argument policy-evaluation research is designed to vindicate, but the realized enforcement regime was soft enough that neither ordinance substantially moved the margin it was legislated against. This is nothing but political theatre wearing the sheepskin of policy, and the cost is twofold: residents do not get the housing relief they were promised, and the researcher loses the identifying variation that would have allowed them to distinguish a small structural price elasticity from a large but unenforced policy instrument. The harder data-driven policymaking becomes, the more surprising it is when the political system claims it.

The conventional wisdom has long been that regulators lag industry–that enforcement is always the cop chasing the criminal. In the last five years, that calculus has shifted. Large language models and cloud-hosted scrapers that cost roughly \$100-\$200 per month to operate now make it technically trivial to identify non-compliant listings at the parcel level, in near-real-time, across an entire island. The enforcement gap we document at length in section 6.3 is, in 2026, a choice rather than a constraint. What is missing is not the tooling–which is cheap, commodified, and in many cases a single API call away–but the political will to deploy it and the public awareness to call for it. Hawaii has fantastic tools at its disposal, though the question of what kind of workman we will be is still in question.

The terms on which Hawaii's housing stock is allocated between residents and visitors are not a narrowly econometric question but a civic and cultural one. Our evidence is one piece of one slice of how that relationship is evolving, and the null we report is as much an indictment of enforcement as it is of policy. The caution we leave with policymakers is this: Honolulu has the capacity to enforce what it has already legislated, and at a cost that is small relative to the housing market stakes at issue. Hawaii's housing debate no longer changes on the order of decades; it changes on the order of years, perhaps even months, and it would be a shame to watch another round of political theatre while the tools to address these problems–steadily improving and widely accessible–lie unused on a cloud server billed in hundred-dollar increments.

# A Appendix

## A.1 Legal definitions of transient-vacation rentals in Hawaii

Hawaii's transient accommodations tax broadly provides a foundational statewide definition, which counties then narrow and interpret through their own zoning codes. The state (2025) defines transient accommodations as the "furnishing of a room, apartment, suite, single family dwelling, or the like to a transient for less than 180 consecutive days for each letting in a hotel, apartment hotel, motel, condominium or unit... cooperative apartment, dwelling unit, or rooming house that provides living quarters, sleeping, or housekeeping accommodations, or other place in which lodgings are regularly furnished to transients."

Both Maui and Kauai counties keep the state's minimum occupancy requirement of 180 days. Maui broadens the state's definition and grants definitional exemptions to B&Bs and short-term rental homes (STRH): "Transient vacation rentals or use means occupancy of a dwelling or lodging unit by transients for any period of less than 180 days," (County of Maui, 1980). Maui maintains the same minimum occupancy requirement for STRHs (permitting at most one single-family home to be rented per lot) and B&Bs but requires the owner of the property to be "physically present at the short-term rental home within one hour following a request by a guest, a neighbor, or a county agency, and having an office or residence within thirty driving miles" (2016a; 2016b). Kauai (2008) uses a slightly generalized version of the state's definition, referencing generally a "dwelling unit" occupied for fewer than 180 days.

The City and County of Honolulu and Hawaii County lower the state's minimum occupancy requirement from 180 days to 30 days. Honolulu is more explicit in its definition of TVRs, specifying both a minimum occupancy requirement and a means of exchange: "transient vacation unit means a dwelling unit or lodging unit that is advertised, solicited, offered, or provided, or a combination of any of the foregoing, for compensation to transient occupants for less than 30 consecutive days, other than a bed and breakfast home... compensation includes, but is not limited to, monetary payment, services, or labor of transient occupants," (City and County of Honolulu, 2019). The city's definition of TVR has been in place since 1989.

Hawaii County's looser definition strays slightly from Honolulu's: "short-term vacation rental means a dwelling unit of which the owner or operator does not reside on the lot site, that has no more than five bedrooms for rent on the building site, and is rented for a period of 30 consecutive days or less," (County of Hawaii, 2018). Notably, Hawaii County restricts the number of rentable units per lot to five–relative to Maui's one.

## A.2 State-level land use designations

The state classifies land into four different designations based on current or prospective use. By area, most of the land is designated for conservation (48%) or agriculture (47%), followed far behind by urban use or development (5%) and rural use (0.5%). The state classifies any land with high capacity for cultivation as agricultural land, while land occupied by forests, water reserves, or coastal zones (among others) as conservation. Both are overseen by state-level bodies, the Land Use Commission and Board of Land and Natural Resources, respectively. Lands in existing urban use or land deemed ripe for future urban development are classified as urban,

## A.3 A history of county land use and TVR regulation

In 1961, the Hawaii State Legislature passed Act 187 (S.L.H 1961; HRS Chapter 205), establishing one of the first statewide zoning systems administered by a dedicated commission. The legislature's rationale was that "a lack of adequate controls had caused the development of Hawaii's limited and valuable land for short-term gain for the few while resulting in long-term loss to the income and growth potential of our State's economy." At the time, the legislature worried the state's growing population would convert prime agricultural land into scattered residential subdivisions. Seven years later, the 1968 constitutional convention amended the state's constitution to grant counties the power of self-governance, and thus the authority to regulate land use.

### A.3.1 City and County of Honolulu

Enacted in 1969, the Revised Ordinances of Honolulu (ROH) was the county's first zoning code. It has been amended and supplemented numerous times since its codification nearly sixty years ago. In 1986, the city's Department of Planning and Permitting began issuing non-conforming use certificates (NUCs) to properties outside resort zones as a temporary measure to legalize their use as TVRs or B&Bs. Three years later in 1989, the city's Department of Planning and Permitting enacted Ordinance 89-154, which prohibited the operation of all STRs outside of resort zones—in addition to a few mixed-use zones with exceptions—without an NUC. Those using their properties as STRs were permitted to continue operating if they proved use preceded enactment of the ordinance (December 28, 1989) and obtained an NUC within nine months. The city stopped issuing NUCs in September 1990, at which point there were 793: 759 TVRs and 34 B&Bs.

### A.3.2 County of Maui

Maui county's first zoning regulation, Ordinance 267, precedes Hawaii's statehood, but the ordinance was interim and there is a lack of consensus on its specific enactment year. Its first zoning ordinance post-statehood was enacted in 1960. Ordinance 268, originally applied only to the island of Maui and not to Molokai and Lanai—both under the jurisdiction of Maui county—and did not phase out the pre-statehood zoning designations. Despite changes throughout the years, zoning designations for parcels are determined jointly by county ordinances, individual zoning changes, and pre-statehood interim zoning assignments, though the system is undergoing a complete overhaul following an audit commissioned in 2018.

Maui was the first to officially incorporate TVR-specific regulation into its zoning code in 1980, when it defined TVRs as units rented for less than 30 days continuously and restricted their operation to apartment and hotel districts. In 1989, the same year as Honolulu's Ordinance 89-154, Maui eliminated TVRs from apartment districts entirely by mandating continuous occupancy for a minimum of six months. A list of exempt units, called the "Minatoya list," were granted informal NUCs and permitted to continue operating.[34] Units on the Minatoya list were officially legally permitted to operate as TVRs following exemptions codified in 2014. The county planning department regulates whole-property rentals and B&Bs outside hotel zones separately through a limited and stringent permitting system introduced in 2016.

### A.3.3 County of Kauai

The foundation for Kauai's current zoning code was introduced in 1972, and later revised in 1976, in Chapter 8 of Ordinance 164. Prior to 2000, the county's comprehensive zoning ordinance only discussed TVRs in the context of multi-unit buildings and permitted them in visitor designated areas (VDAs), which were designated areas layered atop collections of differently zoned parcels. Operators of rentals of single family homes, also called whole-property rentals, interpreted this regulatory inattention as permission to operate without NUCs. The 2000 Kauai General Plan closed this gap by introducing the concept of alternative visitor accommodations to formally include B&Bs and single family TVRs in the zoning code.

In 2008, in response to the rapid spread of vacation rentals in areas beyond VDAs, the Kauai County Council enacted Ordinance 864, which restricted all new TVR operation to VDAs and required single family TVRs operating outside VDAs before the ordinance to obtain an NUC by March 2009. Multi-family TVRs predating September 1982, a separate, earlier grandfathered cutoff, were permitted to continue operating outside VDAs. The stipulation and wording of Ordinance 864 were nearly identical to Honolulu's Ordinance 89-154 and Maui's Ordinance 1797. The prohibition of single family home TVRs outside VDAs without an NUC was explicitly amended to the zoning code in 2012.

### A.3.4 County of Hawaii

Relative to its peers, the County of Hawaii was the last to move on zoning and TVR regulation. The Hawaii County Code 1983 was the first officially compiled zoning code for Hawaii County, with the 2016 edition incorporating all ordinances issued through June 2016 and semi-annual supplements thereafter. Prior to 2018, TVRs in Hawaii County existed in a regulatory vacuum, though some realty sources claim they were historically only

[34] The Minatoya list has been a topic of controversy in recent years, particularly following a reinvigorated policy discussion of affordable housing on Maui following the 2023 Lahaina Wildfires. In December 2025, Mayor Richard Bissen signed Bill 9 into law, which would phase out the TVR exemption for some 6,000 apartment units on the Minatoya list. The law will phase out exemptions on West Maui by December 31, 2028 and the rest of the county by December 31, 2030. It will no doubt be an interesting empirical setting for further TVR-related research.

permitted in resort and special commercial zones.

### A.4 Imputed TVR activity measures

Inside Airbnb provides review timestamps but not calendar availability or booking records. Reviews are thus a lower bound on activity: a guest may stay without leaving a review, or a listing may be available but unbooked. The fundamental problem is undercounting–in missing active but unreviewed months–and overcounting–imputing activity during genuine dormancy attributable to seasonal closures, renovation, exit from market, etc. Rather than adjudicating this ex ante, we construct four measures spanning a plausible range of conservativeness and treat the choice as a specification decision. Table A1 summarizes the activity measure logic.

#### Review-only

The review-only measure registers a listing as active only in months where it received at least one guest review. This understates true activity due to Airbnb's review rate being well below 100%: Zervas et al. (2017) cite a deprecated 2014 paper by Andrey Frandkin and colleagues "reporting 67% of Airbnb guests left a review about their stay across their large dataset." Indeed, a listing with steady bookings may show sporadic review gaps of one or two months purely from random review behavior.

#### 3-month and 6-month imputation

The imputed measures acknowledge short gaps between reviews are more likely to reflect missing reviews than genuine market exit. The rational owner/operator who has received a review in both March and July, for instance, is unlikely to unlist and relist their TVR unit in the period between March and July. The 3-month window is calibrated to seasonal booking troughs (e.g., a listing might receive no winter bookings but remain listed), while the 6-month window accommodates longer low seasons or properties with lower turnover (e.g., entire-home listings with fewer but longer stays).

#### Span continuous

The span continuous measure, the most liberal, assumes the TVR unit is never delisted between the earliest and latest dates it received a review. It is motivated by an assumption on market participation: that is, once a property enters the short-term rental market, it represents a unit of housing supply diverted from the long-term market until it exits. Even during months without bookings, the property is listed and therefore unavailable for long-term tenants (one can assume the demands of renting short-term–such as acquiring permits, furnishing the property to cater to transients, etc.–makes the decision to switch tenure choices costly). This measure is most appropriate for the housing supply channel hypothesis we discuss in section 4.1, where the relevant margin is the stock of properties committed to being listed on the short-term rental market rather than being able to flow in and out of being an active unit.

The review-only measure provides a strict lower bound on TVR activity, while the span measure provides an upper bound. The imputed measures occupy the interior of this range. Figure A1 illustrates the quantitative differences across the four methods using Honolulu as an example. We test the robustness of our results to noisier measures of TVR density (e.g., three month against span continuous imputation) in appendix A.10.

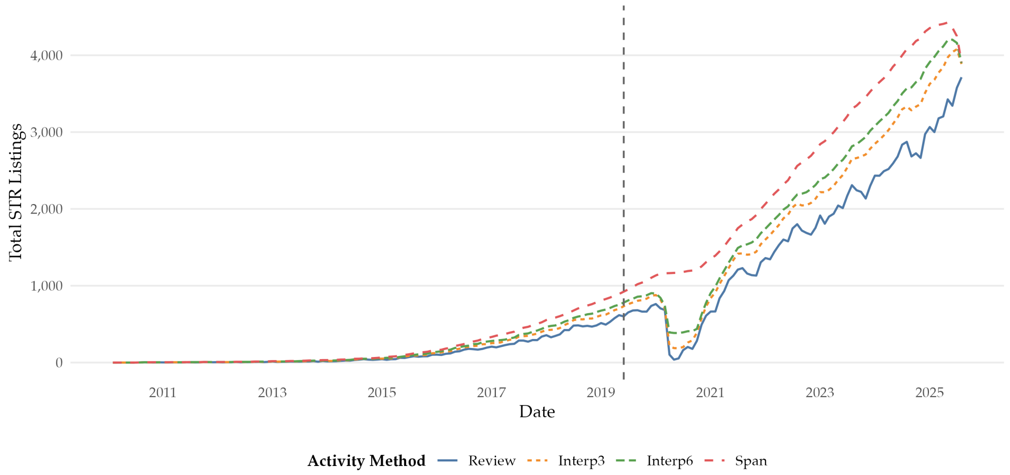



Figure A1: Monthly active Airbnb listings in Honolulu County, 2010–2025, under four alternative activity definitions[35]

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| Measure | Inclusion Rule | Assumption | Bias |
|---|---|---|---|
| **Review-only** | Active iff $\geq$ 1 review in month $t$ | Reviews perfectly proxy activity | Understates (misses unreviewed stays) |
| **Interp-3** | Fill gaps $\leq$ 3 months between reviews | Short gaps = stochastic review absence | Mild overstatement if listing genuinely paused |
| **Interp-6** | Fill gaps $\leq$ 6 months between reviews | Seasonal troughs $\neq$ market exit | Overstates if listing exited and re-entered |
| **Span** | Active every month, first to last review | Market participation $\approx$ supply diversion | Overstates if listing had extended dormancy |

Table A1: Comparison of activity measures and behavioral assumptions

[36] The dashed vertical line marks the June 2019 enactment of Honolulu Ordinance 19-18 (Bill 89). Review requires a review in the given month; Interp3 and Interp6 impute activity within $\pm$3 and $\pm$6 months of a review; Span treats a listing as active in every month between its first and last observed review.

## A.5 Data processing

The analysis panel is constructed from the ATTOM records and tax assessor databases, which together record approximately 207,000 transactions across the four counties between 2013 and 2025. Below we document the screening rules, coordinate reconciliation, zoning assignment, and measurement decisions we apply before estimation. Each subsection explicates the decision and, where relevant, the bias direction it may induce.

### A.5.1 Transaction data screening

We restrict the panel to transactions plausibly corresponding to arm's-length residential sales in three passes. First, we drop records with a transacted price of zero, which in the ATTOM feed are almost exclusively administrative entries–deed revisions, title corrections, and intra-family transfers recorded for legal continuity rather than market pricing. Second, we retain only transactions recorded under one of four deed instruments: warranty deeds (`DTWD`), grant deeds (`DTGD`), limited warranty deeds (`DTLW`), and trustee deeds (`DTTD`). These four instruments together capture the overwhelming majority of arm's-length residential sales while excluding the quitclaim deeds and deed-in-lieu transfers that populate the non-arm's-length margin. Third, we remove pairs of same-property transactions recorded within 30 days of one another, which are overwhelmingly administrative re-recordings of a single underlying sale rather than genuine market events. The 30-day deduplication iterates chronologically within `property_id` $\times$ `year` groups, so a December sale and a January resale on the same parcel are both retained even though fewer than 30 days separate them: the within-year scope is appropriate because the analysis panel is annual and cross-year, same-property transactions carry genuine price change information that belongs in the panel.

Three additional screens govern the scope of the analysis sample. We restrict the sample to properties whose build year field lies at or before the study start, rather than using the assessed tax year as the tax-year field records the most recent billing update rather than the year of first assessment and would spuriously drop properties with frozen tax bases. Building year also captures the margin we care about: properties with a post-2019 build year may have been significantly renovated or newly constructed in the post-ordinance period, and including them would contaminate the comparison. We further exclude vacant lots, commercial parcels, and non-residential tenure categories (we retain fee simple, leasehold, and agreement-of-sale), and we exclude transactions in zoning categories where TVRs were always prohibited (agricultural, industrial, preservation, conservation, and federal zones). These parcels are definitionally outside the variation the zoning ordinances induce and cannot contribute either a treated or a control observation to the design. The residential-zone restriction is implemented at the zone-treatment classification step rather than as a hard filter, so that the zone-panel specifications in A.9 retain the same sample as the transaction-level specifications.

### A.5.2 Coordinate deduplication and merging

Because each property may have been geocoded twice, a deduplication step selects the best available coordinate pair for each property. For consistency, we re-geocode all properties rather than relying on ATTOM-provided geocoding. The duplication thus comes from the possible conflict between the Census Geocoder and the Google Maps API. The deduplication procedure is the following:

1. Construct a dataframe of all geocoded records with a binary indicator for whether the record has a valid (non-null) latitude.
2. Sort by property identifier, then by the valid-latitude indicator (descending), so that records with coordinates precede those without.
3. Deduplicate by property identifier, retaining the first (coordinate-bearing) record.

When both the Census Geocoder and the Google Maps API return valid coordinates for the same property, the Census result is retained. When the Census Geocoder returns `No_Match` but the Google Maps API succeeds, the Google Maps result is used. The ordering is deliberate: the Census Geocoder is built directly on TIGER/Line parcel boundaries and therefore produces the cleanest joins onto the county zoning polygons that anchor the treatment indicator, whereas the Google Maps API geocodes against a proprietary rooftop database and occasionally resolves to centroid points that lie outside the parent parcel polygon. After geocoding, coordinate coverage is effectively 100% across all four counties.

### A.5.3 Zoning assignment and nearest-neighbor imputation

Every geocoded property is assigned a zoning polygon through a spatial point-in-polygon join against the county zoning GIS layer, which is the foundation for both the treated-within-county and treated-across-county indicators. For three of the four counties the join is essentially complete. Kauai's zoning GIS file is the exception: the layer is built around actively developed parcels and leaves large stretches of agricultural and conservation land unmapped, so a strict within-polygon join fails for a non-trivial share of rural Kauai observations. We resolve these failures with nearest-neighbor imputation: an unmatched point is assigned the zoning classification of the nearest mapped polygon. The imputation is justified by the spatial structure of Hawaii zoning (parcels are contiguous and rural gaps are typically surrounded by the same agricultural or conservation classification) and is conservative for our design because unmatched Kauai points are almost definitionally non-resort, so the nearest-polygon fallback almost never misclassifies a property across the resort/non-resort boundary that identifies the Wald-DID.

### A.5.4 Housing stock denominator and STR density construction

The primary STR supply measure is a zone-quarter density: Airbnb listings active under the imputation rule described in Appendix A.4, divided by the total housing stock of the zone. The natural denominator is Census housing units, which we apportion to zoning polygons via areal imputation using 2020 Decennial Census block shapes and the 2020 housing stock count (Goodchild and Lam, 1980). For each county we reproject blocks and zones to a county-specific NAD83 State Plane CRS, intersect them, and allocate each block's housing units to its zone-overlap polygons in proportion to shared area. Allocation rates exceed 98% in three counties and reach 89.7% in Kauai, where the same GIS-coverage gaps discussed above suppress a portion of rural-block coverage. Zones with recorded transactions but no Census overlap are floored at one housing unit (127 zones, 728 transactions). Time variation is introduced by scaling the 2020 base by county-level ACS five-year growth rates (via census variable `B25001_001E`) with clamping to 2015 on the low end and 2023 on the high end. Within-county spatial variation in post-2020 construction is suppressed, a limitation that zone fixed effects partially absorb.

#### A.5.5 Structural covariate treatment

Two hedonic covariates (building area and lot area) have extreme right tails in the raw ATTOM data feed driven by mislabeled units and subdivided-parcel reporting quirks. We winsorize both at the 1st and 99th percentiles (cutoffs 0 to 6,075 sqft and 0 to 1.103,495 sqft, respectively) before computing log transforms, and we top-code property age at 120 years (capping 30 rows). Log transforms use $\log(1+x)$ to remain well-defined at the zero boundary for condominium units recorded with zero lot area. ACS demographic controls are aggregated from Census block-group estimates to zones via the same areal-imputation infrastructure as the housing stock denominator, with vintage routing (2010 TIGER for 2015–2019 ACS, 2020 TIGER for 2020–2023 ACS) to respect the Census's decennial geography revision. Variables plausibly determined in the same housing market equilibrium as prices (median rent, vacancy rate, and owner-occupancy rate) are retained in the output panel for possible outcome use but are explicitly excluded from the control vector in all causal specifications to avoid controlling on post-treatment variables.

#### A.5.6 Exclusion of zero TVR zones

There are many treated zones which have no STRs before the policy is implemented and continue (as one might expect) to have no STRs after the policy is enacted. Our specification registers this as the policy $Z_{j(i)t}$ having less of an association with STR density $D_{j(i)t}$ than is actually the case. First stage results are negative (as we would expect) and significant, but not at as low of a significance level we might get if these zones were excluded. To address this shortcoming, we only subset the transaction and zone data to zones which had non-zero TVR density before their corresponding treatment date.

It also does not make much sense for us to consider these zones in our treatment group. We justify this by considering the following (silly) statisticaly thought exercise: suppose we are interested in the effect of weed killer on weed density, and we have three types of fields which are spatially noncontiguous. The first type has lots of weeds; the second type has some weeds; and the third type has no weeds. For some reason, we only decide to treat, at random, plots with some weeds or no weeds–maybe we find it to be a lost cause or too costly to treat the plots with lots of weeds. In estimating the causal effect of our weed killer, it does not make sense for us to consider the plots with no weeds in our treatment group because they had no weeds to begin with. They are also, perhaps, drawn from a different distribution than the plots with lots of weeds or some weeds.

### A.6 Predictions of the one-shot auction model of the housing market

Our toy posed in section 4.1 does not account for within-neighborhood geography, so the intuition provided by the model and our empirical specifications treat treatment responses as being spatially uniform within neighborhoods. However, one might reasonably assume operator margins on a TVR decay with distance from amenities (e.g., ABC stores, nearest beach access, etc.), but enforcement of the zoning policies does not. This combination generates heterogeneity in compliance and price response our simple model overlooks.

Equivalently, two relevant amenity distances act separately: a property close to resort amenities or close to the beach has a strong TVR margin, even if the other distance is large. A windward oceanfront unit in Kaneohe (far from resort zones but close to the beach) and an apartment in a zone bordering a resort area in McCully (close to resort zones but moderately far from the beach) are both not marginally affected by the enforcement penalty while a property in the suburbs in Pearl City (far from both resort zones and the beach) is the marginal complier. We therefore want to allow for proximity to either amenity to be sufficient to raise TVR profitability.

We present an alternative model in which we consider "bidders" for properties, which may be investors or owners.[37] Consider a neighborhood $n$ with a relatively fixed stock of housing units indexed by $j$. Units are purchased in one-shot auctions by bidders of two types: owner-occupiers, indexed $o \in \mathcal{O}$, and investors, indexed $i \in \mathcal{I}$. A unit is sold to the bidder with the highest willingness to pay. Let $\boldsymbol{X}_j$ denote the vector of hedonic characteristics of property $j$, $A_n$ the intensity of TVR activity in neighborhood $n$, and $P_j$ the transacted price of property $j$.[38] Each property $j \in n$ has a location $x_j \in \mathbb{R}^2$ and an associated distance vector:

$$\boldsymbol{d}_j \equiv (d_j^R, d_j^B, d_j^C)' \in \mathbb{R}_+^3,$$

where $d_j^R \equiv \min_{r \in \mathcal{R}} \|x_j - r\|$ is the distance to the nearest point in the union $\mathcal{R}$ of resort-district polygons, $d_j^B \equiv \min_{b \in \mathcal{B}} \|x_j - b\|$ is the distance to the nearest point in the set of all beach access points $\mathcal{B}$, and $d_j^C$ is the distance to the nearest county urban core.

In line with the motivating intuition discussed earlier, we define the "maximum amenity" kernel for our investigation, which weights according to whichever amenity is closer:

[37] We provide the following conceptual results to support interpretation of our empirical findings but do not solve the model or comment on the behaviors of buyers (i.e., we are not proposing a novel auction model nor applying, calibrating, or estimating an existing one). We leave that as an open end of our investigation perhaps worthy of future research.

[38] We can also write $X_{nj}$ so the hedonic characteristics capture both land and structural features, but we consider only $\boldsymbol{X}_j$ as we fixed a neighborhood $n$ and make comparisons across the fixed neighborhood.

$$K(\boldsymbol{d}) = \exp(-\lambda_C d^C) \max\{\exp(-\lambda_R d^R), \exp(-\lambda_B d^B)\}.$$

The primary mathematical advantage of the maximum amenity kernel is that the level set $\{\boldsymbol{d} : K() = \rho\}$ contains two geometrically distinct regimes–one which is resort-binding and another that is beach-binding–which are exchangeable. Note the "additive amenity" kernel $K(\boldsymbol{d}) = \exp(-\boldsymbol{d}'\boldsymbol{\lambda})$ does not align with the thought experiment we pose as it forces distance to resort and beach amenities to be small to keep $K(\cdot)$ large and thus misrepresents the structure of TVR operator behavior.[39]

Owner-occupier $o$ views property $j$ as having value:

$$V_{oj}^{\text{own}} = \boldsymbol{X}_j'\boldsymbol{\theta}_o + \eta_{on} - \phi_o A_n + \psi(\boldsymbol{d}_j) + \varepsilon_{oj},$$

where $\boldsymbol{\theta}$ prices the owner-occupier's preference for certain structural features, $\eta_{on}$ their taste in the specific neighborhood, $-\phi_o$ their disutility from TVRs, $\psi(\boldsymbol{d}_j)$ their preferences for location, and $\varepsilon_{oj}$ some idiosyncratic preference.

The investor $i$ views the same property as having value given by:

$$V_{ij}^{\text{inv}} = \underbrace{\pi_0(A_n)K(\boldsymbol{d}_j)h(\boldsymbol{X}_j)}_{\equiv \Pi_j(A_n, \boldsymbol{X}_j, \boldsymbol{d}_j)} - c_{ij},$$

where $\pi_0(A_n)$ is a neighborhood-level factor which captures perceived increases in profitability as a function of observed $A_n$, $h(\boldsymbol{X}_j) > 0$ a hedonic pricing function, $K : \mathbb{R}_+^3 \to [0,1]$ the amenity kernel, and $c_{ij}$ the investor's specific pricing of operation, compliance, or conversion costs. Denote the product term to be $\Pi_j(A_n, \boldsymbol{X}_j, \boldsymbol{d}_j)$, which captures the discounted expected profit from owning $j$ as a TVR asset, and $\rho_j \equiv K(\boldsymbol{X}_j)$ to be the property's amenity proximity index. Further assume $\frac{\partial \Pi_j}{\partial A_n} > 0$ for all $j$; that is, the investor perceives more TVR activity as a signal of greater profitability potential for their property.

We cannot make any sign assumptions on $\psi(\cdot)$. Owner-occupiers may prefer beach proximity (i.e., $\frac{\partial \psi}{\partial d^B} \leq 0$) but may dislike resort proximity and tourism-induced congestion (i.e., $\frac{\partial \psi}{\partial d^R} \geq 0$). Still, it is not difficult to see from this functional form that higher TVR activity lowers residential willingness to pay if local buyers dislike the noise, turnover, or reduced residential character of the neighborhood. We only require owner-occupiers value amenities less strongly than investors and formalize this in the assumption:

**Assumption 1** (Operator-margin dominance). $\left|\frac{\partial \psi}{\partial d^k}\right| < \pi_0(A_n)h(\boldsymbol{X}_j)\left|\frac{\partial K}{\partial d^k}\right|$ *for* $k \in \{R, B\}$ *for all* $\boldsymbol{d} \in \mathbb{R}_+^2$.

This assumption is admittedly quite strong but permits us to draw conclusions on spatial variation in compliance. An intuitive interpretation of the result is the following: units which are close to resort amenities or the beach will be valued very highly due to the sub-

[39] More general kernels also may suffice in our analysis under some additional regularity conditions (e.g., normalized to its maximum at the resort or beach amenity and sufficient in either of them, with some further assumption on differentiability and continuity). We leave that as a thought exercise for the reader.

stantial increase in profit expected by the investor, and this change in valuation by the investor exceeds that of the local buyer.

The zoning restriction enters the model as a statutory enforcement cost $\tau \geq 0$ added to investor cost $c_{ij}$ in non-resort zones post-treatment. The regulatory cost for property $j$ is a product of the per-violation penalty schedule (e.g., in Honolulu, daily fines under Bill 89 reach up to \$10,000/day) and the probability the non-compliant operator is detected by the county complaint-and-citation process within a given period. If $t^\star$ is the ordinance enactment date, then the post-ordinance investor enforcement cost is:

$$\tau_j \equiv \tau \mathbf{1}\{j \in \mathcal{R}\}\mathbf{1}\{t \geq t^\star\}.$$

Because $\tau$ does not depend on $\boldsymbol{d}_j$ but $\Pi_j$ does, the ordinance has different effects across locations despite enforcement intensity (theoretically) being spatially uniform. Equivalently, we assume enforcement attention and violation penalties to not be applied proportionally to TVR density or activity across zones.

We define the property-level operator margin to be the gap between the highest investor bid and owner-occupier bid:

$$M_j \equiv [\Pi_j - c^{(0)}_{i^\star j}] - V^{\text{own}}_{o^\star j}, \quad o^\star \in \arg\max_{o \in O} V^{\text{own}}_{oj},\ i^\star \in \arg\max_{i \in \mathcal{I}} V^{\text{inv}}_{ij}$$

Pre-ordinance, the investor wins the aunction when $M_j > 0$; post-ordinance in non-resort zones, the investor wins if $M_j > \tau_j$. The highest bid for property $j$ is defined to be $B^{\text{own}}_j = V^{\text{own}}_{o^\star j}$ and $B^{\text{inv}}_j = V^{\text{inv}}_{i^\star j}$, respectively. The transacted price is thus:

$$P_j = \max\{B^{\text{own}}_j, B^{\text{inv}}_j\}.$$

The fact equilibrium in the housing transaction market in neighborhood $n$ reflects competition between two sources of demand has several implications for housing affordability and the debate on "pricing out," but that is beyond the scope of this paper. That is, it is not difficult to see in a more nuanced version of the model where each type of buyer has a budget that owner-occupiers can be priced out in neighborhoods where TVR activity is very high (i.e., very high profit generating potential for the investor) or in places where effective income (e.g., post cost-of-living adjusted income) of resident buyers is very low. This result points toward there being general equilibrium consequences of TVR restriction in locations with tourism-dependent economies.

Three regimes emerge under this framework. First, inframarginal investors (characterized by $M_j > \tau$ with small $d^R$ or $d^B$) win auctions before and after the zoning restrictions, with the maximum amenity kernel producing a relatively large set in this group (e.g., nearest amenity suffices). Examples of these units are those outside the fringe of resort zones or the oceanview home. Second, compliers ($0 < M_j < \tau$ with intermediate distances $d^R$ and $d^B$) win before the zoning restriction but exit the TVR market after enactment. Transacted prices drop from the investor's previously high bid to the owner-occupier's lower bid, which is a property-level price decrease of $|M_j| \in (0, \tau)$. Third, "no weed" units $(M_j < 0)$

are properties for which owner-occupiers win before and after the zoning restrictions. The property never was and never would have been a TVR; these are precisely the properties we justify filtering out in section A.5.6. Examples of these properties might be multi-family homes in Waianae or rural farmhouses in Kau. For notational convenience, call the set of complier properties $\mathcal{C} \equiv \{j : 0 < M_j < \tau_j\}$. Empirically, our first stage coefficient being the change in TVR status across the ordinance should only be non-zero on $\mathcal{C}$.

Randomness in $M_j$ is induced from the property draw $(\boldsymbol{X}_j, \boldsymbol{d}_j, A_{n(j)}, \mathbf{1}\{j \in \mathcal{R}\}, \varepsilon_{oj})$. Observe the conditional expectation function $\bar{M}(\boldsymbol{d}) \equiv \mathbb{E}[M_j|\boldsymbol{d}_j = \boldsymbol{d}]$ is monotone in both amenity proximities holding the other fixed. More simply, we can consider the average operating margin among properties at amenity proximity level $\rho$ (integrated over the conditional distribution of $\boldsymbol{d}$ given $\rho$) $\bar{M}(\rho) \equiv \mathbb{E}[M_j|K(\boldsymbol{d}_j) = \rho]$ with the following regularity condition:

**Assumption 2** (Sufficiency of proximity)**.** *The conditional distribution of $(\boldsymbol{X}_j, \boldsymbol{d}_j, A_{n(j)}, \mathbf{1}\{j \in \mathcal{R}\}, \varepsilon_{oj})$ and any statistical functional of it given $K(\boldsymbol{d}_j) = \rho$ does not depend on d beyond what is captured by $\rho$.*

The following formal results provide some additional insight into the qualified null and supporting evidence we present in section 5. In short, the structural auction model predicts our results follow from the set of compliers being small relative to the set of inframarginal investors and, when price changes are realized, we observe them to be negative and to be bounded by the enforcement penalty.

**Proposition 1.** *Suppose $M_j$ admits a proximity shift $M_j = \bar{M}(\rho_j) + \varepsilon_j$, where $\varepsilon_j$ has bounded, continuous density $g$ which is symmetric and unimodal about zero. Then the conditional complier probability $f_C(\rho) \equiv \mathbb{P}(j \in \mathcal{C}|\rho_j = \rho)$ is also unimodal in $\rho$ with limits*

$$\lim_{\rho \to 1} f_C(\rho) = \lim_{\rho \to 0} f_C(\rho) = 0$$

*and a unique maximum attained on its interior at $\rho^\star = \bar{M}^{-1}(\tau/2)$.*

*Proof.* Under the proximity shift assumption, $\mathbb{P}(M_j < t|\rho) = G(t - \bar{M}(\rho))$, where $G$ is cumulative distribution function of $\varepsilon_j$. It follows from the Fundamental Theorem of Calculus that

$$f_C(\rho) = G(\tau - \bar{M}(\rho)) - G(-\bar{M}(\rho)).$$

Let $u \equiv \bar{M}(\rho)$ and write $\tilde{f}_C(\rho) = G(\tau - u) - G(-u)$. Then

$$\frac{d\tilde{f}_C}{du} = g(-u) - g(\tau - u).$$

Setting the above first order condition to zero yields $g(-u) = g(\tau - u)$, and it follows from symmetry that $g(-u) = g(u), g(\tau - u) = g(u - \tau)$. It remains to show $g(u) = g(u - \tau)$. Unimodality at zero implies the unique critical point $u = \tau/2$. To show the critical point is a maximizer, the second derivative gives concavity:

$$\frac{d^2 \tilde{f}_C}{du^2}\Big|_{u=\frac{\tau}{2}} = g'\left(\frac{\tau}{2}\right) - g'\left(-\frac{\tau}{2}\right),$$

which is strictly negative because $g'$ is non-positive to the right of zero and non-negative to the left. Finally, by the chain rule:

$$\frac{df_C}{d\rho} = \frac{d\tilde{f}_C}{du} \cdot \frac{du}{d\rho}.$$

The sign of $\frac{df_C}{d\rho}$ must match the sign of $\frac{d\tilde{f}_C}{du}$ since $\frac{du}{d\rho} > 0$ by the fact $\bar{M}(\rho)$ is increasing in $\rho$ everywhere by operator-margin dominance.

For the boundary limits, see that as $\rho \to 1$, $\bar{M}(\rho) \to \bar{M}(1) > \tau$ under the assumption operator margin at maximal amenity proximity exceeds the enforcement penalty (this must hold to accommodate the empirical observation TVRs persist in unpermitted beachfront areas like Kailua). Thus $G(\tau - u) - G(-u) \to G(\tau - \bar{M}(1)) - G(-\bar{M}(1))$, which both have arguments large and negative and thus distribution functions near zero. Together, $f_C \to 0$. Conversely, $\bar{M}(\rho) \to +\infty$ under the property of the maximum amenity kernel that the operator margin vanishes far from any amenity, so both arguments are large and positive and, by similar argument as for $\rho \to 1$, $f_C \to 0$. □

**Proposition 2.** *Under the maximum amenity kernel, the compliance set in $(d^R, d^B)$-space conditional on $d^C$ is the L-shaped region defined:*

$$L(\rho^\star) \equiv \left\{ d^R = \frac{-\log \rho^\star}{\lambda_R}, d^B \geq \frac{-\log \rho^\star}{\lambda_B} \right\} \cup \left\{ d^R \geq \frac{-\log \rho^\star}{\lambda_R}, d^B = \frac{-\log \rho^\star}{\lambda_B} \right\},$$

*with corner at $(-\log \rho^\star/\lambda_R, -\log \rho^\star/\lambda_B)$. The inframarginal region is a union of two perpendicular strips:*

$$I(\rho^\star) \equiv \left\{ d^R \leq \frac{-\log \rho^\star}{\lambda_R} \right\} \cup \left\{ d^B \leq \frac{-\log \rho^\star}{\lambda_B} \right\},$$

*and the "no weed" region is the open quadrant:*

$$N(\rho^\star) \equiv \left\{ d^R > \frac{-\log \rho^\star}{\lambda_R} \right\} \cup \left\{ d^B > \frac{-\log \rho^\star}{\lambda_B} \right\}.$$

*Proof.* We solve $K(\boldsymbol{d}) = \rho^\star$ explicitly. Fix $d^C = 0$ for visualization, the equation $\min\{\lambda_R d^R, \lambda_B d^B\} = -\log \rho^\star$ has solution set $L(\rho^\star)$ given above. The inframarginal region $\{K \geq \rho^\star\} = \{d^R \leq -\log \rho^\star/\lambda_R\} \cup \{d^B \leq -\log \rho^\star/\lambda_B\}$ is indeed the union of two perpendicular strips, and the never-taker region is the remaining open quadrant beyond the corner. Note, the kink at the corner is a set of measure zero for which $\lambda_R d^R = \lambda_B d^B$. □

Critically, compliance is concentrated on a band in the resort and beach amenity set for any given distance to the urban center(s). Properties strictly inside the band have such large margins the zoning restriction is ignored under legally reasonable violation penalties (e.g., trivially $\tau = +\infty$ eliminates the inframarginal group). The maximum kernel also places an oceanfront unit far from any resort district in the set of inframarginal properties despite its large value for $d^R$. Proposition 2 thus formally highlights the importance of strict enforcement, the absence of which we comment on in section 6.3 (e.g., a daily maximum violation penalty is $10,000 imposed continuously over extended periods of time is the closest a policymaker can get to $\tau \to +\infty$).

**Proposition 3.** *For any complier $j \in \mathcal{C}$, the property-level price change induced by the zoning restriction is*

$$\Delta P_j = -M_j \in (-\tau, 0).$$

*The expected price change conditional on compliance and on amenity proximity is subject to the bounds:*

$$-\tau < \mathbb{E}[\Delta P_j | j \in \mathcal{C}, \rho_j = \rho] < 0,$$

*with magnitude approaching $\tau/2$ at the modal proximity $\rho^\star$ and approaching either $0$ or $\tau$ near the band edges.*

*Proof.* Pre-ordinance, the winning bid for a complier property is $P_j^{\text{pre}} = \Pi_j - c_{ij}^{(0)} = V_{o^\star j} + M_j$. Post-ordinance, the winning bid is the owner-occupier's bid of $P_j^{\text{post}} = V_{o^\star j}$. The change in price for complier properties is then $\Delta P_j = P_j^{\text{post}} - P_j^{\text{pre}} = -M_j$. By the definition of the complier set $\mathcal{C}$, $M_j \in (0, \tau)$ and $-\tau < \Delta P_j < 0$. The conditional expectation statement follows from the same identity averaged over the complier conditional distribution, with the magnitude statement following from proposition 1 characterization of $f_C(\rho)$. □

This is a signed prediction at the property-level which does not inherit the sign-ambiguity of the demand-supply model's $\frac{\partial P}{\partial A}$ because our auction model isolates the single mechanism of the supply-side displacement of the marginal bidder. The disamenity function $E(\cdot)$ and capitalization function $I(\cdot)$ channels of the demand-supply model operate at the neighborhood level via general equilibrium effects on $\bar{D}^{\text{own}}(\cdot)$ and $\bar{D}^{\text{inv}}(\cdot)$, which the auction model sidesteps.

Observe also the magnitude of $\Delta P_j$ depends on which side of the compliance band property $j$ sits on. A complier with $M_j$ near the lower edge of the band ($M_j \approx 0$) epxeriences a near-zero price drop because the investor was alrady nearly indifferent to the owner-occupier. Conversely, a complier with $M_j$ near the upper edge ($M_j \approx \tau$) experiences a price drop of nearly $\tau$. Since the distribution of compliers is concentrated at $\bar{M}(\rho^\star) = \tau/2$ following proposition 1, the model complier experiences a price drop of approximately $\tau/2$.

An important point of caution in matching these conceptual findings to our empirical results is to remember proposition 3 is a statement of property-level price changes and not neighborhood-level price changes. The complier property's price falls by $0 < |M_j| < \tau$ but the complier's TVR status also falls (from 1 to 0 so $\Delta A_j = -1$). The Wald-DID estimate on

TVR density for property $j$ is then

$$\beta^{\text{WDID}} \approx \frac{\Delta P_j}{\Delta A_j} = +M_j,$$

which is positive: a one-unit increase in TVR density at the complier's location is associated with a price increase of $M_j$. The empirical Wald-DID coefficient should therefore positive at points $\rho$ where compliance density is non-negligible, with magnitude bounded by $\tau$. This is a formal statement of what we refer to throughout the paper as the "policy-maker's hypothesis."

### A.7 Reconciling cohort 1 first stage estimates with first stage event study diagnostics

Three features of the cohort 1 specification make the event study structurally underpowered relative to the pooled estimator. First, the post-treatment window for Hawaii county is short. Because Bill 108 became legally enforceable in April 2019 and county enforcement demonstrably lagged enactment, the quarterly window extends only five windows past treatment and is too narrow to capture the full dynamic response. We reconcile enforcement intensity and timing in 6.3. The post-treatment coefficients trend downward at event time $E_t^q = 3$ and $E_t^q = 4$, but the series is truncated before the effect can fully materialize. Even after changing the treatment date from the fourth quarter of 2018 to the second or third quarter of 2019 to align either immediately with the bill's enactment date or the quarter, results are still lackluster. We provide possible explanations for this in section 6.3.

Table A2: Sensitivity of Wald-DID estimates to post-period cutoff definitions

| | Wald-DID with $t^\star$ =2019Q2<br>(1) | Wald DID with $t^\star$ =2019Q3<br>(2) |
|---|---|---|
| | *Panel A: Two-Stage Least Squares* | |
| TVR density | 0.130 | 0.054 |
| | (0.331) | (0.327) |
| AR 95% CI | $[-\infty, +\infty]$ | $[-\infty, 4.39]$ |
| | *Panel B: First Stage for TVR Policy* | |
| Zoning policy | -0.166 | -0.168 |
| | (0.103) | (0.096) |
| First stage $F$ | 2.61 | 3.02 |
| $N$ observations | 4,954 | 4,954 |

*Notes:* Standard errors clustered at the zone level are reported in parentheses. AR 95% CI denotes the Anderson–Rubin confidence intervals.

Second, statistical power is split across event-time bins. The pooled Wald-DID estimator assigns every post-treatment observation to a single cohort-treatment-post coefficient which effectively averages quarterly effects. The average is well identified because it pools the full post-treatment sample. The event study parameterization, by contrast, estimates five separate post-treatment coefficients, each of which compares a substantially smaller slice of transactions to the pre-treatment reference bin. Standard errors, and therefore confidence bands, widen due to the bins corresponding to $E_t^q = 3$ and $E_t^q = 4$ being thinly populated for Hawaii county.

Third, it is widely documented event study coefficents are sensitive to the choice of reference period. For our illustration, we normalize every coefficient to $E_t = -1$. If the chosen reference bin is noisy, which is true given the thinness of the within-cell sample at quarterly frequency, the early post-treatment coefficients are mechanically pulled toward zero relative to the true pooled level difference, and the plotted series understates the average post-treatment gap the pooled estimator recovers. Indeed, this pattern is standard in fixed-effects event studies on short panels. The pooled Wald-DID is the estimate for the object of interest. The event study principally serves as a common trends diagnostic rather than as a post-treatment point-estimate series, and the yearly panel is the appropriate angle from which to interpret cohort 1's first stage results.

## A.8 Joint pre-trends tests for the six cohort-specific event studies

The event studies in Figures 2, 3, and 4 each rest on a joint common-trends assumption: absent the zoning ordinances, the treated and control series would have evolved in parallel in both the reduced form and first stage arms of the Wald ratio. Equivalently, treated and control arms both share common trends in TVR density and transacted log land values. We supplement the visual diagnostic with a joint Wald $F$-test of the null hypothesis that all pre-period event study coefficients are simultaneously zero.

Table A3 reports the test for each of the six cohort-specific event studies underlying the headline Wald-DID columns of Tables 4 and 5, along with the stacked TSLS aggregator. For each cohort and frequency we report three models: the first stage (TVR density on the zoning instrument); the transaction-level reduced form (log land value with hedonic controls and property-type fixed effects, matching the primary specification); and the zone-panel reduced form (zone-time mean log land value with zone and time fixed effects only, matching the zone-panel appendix specification in A.9).

Across all six cohort 1 specifications at quarterly frequency, and across their yearly analogs, the joint null of zero pre-trend difference is not rejected at any conventional significance levels with the yearly reduced form being the sole exception, rejecting at the 0.1 level. Indeed, cohort 1's identification under the joint common trends required of TVR density and log transacted price is consistent with the data. For cohort 2, the joint test rejects in one or more of the six specifications. Rather than treat this rejection as a complete collapse of common trends beyond the concession made in section 5.1, we present an alternative reading which does not invalidate the qualified null interpretation our headline cohort 2 Wald-DID estimates.

Naively, beyond the failure of the joint pre-trends testing documented in section 5.1 which returns as significant at the 0.01 level in table A3, half of the rejection of joint pre-trends are marginally significant at the 0.1 level, with two rejections at the 0.05 level. We focus our discussion, then, on the two joint test rejections at the 0.05 level.

Table A3: Joint pre-trends Wald tests across all event-study specifications — residualized log land value.

| Estimator | Cohort-Window | $N$ | $df_1$ | $df_2$ | $F$ | $p$ |
|---|---|---|---|---|---|---|
| First stage | Cohort 1 Q | 3,047 | 9 | 2757 | 0.548 | 0.840 |
| Reduced form, transaction-level | Cohort 1 Q | 3,047 | 9 | 2764 | 0.724 | 0.687 |
| Reduced form, zone-panel | Cohort 1 Q | 2,879 | 9 | 2477 | 1.048 | 0.399 |
| First stage | Cohort 1 Y | 8,071 | 4 | 7731 | 1.632 | 0.163 |
| Reduced form, transaction-level | Cohort 1 Y | 8,071 | 4 | 7738 | 2.196* | 0.067 |
| Reduced form, zone-panel | Cohort 1 Y | 2,102 | 4 | 1779 | 1.389 | 0.235 |
| First stage | Cohort 2 Q | 9,456 | 7 | 9193 | 1.999* | 0.051 |
| Reduced form, transaction-level | Cohort 2 Q | 9,456 | 7 | 9200 | 2.086** | 0.042 |
| Reduced form, zone-panel | Cohort 2 Q | 1,794 | 7 | 1545 | 2.081** | 0.043 |
| First stage | Cohort 2 Y | 26,596 | 4 | 26345 | 3.569*** | 0.006 |
| Reduced form, transaction-level | Cohort 2 Y | 26,596 | 4 | 26352 | 2.063* | 0.083 |
| Reduced form, zone-panel | Cohort 2 Y | 1,698 | 4 | 1464 | 2.002* | 0.092 |
| First stage | STS Q | 16,813 | 7 | 15775 | 2.029** | 0.048 |
| Reduced form, transaction-level | STS Q | 22,615 | 7 | 21959 | 1.560 | 0.142 |
| Reduced form, zone-panel | STS Q | 4,538 | 7 | 3921 | 1.622 | 0.124 |
| First stage | STS Y | 10,373 | 4 | 9399 | 2.201* | 0.066 |
| Reduced form, transaction-level | STS Y | 29,929 | 2 | 29215 | 1.448 | 0.235 |
| Reduced form, zone-panel | STS Y | 4,768 | 4 | 4108 | 3.283** | 0.011 |

*Notes:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

For one, the joint test rejects an exact zero restriction on the pre-period coefficient vector but not a small deviation from it. The event study coefficients in figures 4 and 2 are small in absolute value relative to the the magnitude of the post-enactment responses the Wald-DID captures (i.e., the $-0.2043$ pooled first stage in table 4 and the $20-22\%$ first stage decline we document in section 5.1). Cohort 2 carries $N = 12,049$ non-resort transactions against cohort 1's $N = 3,047$, which increases the precision of the pre-period coefficient vector by a factor of two in standard-error terms. We interpret the cohort 2 rejection and cohort 1 non-rejection as not being directly comparable as evidence of relative identification strength but rather differing precision with which each cohort estimates its pre-period coefficients.

A common trends violation that is non-zero in the pre-period also contaminates the post-treatment reduced form coefficient by continuation of the pre-trends into the post-period. Whether this contamination biases the estimate toward or away from the policy-maker's hypothesis precisely depends on the direction of the continuation. For cohort 2, the policymaker's null is a negative reduced form coefficient on TVR density (i.e., Bill 89 aims to soften non-resort transacted prices through reduced TVR supply). The Honolulu reduced form event study plot in figure 4 straddles zero in the pre-period with a slight positive tilt, which is consistent with the urban economic prediction that Honolulu's non-resort prices (the prices of homes in an urban part of Hawaii) grow moderately faster than Maui and Kauai in 2017 through 2019. If the pre-period pattern is taken at face value as an underlying continuation of pre-trends, the unadjusted post-treatment reduced form coefficient inherits an upward bias, making the observed flat post-enactment reduced form more suggestive of a negative causal effect masked by the cross-county differences in price growth. Cohort 2's joint rejection biases our headline result toward the null of zero. We may even read cohort 2's Wald-DID estimate as a conservative underestimate of the true causal effect we are trying to identify.

Admittedly, it is not implausible for there to be anticipatory effects among both TVR owner-operators and visitors who would have considered staying at a TVR absent the policy in the months leading up to the passage of Bill 89.[40] Public mention and discussion of the stipulations contained within Bill 89 and related policies began as early as mid-2018, providing skiddish operators ample time to exit, reclassify, or relocate their rentals before the policy was fully implemented a year later. Additionally, reports on the vacation rental market from late 2018 through pre-pandemic 2020 from the Hawaii Tourism Authority and the State's Department of Business, Economic Development, and Tourism reveal moderate and observable non-seasonal declines in supply and demand for TVR unit-nights beginning in early 2019, when public discussion of Bill 89 became increasingly heated and emotional. We speculate this anticipatory effect may reflect a hesitant visitor response to hostile local sentiment toward visitors staying at vacation rentals in the period preceding the restriction's passage and enactment.

Still, we maintain caution on the joint pre-trends test as a sharp identification diagnostic.[41] Joint pre-trends have low power against smooth, slowly evolving trends on short pre-period samples, so non-rejection (as in cohort 1) does not validate parallel trends any more than rejection (as in cohort 2) invalidates them Roth (2022). Indeed, the tests are most informative with an understanding of how units select into treatment and the economic rationale behind that selection Ghanem et al. (2026).[42]

Taken together, we present table A3 as partial evidence supporting our identification of the cohort 1 Wald-DID, however uninformative, while the identification of the cohort 2 Wald-DID rests on an argument requiring a bit more rhetoric. The cohort 2 rejection reflects the pre-existence of small across-county trend differences, which our design partially absorbs through fixed effects, thereby biasing our headline result toward the null rather than away from it. We also acknowledge the possible anticipatory effects among vacation rental investors and visitors leading up to passage of Bill 89, provide a possible explanation for the pre-treatment drift away from zero. Our qualified null interpretation of the headline cohort 2 estimates in section 5 is framed to accommodate this residual uncertainty, and we lean on Miyaji's 2024 STS aggregator, whose pre-trends are much more suggestive of clean identification.

[40]I owe it to Paul Brewbaker, in all the work he does on housing and other economic topics in Hawaii, for bringing this response–and many other narrative explanations for my quantitative results–to my attention.

[41]We do not use the term "sharp" here in the statistical sense of identifying exactly the parameter of interest; we use it instead as an adjective which illustrates our hesitance to take our interpretation of the joint pre-trends test as cleanly supporting our identification strategy.

[42]After all, every good applied economist is a good storyteller!

## A.9 Additional results from the headline specification using zone-quarter and zone-year panels

The four main specifications reported in tables 4 and 5 operate on the transaction-level repeated cross section, which is our primary data structure. A parallel set of estimates, reported in tables A4 and A5, collapses the transaction panel to zone-quarter and zone-year cells before estimating the headline specifications. One cell in that appendix table warrants additional attention, as it is the only Wald-DID estimate in the paper whose Anderson-Rubin confidence interval excludes zero.

As we discussed in section 4.2, the residualization of structural covariates is critical to the zone-panel methodology due to mean aggregation. We construct the outcome variable, mean log land value, by first estimating the log land value for each property $i$ in zone $j$, then taking the simple average of the properties to obtain the mean log land value.

$$\overline{\widehat{\log \ell_{it}}} = \frac{1}{n_j} \sum_{i=1}^{n_j} \widehat{\log \ell_{it}} \tag{9}$$

We then proceed with the same outcome regressions as in equation 4 with the mean log land value in place of log land value at the zone-level.

The Honolulu zone-year Wald-DID returns a coefficient of 0.4917 on TVR density (standard error 0.2797, first stage $F = 17.6$), significant at the 0.1 level, with AR confidence interval $[-2 \times 10^{-4}, 1.2647]$. We propose two considerations to inform its interpretation. First, the AR interval technically containing zero results in marginal significance: repeated Rademacher resampling for AR interval estimation would likely produce intervals which more generously contain zero. Second, collapsing the panel to zone-year cells weights zones uniformly rather than by the number of transactions or housing units they contribute, and the collapse converts an identification that exploits within-zone transaction variation into one that exploits the zone-level time series alone. The estimand is therefore not a direct analog of the transaction-level Wald-DID, and the two objects are not strictly comparable even in sample. The positive point estimate is directionally consistent with the investor capitalization of the model in section 4.1 (i.e., greater investor demand for TVR-convertible properties raises transacted prices) and inconsistent with the policymaker hypothesis that TVR proliferation inflates housing affordability through stock withdrawal.

We therefore do not treat this single specification as decisive. Because our pre-registered primary specification operates at the transaction level–and because that specification, in both its Wald-DID and stacked forms, returns AR confidence intervals that contain zero–we characterize the full body of evidence as mostly limited but suggestive, as is the theme throughout the paper, rather than as a confirmed positive effect.

As a companion to the body's lagged-density exercise in section 6.1, we also re-estimate the zone-panel specifications of tables A4 and A5 with an identical lagged-density augmentation. Results appear in tables A6 and A7. The narrative is the same as in the transaction-level panel but sharper in one place. The Honolulu zone-year Wald-DID (the only specification in the paper whose AR interval narrowly excludes zero in the base table)

Table A4: Effect of TVR Density on Housing Prices (Zone-Panel, Pre-Stage Residualized) — Quarterly, Residualized Outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | 0.2287 | 0.3151 | 0.1667 | 0.5503* |
| | (0.2011) | (1.0785) | (0.2958) | (0.2946) |
| AR 95% CI | [-0.1452, 0.7239] | $[-\infty, +\infty]$ | [-0.5319, 0.9842] | [0.2394, 7.0193] |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.2436*** | 0.0460 | -0.1279*** | -0.2476** |
| | (0.0563) | (0.0439) | (0.0430) | (0.1130) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 18.7 | 1.1 | 8.8 | 4.8 |
| $R^2$ | 0.451 | 0.555 | 0.546 | 0.377 |
| N observations | 2,557 | 5,260 | 5,271 | 9,462 |

*Notes:* The "Hedonic controls" row indicates whether the upstream residualization is applied (Y) or omitted (N). Instrumented variable and instruments follow the headline table. Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects. Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Effect of TVR Density on Housing Prices (Zone-Panel, Pre-Stage Residualized) — Yearly, Residualized Outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | 0.4917* | -0.4188 | 0.3889 | 0.5218 |
| | (0.2797) | (5.4700) | (0.7670) | (0.3576) |
| AR 95% CI | [-0.0002, 1.2647] | $[-\infty, +\infty]$ | $[-\infty, +\infty]$ | $[0.1085, +\infty]$ |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.2183*** | -0.0127 | -0.0771 | -0.2611* |
| | (0.0520) | (0.1171) | (0.0886) | (0.1426) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 17.6 | 0.0 | 0.8 | 3.4 |
| $R^2$ | 0.592 | 0.667 | 0.655 | 0.471 |
| N observations | 905 | 1,678 | 2,583 | 3,893 |

*Notes:* The "Hedonic controls" row indicates whether the upstream residualization is applied (Y) or omitted (N). Columns (1), (2), and (4) use the residualized outcome $\log(\widetilde{\text{price}})$ on the collapsed panel; column (3) reports the Stacked 2SLS with cohort $\times$ tract and cohort $\times$ time fixed effects absorbing unit and time variation. Instrumented variable and instruments follow the headline table. Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects. Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

remains positive after conditioning on zone TVR density measured one year prior (2.897, with AR CI $[0.1053, 18.511]$), and the point estimate grows relative to the unconditional 0.4917 reported above. The first stage $F$ falls from 17.6 to 4.2, consistent with the attenuation mechanism developed in section 6.1. That is, a large share of the cross-sectional variation in Honolulu's non-resort zoning instrument is collinear with pre-existing STR stock which the lag absorbs. Outside column (1), the stacked 2SLS and TWFE-IV aggregators collapse to unbounded AR intervals at both frequencies, mirroring the corresponding columns of the transaction-level robustness tables.

Table A6: Effect of TVR Density on Housing Prices (Zone-Panel, Lagged TVR Density Control) — Quarterly, Residualized Outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | 1.5287 | 0.1551 | 1.2730 | -40.2425 |
| | (1.5760) | (1.7613) | (2.9495) | (327.9946) |
| AR 95% CI | [-1.4015, 6.5181] | $[-\infty, +\infty]$ | $[-\infty, +\infty]$ | $[-\infty, +\infty]$ |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.0324$^{***}$ | 0.0288 | -0.0143 | 0.0031 |
| | (0.0109) | (0.0343) | (0.0160) | (0.0249) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 8.8 | 0.7 | 0.8 | 0.0 |
| $R^2$ | 0.451 | 0.555 | 0.546 | 0.361 |
| N observations | 2,557 | 4,409 | 5,271 | 8,619 |

*Notes:* The "Hedonic controls" row indicates whether the upstream residualization is applied (Y) or omitted (N). Columns (1), (2), and (4) use the residualized outcome $\log(\widetilde{\text{price}})$ on the collapsed panel; column (3) reports the Stacked 2SLS with cohort $\times$ tract and cohort $\times$ time fixed effects absorbing unit and time variation. Instrumented variable and instruments follow the headline table. Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects. Standard errors clustered at the zone level in parentheses. $^{***}$ $p < 0.01$, $^{**}$ $p < 0.05$, $^{*}$ $p < 0.1$.

We include these results for completeness but do not redraw the interpretive conclusion of section A.9. The narrative is the same in both the transaction-level and zone-panel lag robustness exercises. Conditioning on observable selection on pre-existing TVR density does not overturn the headline reading of the paper, and, if anything, we believe it reinforces the one anomalous positive zone-panel cell. Whether the surviving anomaly reflects a genuine investor-capitalization channel, the zone-panel uniform-weighting artifact described above, or residual unobservable selection on trend is an area for future work.

## A.10 Robustness to choice of TVR density measure: 3-month imputation v.s. span-continuous

Per our discussion in appendix A.4, we may be curious how sensitive our results are to different measures of TVR density. Span-continuous imputation provides a smoothed summary of TVR density in zones (this smoothing behavior is visible in figure A1) relative to the more granularly imputed density measures. We test this sensitivity by re-running our quarterly specifications–both the headline and lagged–using the 3-month imputed density measure in place of the span-continuous one. We exclude the yearly specifications in this

Table A7: Effect of TVR Density on Housing Prices (Zone-Panel, Lagged TVR Density Control) — Yearly, Residualized Outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | 2.8971 | 0.1822 | -2.2221 | -6.0337 |
| | (2.0577) | (2.4670) | (5.6432) | (8.9040) |
| AR 95% CI | [0.1053, 18.5106] | $[-\infty, +\infty]$ | $[-\infty, +\infty]$ | $\emptyset$ |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.0380** | 0.0215 | 0.0149 | 0.0203 |
| | (0.0186) | (0.0345) | (0.0330) | (0.0288) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 4.2 | 0.4 | 0.2 | 0.5 |
| $R^2$ | 0.592 | 0.668 | 0.656 | 0.461 |
| N observations | 905 | 1,678 | 2,583 | 3,567 |

*Notes:* The "Hedonic controls" row indicates whether the upstream residualization is applied (Y) or omitted (N). Columns (1), (2), and (4) use the residualized outcome $\log(\widetilde{\text{price}})$ on the collapsed panel; column (3) reports the Stacked 2SLS with cohort $\times$ tract and cohort $\times$ time fixed effects absorbing unit and time variation. Instrumented variable and instruments follow the headline table. Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects. Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

exercise due to the fact the 3-month imputation and span-continuous measures are approximately equivalent when the 3-month imputed measure is aggregated up to the year level.

Tables A8 and A9 below present the results from the described robustness exercise. Across the four transaction-level specifications in table A8, signs match between the 3-month and span-continuous imputed measures of TVR density. However, the magnitude of estimates are considerably smaller for the 3-month measure. First stage point estimates are smaller by factors of two, which we might interpret as the zoning policy moving 3-month imputed TVR density only half as much as the span-continuous density. It is from these attenuated estimates we observe inflated Wald-DID point estimates, none of which are statistically significant at conventional levels. Similar observations apply to the lagged estimates in table A9, were one column deviates in sign.

Observe the span-continuous STS specification in table 6 goes from $-1.14$ with no lag to $-30.28$ with lag and cluster-robust first stage $F$-statistic $F = 0.3$, suggesting the first stage under the span-continuous measure picks up mostly noise once the four quarter lag is added to the set of covariates. Conversely, the 3-month imputed STS specification in table A9 shifts only slightly from $-2.66$ with no lag to $-3.47$ with lag and cluster-robust first stage $F$-statistic $F = 9.1$ (conveniently, similar magnitude, sign, and $F$-statistic). The comparison of these changes may suggest the cleaner activity 3-month imputed measure and its four quarter lag are less collinear than the span-continuous measure and its four

Table A8: Effect of TVR Density on Housing Prices (Transaction-Level) — Quarterly, Residualized Outcome (interp3 robustness)

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| *Panel A: Two-Stage Least Squares* | | | | |
| TVR density | -0.7493 | 2.3772 | -2.6589 | 0.3000 |
| | (1.4105) | (5.0127) | (1.9263) | (0.4351) |
| AR 95% CI | [-4.9312, 2.0150] | $[-\infty, +\infty]$ | [-12.2421, 0.5355] | [-0.6402, 1.5464] |
| *Panel B: First Stage for TVR Policy* | | | | |
| Zoning policy | -0.1102$^{***}$ | -0.0429 | -0.1025$^{***}$ | -0.2859$^{***}$ |
| | (0.0406) | (0.0886) | (0.0389) | (0.0744) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 7.4 | 0.2 | 7.0 | 14.8 |
| $R^2$ | 0.237 | 0.465 | 0.333 | 0.282 |
| N observations | 12,049 | 3,047 | 27,626 | 44,318 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. The 5 structural variables (building area, age, pool, baths, bedrooms) are absorbed by the residualization step that constructs $\widetilde{\ell}$ and so are omitted from the second-stage RHS by Frisch–Waugh–Lovell. Stacked 2SLS omits all hedonic regressors; cohort $\times$ property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + year-quarter. Property-type fixed effects are absorbed by the residualization step that constructs $\widetilde{\ell}$. Fixed effects for (3): cohort $\times$ tract + cohort $\times$ time + cohort $\times$ property type. Standard errors clustered at the zone level in parentheses. $^{***}$ $p < 0.01$, $^{**}$ $p < 0.05$, $^{*}$ $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

quarter lag.[43].

## A.11 More on the enforcement failure of Hawaii county's Bill 108 and Honolulu's Bill 89

Here we expand the compliance diagnostics of section 6.3 with direct documentation of the Bill 108 and Bill 89 enforcement regimes. The central claim we advance is that these ordinances were not the clean supply restrictions the Wald-DID design demand rquires. Each layers statutory exceptions, administrative bottlenecks, and court-ordered compromises onto a nominal prohibition in ways that materially attenuate the first stage. We organize the material around the four institutional pillars introduced in section 6.3 (the grandfather flood, enforcement asymmetry, hosted/unhosted scope gap, and sticky operator dynamics) then enumerate the operative provisions and cite the primary sources for each, and close with a comparative table.

**The grandfather flood** Ordinance 18-114 (Bill 108) was signed December 5, 2018 and took effect April 1, 2019 (Hawai'i County Planning Department, 2024). The ordinance restricts

[43] The "inertia" of the span-continuous measure is highly correlated with the lagged span measure, leading to important variation being partialled out in the regression when the lag is added to the set of covariates

Table A9: Effect of TVR Density on Housing Prices (Transaction-Level, Lagged TVR Density Control, interp3) — Quarterly, Residualized Outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | | *Panel A: Two-Stage Least Squares* | | |
| TVR density | -0.8164 | -8.3858 | -3.4692 | 0.3793 |
| | (1.3962) | (32.3039) | (2.3085) | (1.8312) |
| AR 95% CI | [-4.8156, 1.7795] | $[-\infty, +\infty]$ | [-11.9375, 0.5909] | [-3.3935, 4.8881] |
| | | *Panel B: First Stage for TVR Policy* | | |
| Zoning policy | -0.1100$^{***}$ | 0.0131 | -0.0802$^{***}$ | -0.0654$^{***}$ |
| | (0.0382) | (0.0473) | (0.0265) | (0.0193) |
| Fixed effects | Y | Y | Y | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 8.3 | 0.1 | 9.1 | 11.5 |
| $R^2$ | 0.237 | 0.465 | 0.333 | 0.244 |
| N observations | 12,049 | 3,047 | 27,626 | 38,950 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. The 5 structural variables (building area, age, pool, baths, bedrooms) are absorbed by the residualization step that constructs $\widetilde{\ell}$ and so are omitted from the second-stage RHS by Frisch–Waugh–Lovell. Stacked 2SLS omits all hedonic regressors; cohort $\times$ property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + year-quarter. Property-type fixed effects are absorbed by the residualization step that constructs $\widetilde{\ell}$. Fixed effects for (3): cohort $\times$ tract + cohort $\times$ time + cohort $\times$ property type. Standard errors clustered at the zone level in parentheses. $^{***}$ $p < 0.01$, $^{**}$ $p < 0.05$, $^{*}$ $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

new unhosted short-term vacation rentals (STVRs) to V (Resort), CG (General Commercial), and CV (Village Commercial) districts, RM (multi-family) zones inside designated resort nodes, and General Plan Resort/Resort Node areas. Under the Hawaii County Code, an STVR is a dwelling whose owner or operator does not reside on the building site (Hawai'i County Planning Department, 2024). Any STVR operating outside the permitted zones before April 1, 2019 had 180 days–until September 28, 2019–to apply for a NUC. Qualification required demonstrated pre-April 1 operation, clean General Excise Tax and Transient Accommodations Tax records, a \$250 annual renewal fee, and a \$500 one-time registration fee.

The NUC pool so generated is large enough to mechanically absorb most of the Airbnb observable non-resort margin. Burnett (2021) reports approximately 988 initial NUC applicants at the close of the September 2019 window, declining to roughly 768 by April 20210–an attrition rate of 22.3% over 18 months that is slower than the Inside Airbnb span panel suggests and leaves the bulk of the ex ante legal stock intact through our estimation window. In the same county, the Inside Airbnb span panel observes only 381 active listings in non-resort zones in 2019, an Airbnb-to-NUC ratio of approximately 0.39.

At first read, two puzzling patterns emerge. First, the legal pool alone exceeds the set of listings our first stage could plausibly move. Even an enforcement regime that removed every observable Airbnb unit would have left several hundred legally operating non-resort

TVRs in the county. Second, because Airbnb undercounts total TVR activity (Hunden Partners (2025) estimate 8,008 active Hawaii Island listings in March 2025, roughly three times the contemporaneous Airbnb scrape) our measured first stage is an upper bound on the true post-ordinance decline in non-resort TVR density, biased toward zero by the platform coverage gap.

Honolulu's grandfathered pool is substantially smaller. The city stopped issuing new TVU NUCs in 1989 under an administrative moratorium and maintained this posture through Bill 89, so the pre-2019 grandfathered count for the entire island had fallen below one thousand by the time Ordinance 19-18 took effect (Honolulu Magazine, 2020; Hawai'i Appleseed Center for Law & Economic Justice, 2019). The DPP public ArcGIS roster lists 750 pre-1989 TVU NUCs as of November 2022 (Honolulu Department of Planning and Permitting, 2022), alongside 1,378 STR Certificates of Registration under Ordinance 19-18 as of January 2025: a combined legal pool of roughly 2,128 units against an estimated 9,000-plus pre-ordinance active listings island-wide (Hawaii Public Radio, 2019; Honolulu Department of Planning and Permitting, 2025). The ratio of grandfathered to total pre-ordinance stock is therefore an order of magnitude smaller in Honolulu than in Hawaii County, and this is part of why Bill 89's first stage bites at all.

**Enforcement asymmetry**

Honolulu's Department of Planning and Permitting (DPP) mailed approximately 5,000 pre-enforcement warning letters to suspected Bill 89 violators in July 2019, then began issuing Notices of Violation (NOVs) and Notices of Order (NOOs) immediately on the August 1 effective date (Honolulu Star-Advertiser, 2019). The fine schedule is steep: $1,000 initial violation plus $5,000 per day of continued violation, escalating to $10,000 per day (Imanaka Asato LLLC, 2019). By March 2024, DPP had issued 1,957 NOVs (of which 1,953 had been classified "Corrected") and 880 NOOs for Bill 89 related violations (Honolulu Department of Planning and Permitting, 2024).

Hawaii County's enforcement posture is institutionally weaker along every margin. Nominally, fines under the Hawaii County Code can reach $10,000 per day in aggravated cases, but absent a dedicated enforcement funding stream the Planning Department relied on complaint-driven investigation requiring resident reporting (Hawai'i County Planning Department, 2024). The department issued warning letters in isolated cases (e.g., the 2020 Cleveland warning letter posted by the state Land Use Commission is one of the few documented examples (Hawai'i County Planning Department, 2020)) but did not mount a mass compliance campaign comparable to Honolulu's. A long running agricultural zone legal challenge by a group of West Hawaii landowners (the *Rosehill* petitioners) added further friction by producing interpretive ambiguity around agricultural zone STVRs. The Hawaii Supreme Court did not resolve the matter until its unanimous September 2024 ruling upholding Bill 108's agricultural zone prohibition, and administrative action on agriculture zoned STVRs was correspondingly cautious in the intervening years (West Hawaii Today, 2024).

Neither ordinance imposed a workable platform liability regime within our estimation window. Bill 89 did contain nominal platform liability and monthly reporting provisions, but the platform reporting requirement was suspended as a condition of dismissal in the 2019 *Kokua Coalition v. DPP* stipulation (Honolulu Star-Advertiser, 2019; Honolulu Civil Beat, 2025; Kokua Coalition dba Hawaii Vacation Rental Owners Association, 2019), and the city's voluntary November 2020 agreement with Airbnb and Expedia produced, per DPP's own admission, zero monthly reports through 2025 (Honolulu Civil Beat, 2025). Bill 108 contained no platform liability provision at all. Honolulu's Ordinance 25-50 (June 2025), which introduces the first enforceable platform liability regime in the state, lies outside our estimation window.

**The hosted-unhosted scope gap**

Both ordinances restrict unhosted TVRs and leave owner-present hosted B&B-style rentals largely untouched. Bill 108's definition of an STVR explicitly requires that the owner or operator not reside on the building site (Hawai'i County Planning Department, 2024). Hosted rentals are outside the bill's scope. Bill 89 similarly distinguishes unhosted TVUs from hosted B&B Homes, permitting up to 0.5% of the island's dwelling-unit stock (approximately 1,715 units) as newly registered B&Bs (Hawai'i Appleseed Center for Law & Economic Justice, 2019; Imanaka Asato LLLC, 2019). Both scopes leave a low-cost reclassification margin: an operator can remain on-platform by shifting from whole-home to room-only or owner-present hosting and formally exit the regulated category without exiting the market. The Airbnb panel shows this directly. Post-2019Q3 Honolulu non-resort listings exhibit a sharp drop in the entire-home share without a commensurate drop in total listings, consistent with reclassification rather than withdrawal. The Hawaii county non-resort series shows no such shift, consistent with the lighter enforcement posture that made reclassification unnecessary.

The B&B registration cap is also substantially under-utilized. Hawai'i Appleseed Center for Law & Economic Justice (2019) and Honolulu Civil Beat (2025) indicate that new B&B permit issuance has remained well below the roughly 1,715-unit cap likely because the owner-occupancy requirement disqualifies the non-resident investor operators that dominate the non-compliant stock, and partly because the registration fee and reporting burden shift the hosted category closer to long-term rental economics for the remaining eligible operators. The cap is therefore an available but unfrequently used legal exit, and the bulk of the non-compliant non-resort stock appears simply to continue operating without reclassification.

**Sticky operators and the COVID rebound**

The post-pandemic recovery in Hawaii tourism (2021–2022) lifted TVR profitability in both counties, rewarding operators who did not exit under the 2019 ordinances and inducing entry of new operators under nominal non-compliance. Hunden Partners (2025) document Hawaii Island active listing counts rising from the low thousands in 2019 to 8,008 by March 2025 despite Bill 108's prohibition on new non-resort STVRs. On Oahu, the 5,000-to-3,000 sharp drop documented by Honolulu Magazine (2020) in the months immediately following Bill 89's effective date had, by 2023, substantially reversed on the non-resort margin. Honolulu Civil Beat (2025) reports that the bulk of active non-resort listings on major platforms in 2023 were not traceable to a Bill 89 permit. Absent a mass re-audit campaign–which neither county has funded–the marginal cost of continued non-compliance remained low relative to expected revenue, and the stock of non-compliant listings grew with the broader tourism recovery.

**Bill 108 enforcement-gap factors.**

In compact form, the seven operative features of the Bill 108 regime that push the Cohort 1 first stage toward zero are:

1. Grandfather flood: Approximately 988 NUC applicants at the close of the September 2019 window (Burnett, 2021); attrition to ≈768 by April 2021; the NUC pool exceeds the Airbnb observable non-resort pool by a factor of roughly 2.6.

2. No pre-ordinance permit system: Hawaii county had no STVR registry before Bill 108 (Hawai'i County Planning Department, 2024); pre-period compliance counts do not exist, and the first stage denominator is effectively censored.

3. Complaint-driven enforcement: The Planning Department ran no proactive investigation regime and depended on resident complaints, which under the prevailing fiscal posture was effectively a tight rate limit on enforcement actions (Hawai'i County Planning Department, 2024).

4. Agricultural zone legal uncertainty: The *Rosehill* challenge produced interpretive ambiguity on AG-zoned STVRs until the Hawaii Supreme Court's September 2024 ruling upholding the prohibition (West Hawaii Today, 2024); administrative action in agricultural zones was correspondingly cautious in the intervening years.

5. No platform liability: Ordinance 18-114 contains no platform liability provision (Hawai'i County Planning Department, 2024; County of Hawaii, 2018); the only enforcement channel is operator-level.

6. Hosted rentals unregulated: Owner-occupied hosted rentals are outside the Hawaii County Code scope (Hawai'i County Planning Department, 2024; Hawaii Life, 2019), providing a low-cost reclassification exit.

7. COVID-era entry under non-compliance: The 2021-2022 tourism recovery lifted marginal revenue enough to induce new entrants despite the statutory prohibition (Hunden Partners, 2025), and the observed Airbnb non-resort stock in Hawaii County rises rather than falls through 2025.

**Bill 89 enforcement-gap factors.**

The parallel Bill 89 enumeration:

1. Kokua Stipulation weakening: The October 2019 *Kokua Coalition v. DPP* stipulation (Case No. 1:19-cv-00414, D. Haw.) narrowed the city's platform reporting authority and clarified that the 30-day rule does not require physical occupancy, removing the interpretive lever that would have made contract-only 30-day stays prima facie evidence of a violation (Kokua Coalition dba Hawaii Vacation Rental Owners Association, 2019; Honolulu Star-Advertiser, 2019; Honolulu Civil Beat, 2025).

2. Monthly platform reports never materialized: The November 2020 voluntary agreement with Airbnb and Expedia produced zero monthly reports through 2025 per DPP's own statement to Honolulu Civil Beat (2025), eliminating the most powerful enforcement channel contemplated by the ordinance.

3. HILSTRA injunction of the 90-day rule: *HILSTRA v. City & County of Honolulu* (Case No. 1:22-cv-00247, D. Haw.) enjoined the Ordinance 22-7 90-day minimum in October 2022 (Honolulu Civil Beat, 2022), preserving the 30-day threshold and the associated scope gap throughout our estimation window.

4. Under-utilization of the B&B cap: The 0.5% (≈1,715) cap on new B&B Home permits remains well below utilization (Hawai'i Appleseed Center for Law & Economic Justice, 2019; Honolulu Civil Beat, 2025); the hosted reclassification margin is available but lightly exercised.

5. Majority non-compliance in 2023: The bulk of active non-resort listings on major platforms in 2023 were not traceable to a Bill 89 permit (Honolulu Civil Beat, 2025), implying that the DPP enforcement campaign of 2019–2020 had not persisted at anything like its initial intensity.

6. Hosted rentals partially insulated: The B&B category continues to permit hosted short-term occupancy outside the Bill 89 TVU prohibition (Imanaka Asato LLLC, 2019; City and County of Honolulu, 2019), providing the same reclassification exit observed in Hawaii County.

7. Post-pandemic listing rebound: The 5,000-to-3,000 drop documented by Honolulu Magazine (2020) immediately post-enactment substantially reversed on the non-resort margin by 2023-2025 per Honolulu Civil Beat (2025) and the Airbnb panel in Figure 7.

Taken together, the Wald-DID LATET we identify is defined over the narrow set of complier properties whose TVR exposure actually changed because of the ordinance–a set shrunk by the grandfather flood, the scope gap, the sticky operator dynamics, and, in Bill 89's case, the stipulation-era suspension of platform enforcement. The point estimate should be interpreted as an effect for this subsample, not as the average effect of the regulation on the ex-ante-illegal non-resort stock that policymakers targeted.

## A.12 Technical observations on interference and its implications for the LATET

Section 6.2 reports the empirical displacement test and its principal reading. Here we formalize the cross-zone spillover bias the test is designed to detect and its bias direction on the first stage results.

Let $D_{it} \in \{0, 1\}$ instead denote the binary TVR-operating status of property $i$ in zone $j(i)$ at time $t$, and let $Z_{j(i)t}$ be the zoning instrument. The SUTVA the canonical design requires has is two assumptions stitched together. The assumption of no interference requires potential outcomes and potential treatment assignments depend only on a unit's own instrument value: $Y_{it}(Z_{1t}, \ldots, Z_{Nt}) = Y_{it}(Z_{j(i)t})$ and $D_{it}(Z_{1t}, \ldots, Z_{Nt}) = D_{it}(Z_{j(i)t})$. Consistency (or "no hidden versions of treatment") requires the treatment variable admit a single, well-defined version. Spatial displacement violates the no interference: an operator relocating from a treated non-resort zone to an always-allowed resort zone links the potential treatment status of the destination unit to the instrument value of the source zone. The reclassification response we document in section 6.3 instead violates the second component, because the nominally regulated treatment variable, "non-resort TVR operation," admits two distinct versions post-2019 in Honolulu: a regulated, unhosted TVU category that Bill 89 restricts, and an effectively unregulated, hosted B&B-home category that survives the ordinance. Our first stage measures a decline in the first version; the second continues to operate and is not observable. We exclude discussion of it as it is not a primary focus of our analysis.

Let $\pi_{\text{NR}}(z)$ and $\pi_{\text{R}}(z)$ denote the share of unhosted TVR operation in treated non-resort and always-allowed resort zones respectively, and let $\beta_{\text{NR}}$ and $\beta_{\text{R}}$ denote the structural price effects of TVR operation in each zone type. If Bill 89 and Bill 108 induce relocation from non-resort to resort zones within the treated counties, then $\Delta\pi_{\text{NR}} \equiv \pi_{\text{NR}}(1) - \pi_{\text{NR}}(0) < 0$ and $\Delta\pi_{\text{R}} \equiv \pi_{\text{R}}(1) - \pi_{\text{R}}(0) > 0$. Under no-interference the Wald-DID denominator would be the unbiased non-resort contraction $\Delta\pi_{\text{NR}}$; under displacement it measures instead the aggregate net flow $\Delta\pi_{\text{NR}} + \Delta\pi_{\text{R}}$, which is smaller in magnitude. The first stage denominator is thus pushed toward zero, and mechanically pushes $|\beta^W|$ toward zero whenever the numerator channel does not expand by more than the denominator does. The reduced form numerator is simultaneously contaminated by the price response in the resort zones receiving the displaced stock, with sign equal to $\text{sign}(\beta_{\text{R}}) \cdot \Delta\pi_{\text{R}}$. The sign of the net bias on $\beta^W$ is therefore generally unclear without auxiliary restrictions on $(\beta_{\text{NR}}, \beta_{\text{R}})$, but the denominator channel on its own attenuates the Wald ratio magnitude toward zero, consistent with the conservative interpretation we advance in support of our headline results (de Chaisemartin and D'Haultfœuille, 2018).

The displacement test in section 6.2 is designed to identify exactly the second inequality, $\Delta\pi_{\text{R}} > 0$, via the resort-zone first stage event study. A positive post-treatment path on the resort-zone first stage is the direct empirical signature of displacement; a flat or negative path is consistent with $\Delta\pi_{\text{R}} \approx 0$ under the realized enforcement regime, which is what we observe.

The displacement test is, under the realized enforcement regime, an honestly uninformative exercise: we cannot reject the null of no spatial displacement, but we also cannot affirm it, because the COVID-era common shock on resort-zone TVR density dominates any margin the test could have detected. We maintain our interpretation is consistent with, and reinforces, the qualified-null perspective of the headline results in 5, which is that our point estimates are robust to cross-zone spatial displacement of the magnitude the test is powered to detect.

### A.13 Additional results on heterogeneity from yearly specifications

Tables A10 and A11 reproduce the four column heterogeneity results of section 7.1 and 7.2 at annual frequency. The yearly post indicator combines both ordinances into a single year assignment (both Bill 108 and 89 take effect in calendar year 2019), and the TVR density measure aggregates to a span-weighted activity proxy to preserve identifying temporal variation. The estimating equations are identical to the ones we re-estimate in section 7 with time indices replaced by year and the cohort instrument modified accordingly.

Table A10: Effect of TVR density on residualized log land value (transaction-level) — yearly — Condo, residualized outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | -0.0269 | 0.2800 | 0.0917 | 0.8166 |
| | (1.1879) | (0.3128) | (0.9114) | (0.6176) |
| AR 95% CI | [-4.3847, 2.7788] | $[-\infty, +\infty]$ | [-6.5494, 3.9848] | [-0.2697, 2.9580] |
| TVR density (+ dist control) | 0.0850 | -4.4695 | 0.0169 | 0.4439 |
| | (0.5001) | (70.9414) | (0.5624) | (0.3003) |
| AR 95% CI | [-1.0960, 1.2158] | $[-\infty, +\infty]$ | $\emptyset$ | [-0.0238, 1.6359] |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.2249$^{**}$ | 0.2888 | -0.2564$^{*}$ | -0.3996$^{***}$ |
| | (0.0957) | (0.2437) | (0.1377) | (0.1443) |
| Fixed effects | Y | Y | Y (no coh×type) | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 5.5 | 1.4 | 3.5 | 7.7 |
| $R^2$ | 0.239 | 0.369 | 0.240 | 0.276 |
| N observations | 7,240 | 2,531 | 13,766 | 19,059 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: building area, age, pool, baths, bedrooms, lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. Stacked 2SLS omits hedonic regressors; cohort × property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + year. Fixed effects for (3): cohort × tract + cohort × year (cohort × property-type FE omitted because sample is a single property type). Standard errors clustered at the zone level in parentheses. $^{***}$ $p < 0.01$, $^{**}$ $p < 0.05$, $^{*}$ $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

The qualitative patterns documented in section 7.1 and 7.2 survive the chain in frequency in two senses and break down in one. The SFR cohort 2 cell remains the strongest identified cell in the re-estimation at this frequency (−0.207 with cluster-robust first stage

Table A11: Effect of TVR density on residualized log land value (transaction-level) — yearly — SFR, residualized outcome

| | Honolulu Wald-DID (1) | Hawaii County Wald-DID (2) | Stacked 2SLS (3) | Naive TWFE (4) |
|---|---|---|---|---|
| | *Panel A: Two-Stage Least Squares* | | | |
| TVR density | -0.2071 | -2.6870 | -1.1447 | 0.4108 |
| | (0.8006) | (2.0314) | (1.0416) | (0.3421) |
| AR 95% CI | [-2.6613, 1.3619] | [-21.1638, 0.2733] | [-5.5937, 0.5825] | [-0.3627, 1.2531] |
| TVR density (+ dist control) | -0.1642 | -2.8053 | -1.1162 | 0.4706 |
| | (0.7690) | (2.1181) | (1.0303) | (0.3275) |
| AR 95% CI | [-2.4441, 1.4974] | [-28.3469, 0.2813] | [-4.9991, 0.5923] | [-0.2370, 1.3098] |
| | *Panel B: First Stage for TVR Policy* | | | |
| Zoning policy | -0.0965*** | -0.0612** | -0.0816*** | -0.1964*** |
| | (0.0249) | (0.0253) | (0.0260) | (0.0529) |
| Fixed effects | Y | Y | Y (no coh×type) | Y |
| Hedonic controls | Y | Y | N | Y |
| First-stage F | 15.1 | 5.8 | 9.9 | 13.8 |
| $R^2$ | 0.155 | 0.392 | 0.242 | 0.213 |
| N observations | 4,133 | 2,022 | 8,363 | 14,549 |

*Notes:* Bracketed rows report 95% Anderson–Rubin confidence intervals, constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization of outcome, endogenous variable, and instrument against fixed effects and hedonic controls. Hedonic controls: building area, age, pool, baths, bedrooms, lot area, TEZ indicators, elevation, tract BA share, tract mean HH income. Stacked 2SLS omits hedonic regressors; cohort $\times$ property-type FE absorbs property-type variation. Fixed effects for (1), (2), and (4): zone + year. Fixed effects for (3): cohort $\times$ tract + cohort $\times$ year (cohort $\times$ property-type FE omitted because sample is a single property type). Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument, computed under the same zone-level clustering as the coefficient standard errors (`fixest::fitstat(m, "ivwald")`).

$F$-statistic $F = 15.1$ and AR interval $[-2.66, 1.36]$), but the condo cohort 2 estimate attenuates toward zero ($-0.027$ and first stage $F$-statistic $F = 5.5$). As has been the theme across several specifications throughout our paper, the condo arm loses power under aggregation: temporal averaging of a sharply noisy quarterly series compresses the point estimate without correspondingly compressing standard errors.

The sign on the SFR cohort 2 yearly estimates also flip relative to the quarterly version, and we take this (in conjunction with the yearly stacked estimates) as a result of the COVID-19 pandemic post window absorbing structural pressure on SFR prices the quarterly specification can isolate at the third quarter of 2019 but the yearly specification cannot.

Table A12 reproduces the distance band estimates of section 7.2 at annual frequency. Concisely, two patterns are of note. First, the quarterly pattern of the boundary and intermediate first stage ranking inverts mildly. First stage $F$-statistics are weakly smaller across all bands and property types, suggesting boundary cells remain better identified and that identification in the intermediate bands becomes less defensible at annual frequency.

Second, the condo intermediate band yearly estimate delivers a conventionally significant but weak-instrument non-significant Wald ratio of 0.37 with cluster robust standard

Table A12: Cohort 2 Wald-DID on TVR density (residualized log land value), by property type and distance band — yearly

| | Condo | | SFR | |
|---|---|---|---|---|
| | ≤1 km | 1–5 km | ≤1 km | 1–5 km |
| | (1) | (2) | (3) | (4) |
| *Panel A: Two-Stage Least Squares* | | | | |
| TVR density | -1.0258 | +0.3725** | +1.8466 | -0.7329 |
| | (3.4938) | (0.1877) | (3.2260) | (1.0086) |
| AR 95% CI | [-12.7888, 5.1190] | [-0.0896, 2.1362] | [-6.4210, 12.7079] | [-9.5012, 2.0547] |
| *Panel B: First Stage for TVR Policy* | | | | |
| Zoning policy | -0.1130*** | -0.8088** | -0.0705*** | -0.1017** |
| | (0.0329) | (0.3411) | (0.0203) | (0.0473) |
| First-stage F | 11.8 | 5.6 | 12.1 | 4.6 |
| $R^2$ | 0.251 | 0.156 | 0.235 | 0.132 |
| N observations | 5,730 | 1,128 | 715 | 1,502 |
| Clusters (zones) | 72 | 23 | 90 | 59 |

*Notes:* Fixed effects: zone + year. Resort-zone observations are excluded from all cells. Sample restricted to the listed distance band; $>$ 5 km observations are intentionally excluded (chronically weak first stage and dominated by interior non-resort variation). Standard errors clustered at the zone level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First-stage F is the heteroskedasticity- and cluster-robust Wald statistic on the excluded instrument (`fixest::fitstat(m, "ivwald")`). Cells with first-stage F $<$ 16.38 (Stock-Yogo 2005, 5% IV-bias threshold) should be interpreted under weak-instrument inference; bracketed rows report 95% Anderson–Rubin CIs constructed via 1,000 cluster-level Rademacher sign-flip permutations after Freedman–Lane residualization (grid size 200, seed 42). TVR density measure: `str_density_total_span` (annual Inside Airbnb listings per zone, span-weighted; only `span` provides temporal variation at annual resolution).

error 0.188 (significant at 0.05 level) but AR interval $[-0.09, 2.14]$ that does not reject zero. The discrepancy is the standard weak-instrument inference problem. The conventional TSLS standard error understands the uncertainty when the first stage is borderline (first stage $F$-statistic of $F = 5.6$) and randomization inference is the appropriate default for a statistic of this strength. We flag this finding but do not interpret it as a structural signal.

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